NAVAL AIR TRAINING COMMAND

NAS CORPUS CHRISTI, TEXAS
(TW1) Q-2A-1671; Q-2A-1672
(TW2) Q-2A-7167; Q-2A-8167

CNATRAINST 1542.167D
12 Nov 2024

CHIEF OF NAVAL AIR TRAINING

T-45C UNDERGRADUATE STRIKE AND ADVANCED E-2 TRAINING MASTER CURRICULUM GUIDE

2024

DEPARTMENT OF THE NAVY
CHIEF OF NAVAL AIR TRAINING
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CNATRAINST 1542.167D
N715
12 Nov 2024

CNATRA INSTRUCTION 1542.167D

From: Chief of Naval Air Training

Subj: T-45C UNDERGRADUATE STRIKE AND ADVANCED E-2 TRAINING

1. Purpose. To publish the curriculum for training Naval Flight Students in the Undergraduate Strike and Advanced E-2 phases of training.

2. Cancellation. CNATRAINST 1542.167C will be canceled when the last student enrolled completes the respective curriculum or is transitioned to the CNATRAINST 1542.167D. CNATRAINST 1542.167D chapters 1 and 2, syllabus notes, pre-requisites and course flow are authorized for students enrolled in the CNATRAINST 1542.167C upon receipt.

3. Action. This instruction is effective on receipt. No changes will be made without the written authorization by the Chief of Naval Air Training (CNATRA).

4. Records Management. Records created as a result of this instruction, regardless of media and format, must be managed per Secretary of the Navy Manual 5210.1 of September 2019.

5. Review and Effective Date. Per OPNAVINST 5215.17A, CNATRA N7 will review this instruction annually around the anniversary of its effective date to ensure applicability, currency, and consistency with Federal, DoD, SECNAV, and Navy policy and statutory authority using OPNAV 5215/40 Review of Instruction. This instruction will be in effect for 10 years, unless revised or cancelled in the interim, and will be reissued by the 10-year anniversary date if it is still required, unless it meets one of the exceptions in OPNAVINST 5215.17A paragraph 9. Otherwise, if the instruction is no longer required, it will be processed for cancellation as soon as the need for cancellation is known following the guidance in OPNAV Manual 5215.1 of September 2020.

6. Forms. The CNATRA forms required by this instruction are automated in the Training Learning Management System (T/LMS) computer program. Additional CNATRA forms are available on the CNATRA website https://www.cnatra.navy.mil/pubs-forms.asp.

P. N. FLORES
Chief of Staff

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SUMMARY OF CHANGES

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COURSE DATA

1. Course Title. T-45C Undergraduate Strike and Advanced E-2 Training.

2. Course ID Number (CIN)

a. TW-1: Strike, Q-2A-1671; Advanced E-2, Q-2A-1672.

b. TW-2: Strike, Q-2A-7167; Advanced E-2, Q-2A-8167.

3. Location(s). Naval Air Station (NAS) Kingsville and NAS Meridian.

4. Course Status. Active.

5. Course Mission. T-45C Undergraduate Strike and Advanced T-45C E-2 are designed to qualify graduates for follow-on flight training in operational fleet aircraft and to prepare them for their future responsibilities as military officers.

6. Prerequisite Training.

a. Strike.

(1) Successful completion of Intermediate Strike Pilot Training (ISPT): Q-2A-1201 (TW4) or Q-2A-3201 (TW5).

(2) Successful completion of Centrifuge-based Flight Environment Training (CFET) prior to Basic Fighter Maneuvering (BFM) execution.

b. Advanced E-2. Successful completion of Intermediate E-2 Multi-Service Pilot Training System (MPTS): Q-2A-5175 (TW-4).

7. Security Clearance Requirements. None.

8. Follow-on Training. Assigned by the graduate's parent service.

9. Course Length. Overall time to train is calculated as per the CNATRAINST 1550.6G. Training days account directly or provide margin for factors including weather, personnel and equipment availability, briefing and preparation time, and historical delays. Production weeks further account for weekends, holidays, safety stand-downs, and other expected nonworking days throughout the year.

	Training Days	Production Weeks
TW-1		
Strike	241.2	52.0
Advanced E-2	154.0	33.0
TW-2		
Strike	241.2	52.0
Advanced E-2	154.0	33.0

10. Class Capacity. Variable.

11. Instructor Requirements. As established by Chief of Naval Operations (CNO) planning factors.

12. Course Curriculum Model Manager. Commander, Training Air Wing TWO (COMTRAWING TWO).

13. Quota Management Authority. Chief of Naval Air Training (CNATRA).

14. Quota Control. CNO.

15. Course Training Subjects

a. Ground Training

GROUND TRAINING		
Stage	**Symbol**	**Hours**
Administration/Indoctrination	ASI01	4.7
Engineering/Systems	ENG01	18.5
Aerodynamics	AER01	5.0
Meteorological Principles	MET01	3.0
Instrument Navigation	NAV01	8.8
Crew Resource Management	CRM11	3.0
Operational Risk Management	ORM11	1.0
NACES Flight Physiology	SEA11	3.0
Total		**47.0**

b. Flight Support

FLIGHT SUPPORT		
Stage	**Symbol**	**Hours**
Cockpit Orientation	CO11	5.0
Emergency Flight Procedures (EPs)	EP11	14.5
Course Rules	CR11/CR12	4.0
Basic Instrument Flight Procedures	BI11	8.0
Radio Instrument Flight Procedures	RI11	8.5
Familiarization Flight Procedures	FAM11	8.6
Out-of-Control Flight (OCF) Procedures	OCF11	3.0
NATOPS/NATOPS Examinations	NA11	6.0
Night Transition Flight Procedures	NTR11	3.5
Field Carrier Landing Flight Procedures	FCL11	2.5
Carrier Qualification Flight Procedures	CQ11	6.0
Airways Navigation Flight Procedures	AN11	2.0
Instrument Rating Flight Procedures	IR11	3.3
Formation Flight Procedures	FRM11	5.5

FLIGHT SUPPORT (cont.)		
Stage	**Symbol**	**Hours**
Night Formation Flight Procedures	NFR11	2.2
Division Formation Flight Procedures	DIV11	2.5
Tactical Formation Flight Procedures*	TAC11	4.0
Division Tactical Formation Flight Procedures*	DTF11	2.5
Operational Navigation Flight Procedures*	ON11	15.0
Section Low-Level Flight Procedures*	SLL11	2.5
Strike Flight Procedures*	STK11	5.0
Target Area Mechanics Flight Procedures*	TAM11	2.5
Basic Fighter Maneuvering Flight Procedures*	BFM11	5.8
Fighter Intercepts Flight Procedures*	FTX11	2.5
Strike Total		**124.9**
Advanced E-2 Total		**85.1**

* Advanced E-2 do not execute.

c. Device Training Summary. The programmed times for each phase, stage, and media are:

DEVICE/FLIGHT TRAINING SUMMARY								
Flight/Events	**Simulator**				**T-45C**			
	VTD		**OFT**		**Dual**		**Solo**	
	Flts	**Hrs**	**Flts**	**Hrs**	**Flts**	**Hrs**	**Flts**	**Hrs**
CO			4	6.0				
EP			4	6.0				
TR	1	1.0	30	45.0	16	19.2	1	1.2
OCF			2	2.5	2	1.7		
IR			16	23.8	5	7.5		
NTR			2	2.4	2	2.2	2	2.2
FRM	2	2.0	4	5.8	8	10.4	2	2.6
NFR			2	2.4	2	2.6	1	1.2
DIV	1	1.0			4	5.6	1	1.4
FCL			2	2.8	1	0.6	3	1.8
CQ			3	4.3	1	0.6	13	10.8

DEVICE/FLIGHT TRAINING SUMMARY (cont.)								
Flight/Events	**Simulator**				**T-45C**			
	VTD		**OFT**		**Dual**		**Solo**	
	Flts	**Hrs**	**Flts**	**Hrs**	**Flts**	**Hrs**	**Flts**	**Hrs**
AN*			2	3.0	2	2.8	2	2.4
TAC*	1	1.0			5	5.5	4	4.4
DTF*	1	1.0			2	2.2		
ON*			3	3.9	3	3.3		
SLL*	1	1.0			2	2.0		
STK*	2	2.0	7	9.1	4	3.6	2	1.8
TAM*	1	1.0	2	3.0	2	2.2	1	1.1
BFM*	2	2.0			8	7.6	4	3.7
FTX*	1	1.0	1	1.5	2	2.0	2	2.0
Strike Totals	**13**	**13.0**	**84**	**121.5**	**70**	**81.6**	**37****	**33.0**
Advanced E-2 Totals	**4**	**4.0**	**69**	**101.0**	**41**	**50.4**	**23**	**21.2**

* Advanced E-2 do not execute.
** Strike totals reflect waiver of shipboard carrier qualification (CQ4S91) for USN and USMC NFSs.

16. Course Training Allocation by Media

Media	Events	Hours
Strike		
Academic –Lectures, CAIs, MILs, VBT	155	170.0
VTD	13	13.0
OFT	84	120.7
Dual Aircraft	70	81.6
Solo Aircraft	37	33.0
Lead Overhead	55	46.5
Advanced E-2		
Academic – Lectures, CAIs, MILs, VBT	127	130.2
VTD	4	4.0
OFT	69	100.2
Dual Aircraft	41	50.4
Solo Aircraft	23	21.2
Lead Overhead	20	20.4

17. Training Preparation Time. In addition to the hours formally planned for classes, simulators, and flights, significant additional time to prepare and study should be expected outside of scheduled training hours. This range will vary depending on the complexity of the material and individual NFS's needs, and may be up to several hours per event. For simulator and flight events, specific brief and debrief times shall be programmed into the CNATRA approved Training and Learning Management System (T/LMS) and accounted for on the flight schedule, per the following table:

ADDITIONAL TRAINING TIME PER CURRICULUM EVENT			
Media	**Brief**	**Debrief**	**Total**
OFT/VTD	0.5	0.5	1.0
Aircraft	1.75	1.0	2.75
Aircraft (TR4101, FRM4101, TAC4101, ON4101, STK4101, TAM4101, BFM4201, FTX4101)	2.0	1.0	3.0

MINIMUM NFS TURN TIME PER EVENT (End of Debrief to Brief)				
Media	**To Aircraft**	**To OFT**	**To VTD**	**To GBT**
OFT/Aircraft/VTD	1.0	1.0	0.5	0.5
Ground-Based Training (GBT)/Flt Support	1.0	1.0	0.5	0.5

Notes:
-The above planning factors are for event scheduling and may be reduced during execution in the event an NFS becomes delayed due to maintenance, weather, or other unplanned issues. Instructors shall ensure NFSs receive adequate time to rest and prepare for the next event.
-Self-study events and non-IP guided VTDs do not require turn time to or from any event, however, be aware of eye-strain after VTD execution prior to flying.
-Turn times do not apply to out-and-in, cross-country, round-robin, FCL, CQ, or safe-for-solo to solo profiles (provided one of the IPs is from the safe-for-solo flight, if a multi-plane event). In all cases, the instructor shall ensure adequate debrief and brief time is allocated and the quality of NFS training is not compromised.

18. Physical Requirements. As specified in the Manual of Medical Department, Chapter 15, and all applicable anthropometric standards.

19. Obligated Service. Refer to MILPERSMAN for Naval personnel.

20. Primary Instructional Methods. Lect, MIL, CAI, VBT, self- and group-paced study, simulators, and in-flight instruction.

a. The intent of VBT is for NFSs to study required material outlined in this curriculum at their own pace.

b. In-person reviews with instructors are provided through the use of lectures and MILs to ensure NFSs acquire the necessary knowledge provided via VBT.

21. Preceding Curriculum Data. This curriculum replaces CNATRAINST 1542.167C.

22. NFS Performance Measurement and Application of Standards. The standards outlined in Chapter IX, Course Training Standards, are used to evaluate Naval Flight Student (NFS) performance of individual items and maneuvers. Final judgment regarding the satisfactory performance of any flight maneuver rests with the Instructor Pilot who must assess the environmental and systems factors affecting the conditions under which the performance is measured.

23. Summary of Lead Overhead. The summary of the Instructor Lead planning factor hours for the T-45C Strike and Advanced E-2 curricula are tabulated below. Lead Hours per Event are approximate and are derived by subtracting 0.2 hours from the NFS event length for all events except TAC, DTF, SLL, STK, TAM, BFM, and FTX, where Lead hours are 0.1 hours less than NFS event length. This accounts for NFS landings.

LEAD OVERHEAD				
Flight/Event	**# Events**	**Lead Hrs/Event**	**NFSs/Lead**	**Hrs/NFS**
FRM41	8	1.1	1	8.8
FRM42	2	1.1	1	2.2
NFR41	2	1.1	1	2.2
NFR42	2	1.0	1	2.0
DIV41	4	1.4	2.5	2.24
DIV42	1	1.4	2.5	0.56
CQ4S91	1	3.6	1.5	2.4
TAC41*	3	1.0	1	3.0
TAC42*	2	1.0	1	2.0
TAC43*	2	1.0	1	2.0
TAC44*	2	1.0	1	2.0
DTF41*	2	1.0	1	2.0
SLL41*	2	0.9	1	1.8
STK41*	4	0.8	3	1.07
STK42*	2	0.8	3	0.53
TAM42*	1	1.0	2	0.5
BFM41*	1	0.9	1	0.9

LEAD OVERHEAD (CONT.)				
Flight/Event	**# Events**	**Lead Hrs/Event**	**NFSs/Lead**	**Hrs/NFS**
BFM42*	3	0.9	1	2.7
BFM43*	1	0.8	1	0.8
BFM44*	2	0.8	1	1.6
BFM45*	1	0.8	1	0.8
BFM46*	2	0.8	1	1.6
BFM47*	2	0.8	1	1.6
FTX41*	2	0.9	1	1.8
FTX42*	2	0.9	1	1.8
Strike Totals**	**55**	**N/A**	**31.5**	**46.5**
Advanced E-2 Totals	**20**	**N/A**	**10.5**	**20.4**

* Advanced E-2 do not execute.
** Strike Totals no not include CQ4S91.

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ABBREVIATIONS

The following is a list of abbreviations used in the curriculum:

ADI	-	Attitude Director Indicator
A/G	-	Air-to-Ground
AGL	-	Above Ground Level
AOA	-	Angle of Attack
AOB	-	Angle of Bank
AP/IB	-	Area Planning
ASR	-	Airport Surveillance Radar
ATC	-	Air Traffic Control
ATF	-	Aviation Training Form
ATJ	-	Aviation Training Jacket
AW	-	Attack Window
AWE	-	Attack Window Entry
BTX	-	Blown Tire Exercise
CAI	-	Computer Aided Instruction
CCIP	-	Continuously Computed Impact Point
CDI	-	Course Deviation Indicator
CEP	-	Circular Error Probability
CO	-	Commanding Officer
CO-PC	-	Commanding Officer Progress Check
C AUG	-	Control Augmentation
CNAF	-	Commander, Naval Air Forces
CQ	-	Carrier Qualification
CRM	-	Crew Resource Management
CTS	-	Course Training Standard

CV	-	Carrier
CWS	-	Centralized Warning System
DEU	-	Display Electronics Unit
DME	-	Distance Measuring Equipment
ECA	-	Engine Control Amplifier
ECS	-	Environmental Control System
EDP	-	Engine-Driven Pump
EOB	-	End of Block
EP	-	Emergency Procedure
ET	-	Extra Training
FAF	-	Final Approach Fix
FAR	-	Federal Aviation Regulation
FCLP	-	Field Carrier Landing Practice
FIH	-	Flight Information Handbook
FLIP	-	Flight Information Publication
FOD	-	Foreign Object Debris
FPM	-	Feet per minute
FSS	-	Flight Service Station
FTI	-	Flight Training Instruction
FWD	-	Forward
GCA	-	Ground-Controlled Approach
GINA	-	GPS/Inertial Navigation Assembly
GLOC	-	“G” Induced Loss of Consciousness
GSI	-	Glideslope Indicator
GTS	-	Gas Turbine Starter
HSI	-	Horizontal Situation Indicator

HUD	-	Head-Up Display
IAF	-	Initial Approach Fix
ICS	-	Intercommunication System
IFLOLS	-	Improved Fresnel Lens Optical Landing System
IFR	-	Instrument Flight Rules
ILS	-	Instrument Landing System
IMC	-	Instrument Meteorological Conditions
IP	-	Instructor Pilot
IPC	-	Initial Progress Check
KCAS	-	Knots Calibrated Airspeed
KIAS	-	Knots Indicated Airspeed
KIO	-	Knock-it-off
LAR	-	Launch Acceptability Region
LAW	-	Low Altitude Warning
LSO	-	Landing Signal Officer
LV	-	Lift Vector
MAR	-	Minimum Abort Range
MDA	-	Minimum Descent Altitude
MFCD	-	Multi-function Color Display
MIF	-	Maneuver Item File
MIL	-	Mediated Interactive Lecture
MPTS	-	Multi-Service Pilot Training System
MR	-	Mixed-Reality Device
MRT	-	Maximum Rated Thrust
MSL	-	Mean Sea Level
NATOPS	-	Naval Air Training and Operating Procedures Standardization

NFS	-	Naval Flight Student. Includes International Military Student (IMS)
NORDO	-	No Radio
NWS	-	Nose Wheel Steering
OBOGS	-	On-Board Oxygen Generating System
OCF	-	Out-of-Control Flight
OFT	-	Operational Flight Trainer (2F138 - visual)
P/P	-	Partial Panel (Jet), or Pen/Paper Exam
PA	-	Precautionary Approach
PADS	-	Position, Altitude, Distance, and Speed
PAR	-	Precision Approach Radar
POM	-	Plane of Motion
QOD	-	Question of the Day
RAT	-	Ram Air Turbine
RIP	-	Roll-in Point
ROD	-	Rate of Descent
RPM	-	Revolutions per minute
RRU	-	Ready Room UNSAT
RTB	-	Return to Base
SA	-	Situational Awareness
SRT	-	Standard Rate Turn
SSD	-	Snap-shot Drill
SSR	-	Special Syllabus Requirement
STAR	-	Standard Terminal Arrival
TACAN	-	Tactical Air Navigation System
TOT	-	Time on Target
UHF	-	Ultra High Frequency

UNSAT	-	Unsatisfactory
VBT	-	Video-Based Training
VDP	-	Vertical Descent Point
VFR	-	Visual Flight Rules
VHF	-	Very High Frequency
VSI	-	Vertical Speed Indicator
WRT	-	With respect to
VTD	-	Virtual Training Device (Indicates usage of VR and MR training devices)
VMC	-	Visual Meteorological Conditions
VOR	-	VHF Omnidirectional Range
VR	-	Virtual-Reality Device
WEZ	-	Weapons Engagement Zone
WU	-	Warmup
WX	-	Weather
WYPT	-	Waypoint

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GLOSSARY

1. Advancing X. Completed event within normal syllabus flow, or advancing warmup (86).

2. Aviation Training Form (ATF). A grade sheet documenting NFS performance for all categories of training regardless of media, phase, or stage.

3. Aviation Training Jacket (ATJ). The ATJ is the NFS's training record. It contains ATFs, calendar cards, grade reports, and all other associated training information. It is filed in student control and follows the NFS through all phases of training.

4. Block of Training. A sequential series of lessons within a training stage sharing an identical MIF. The second number in the lesson designator identifies a block.

5. Check Event (XX9X). An event check in any stage of training.

6. Class Advisor. An IP assigned to provide counseling and guidance to a specific NFS or a group of NFSs throughout the applicable syllabus.

7. CO Progress Check (CO-PC) (XX89). A special check normally given by the CO or XO. The CO may designate, in writing, CO-PC duty to a qualified O-4 or above. This is only done if the CO or XO is unqualified or unavailable to instruct in the required stage. A satisfactory CO-PC returns the student to normal syllabus flow. An UNSAT CO-PC results in a TRB.

8. Course Training Standard (CTS). A description of required behaviors and standards of performance for a specific maneuver. These standards are in Chapter IX.

a. As NFSs progress through this syllabus, many maneuver items become combined in overarching CTS references. For example, "takeoff" is listed as a separate maneuver item early in training, but will fall under its overarching CTS reference "admin" as training focus shifts toward additional maneuver items.

b. The intent of this consolidation of maneuver items is to reinforce the importance of tying individual skills and maneuvers together into a singular overall effort.

9. Courseware. The technical data, FTIs, audio, video, instructor guides, NFS study guides, and other training material developed to support and implement the syllabus of instruction.

10. End of Block (EOB). Last event in block.

11. Emergency Procedure (EP). Any degradation of aircraft systems or flight conditions requiring pilot action or intervention.

12. Extra Training (XX87). Additional NFS training flights ordered by the Commanding Officer or higher in addition to programmed syllabus events.

13. Flight Training Instruction (FTI). A CNATRA-approved manual describing flight procedures and techniques for each training stage.

14. Hours per X (H/X). Programmed length for an event, rounded to nearest tenth of an hour.

15. Initial Progress Check (XX88). A special check given by the most experienced instructors that have a complete understanding of NATRACOM and PC processes and understand the gravity of their responsibility in helping maintain the standards of Naval Aviation. These instructors should carry the CO's full confidence and shall be designated in writing by the CO. A satisfactory IPC returns the student to normal syllabus flow. An IPC can count as an Advancing "X" at the CO's discretion. An UNSAT IPC results in a CO-PC. IPC events shall be coded as XX88, (e.g., OPS3188).

16. Lesson Designator. All syllabus events have a lesson designator consisting of a stage identifier of up to three letters and an event code of four numbers in the following format:

Char	Meaning	Remarks		
1st - 3rd	Stage	AER - Aerodynamics AN - Airways Navigation ASI - Aviation NFS Indoctrination BFM - Basic Fighter Maneuvering BI - Basic Instruments CO - Cockpit Orientation CR - Course Rules CRM - Crew Resource Management CQ - Carrier Qualification DIV - Division Formation DTF - Division Tactical Formation ENG - Engineering EP - Emergency Procedures FAM - Familiarization FCL - Field Carrier Landing FRM - Formation FTX - Fighter Intercepts IR - Instrument Rating	MET - Meteorology NA - NATOPS NAV - Instrument Navigation NTR - Night Transition NFR - Night Formation OCF - Out-of-Control Flight ON - Operational Navigation ORM - Operational Resource Management RI - Radio Instruments SEA - Seat SLL - Section Low-Level STK - Strike/Air-to-Ground Weapons TAC - Tactical Formation TAM - Target Area Mechanics TR - Transition	
4th	Media*	0 - Ground Training 1 - Flight Support	2 - EP Simulator (OFT) 3 - Simulator (OFT) 4 - Aircraft	6 - Virtual Training Device (VTD)
5th - 7th	Block/ Event/ Check Identifier	Sequential, indicating block, event within block, and other event types as shown below: 84 - Adaptation 85 - Practice Sim 86 - Warmup 87 - Extra Training	88 - Initial Progress Check 89 - CO Progress Check 9X - Check Flight/Exam S - Solo	

* VTD is used to indicate the use of VR or MR training devices.

17. Maneuver Item File (MIF). A listing of all non-demo maneuvers and associated proficiency levels for each block of training. A plus sign (+) indicates the maneuver is required and must be accomplished to the specified standard in that block of training. "1" indicates the item is optional, and if executed, must attain the last MIF level required for that item.

18. Phase of Training. A phase consists of a major division in the course of training. CNATRAINST 1542.167D consists of two phases: Strike, which serves as undergraduate-level jet training for TACAIR NFSs, and Advanced E-2, which serves as undergraduate-level jet training for E-2 NFSs. Upon completion of their respective phase, NFSs will be assigned to the appropriate Fleet Replacement Squadron.

19. Pink ATF. A standard ATF that is printed on pink paper. The pink ATF is used to denote an UNSAT event generating a progress check.

20. Progress Check Pilot. An instructor pilot designated in writing by the CO to administer Initial or CO Progress Checks.

21. Ready Room UNSAT (RRU). An UNSAT grade given for inadequate knowledge of flight procedures, systems, discuss items, EPs, deficient preflight planning, or failure of a non-academic examination (e.g., NATOPS quiz/exam). Missing a brief does not constitute an RRU but shall be documented on a supplemental ATF.

22. Shotgun Solo. Solo flights flown with an IP Safety Observer for **weather** requirements only. It is not a tool to mitigate sub-standard NFS performance. Shotgun solo flight hours do not count toward NFS solo flight hours.

23. Special Syllabus Requirements (SSR). A one-time, ungraded demonstration item(s) or other special training that requires documentation. SSRs are annotated on MIF tables with a "1". While SSRs are recommended for certain events in block, they may be accomplished at any time during the block.

24. Stage of Training. A stage consists of all training of a particular type (Engineering, Transition, Operational Navigation, Formation, etc.) within a phase. The first three letters in the lesson designator identify the stage of each lesson (example: FRM4101 is in the Formation Stage). Refer to the Lesson Designator Table for a complete listing of stages.

25. Supplemental ATF. A form inserted into an NFS's ATJ that contains information on non-standard situations or as directed by this or higher directives.

26. Training Review Board (TRB). A fact-finding board appointed to conduct an administrative review of training following a failed CO-PC. A TRB will generate a CNATRA 1542/1827 TRB Summary Form, which summarizes a specific NFS's progress in a given syllabus and provides detailed information on the application of MPTS training for that NFS. This document indicates whether the quality and continuity of training provided was per the CNATRAINST 1542.167D and CNATRAINST 1500.4L.

Chapter I

General Instructions

1. Syllabus Management

a. Distribution. Participating squadron personnel.

b. Interpretation. The syllabus is directive. Should circumstances create situations not covered within the scope of this syllabus or a specific course of action appears to conflict with other directives, consult CNATRA (N71).

c. Deviations. Document all deviations on the event's ATF or a supplemental ATF if outside of a syllabus event.

d. Changes. Shall be submitted per CNATRAINST 1550.6G.

e. Execution

(1) All flights shall be flown from the front cockpit unless otherwise delineated.

(2) All events shall have access to the Heads-Up Display (HUD) or HUD repeater unless otherwise delineated. Use of partial panel training is encouraged throughout the syllabus, especially during instrument stages of training.

(3) Electronic Kneeboard (EKB) navigation functions shall only be utilized on instrument events (simulator or aircraft) executed outside the local area.

(4) The EKB is primarily a reference tool and shall be properly stowed during simulator and flight events unless specific instances require its temporary usage.

f. Syllabus Description. The T-45C Undergraduate Strike and Advanced E-2 Curriculum consists of undergraduate flight training for USN, USMC, and IMS NFSs, flown in the T-45C aircraft. Each training stage is subdivided into training blocks. The training blocks consist of a specified number of events. Maneuver item files (MIF) identify the acceptable level of performance that must be achieved by the completion of each training block.

g. Grade Calculation

(1) Phase Aggregate Score (PAS). See CNATRAINST 1500.4L.

(2) NSS calculation. See CNATRAINST 1500.4L. All Phase UNSAT events apply toward NSS regardless of event code. However, the following blocks or events graded items shall not count toward NSS calculation:

OCF3XXX	OCF4XXX	FCL31XX	CO31XX	CO32XX
FCL4S01-02	CQ21XX	CQ31XX	CQ4S01-11	

2. Training Management

a. Syllabus Progression. Fly syllabus events within each stage sequentially. Do not start a block without all prerequisites. NFSs may be in different stages or blocks simultaneously. Where applicable, NFSs will be eligible for, and shall be prepared for, more than one syllabus event. NFSs must complete all events except as listed in paragraph 1e. The flowcharts on pages I-5 through I-15 delineate the sequence of flying events and their ground training prerequisites except as listed in paragraph 1e and 2b. System training management is designed to facilitate two graded events (flight, simulator, or exam) per NFS per day.

b. Accelerated Progression

(1) TAP (Training Acceleration Program). NFS with previous flight experience may warrant accelerated progression through the syllabus. TAP is a specifically tailored syllabus deviation, designed and approved by the squadron CO, printed on a Supplementary ATF, placed in the NFS's ATJ, specifying which syllabus events are waived based on the NFS previous flight experience. This policy shall not be used to meet squadron production goals; it is strictly for instances where previous flight experience makes completion of all events within a block of instruction unnecessary. All ATFs for the accelerated NFS will be clearly marked "Proficiency Adv" in the appropriate ATF dropdown box. The squadron CO shall closely monitor the progress of TAP NFS. If performance suffers due to participation in TAP, the NFS shall return to normal syllabus progression.

(2) Proficiency Advance. NFS with demonstrated exceptional performance may warrant accelerated progression, known as "Proficiency Advance." Squadron CO is encouraged to support a squadron program where a NFS may be advanced to the next block of instruction when all required items for the current block meet or exceed performance prerequisites for the follow-on block of training:

(a) The squadron CO shall closely monitor the progress of Proficiency Advanced NFS. If performance suffers, the CO has options to assist the NFS with meeting performance expectations, e.g., awarding ETs or using Proficiency Reallocation.

(b) All ATFs for the Proficiency Advanced events will be marked "Proficiency Adv" in the appropriate ATF dropdown box.

(c) This policy shall not be used as a shortcut to meet squadron production goals; it is strictly for instances where demonstrated proficiency makes completion of all events within a block of instruction unnecessary.

(d) In addition to CNATRAINST 1500.4L requirements, proficiency advancement decisions are delegated to the individual IP. Furthermore, events designated Provisional Safe-for-Solo, such as FRM4107, shall be flown by an "X" or "S" qualified IP.

c. Multiple Stage Enrollment

(1) Other than noted exceptions, syllabus events shall be flown sequentially within each stage. Blocks shall not be started without all prerequisites completed. NFSs must complete all events in their assigned phases unless otherwise noted.

(2) NFSs may be in different stages or blocks simultaneously. Where applicable, NFSs will be eligible for, and shall be prepared for, more than one syllabus event. System training management is designed to facilitate up to two graded events (flight, simulator, exam, or combination thereof) per NFS per day.

(a) NFSs may be enrolled in three stages of training simultaneously, while only scheduled or executing two per day. The third stage of training shall not be executed same day. The intent is to allow both beneficial NFS cross-training and operational flexibility while not overwhelming the NFS with excessive study and training material. See below for additional multiple stage enrollment information.

(b) Events executed within the following stages will update stage event currency across all stages in that group:

1. CO, EP, TR and IR.

2. FRM, NFR, and DIV.

3. STK and TAM. Considered a single stage for enrollment purposes.

4. ON and SLL. Considered a single stage for enrollment purposes.

5. BFM and FTX.

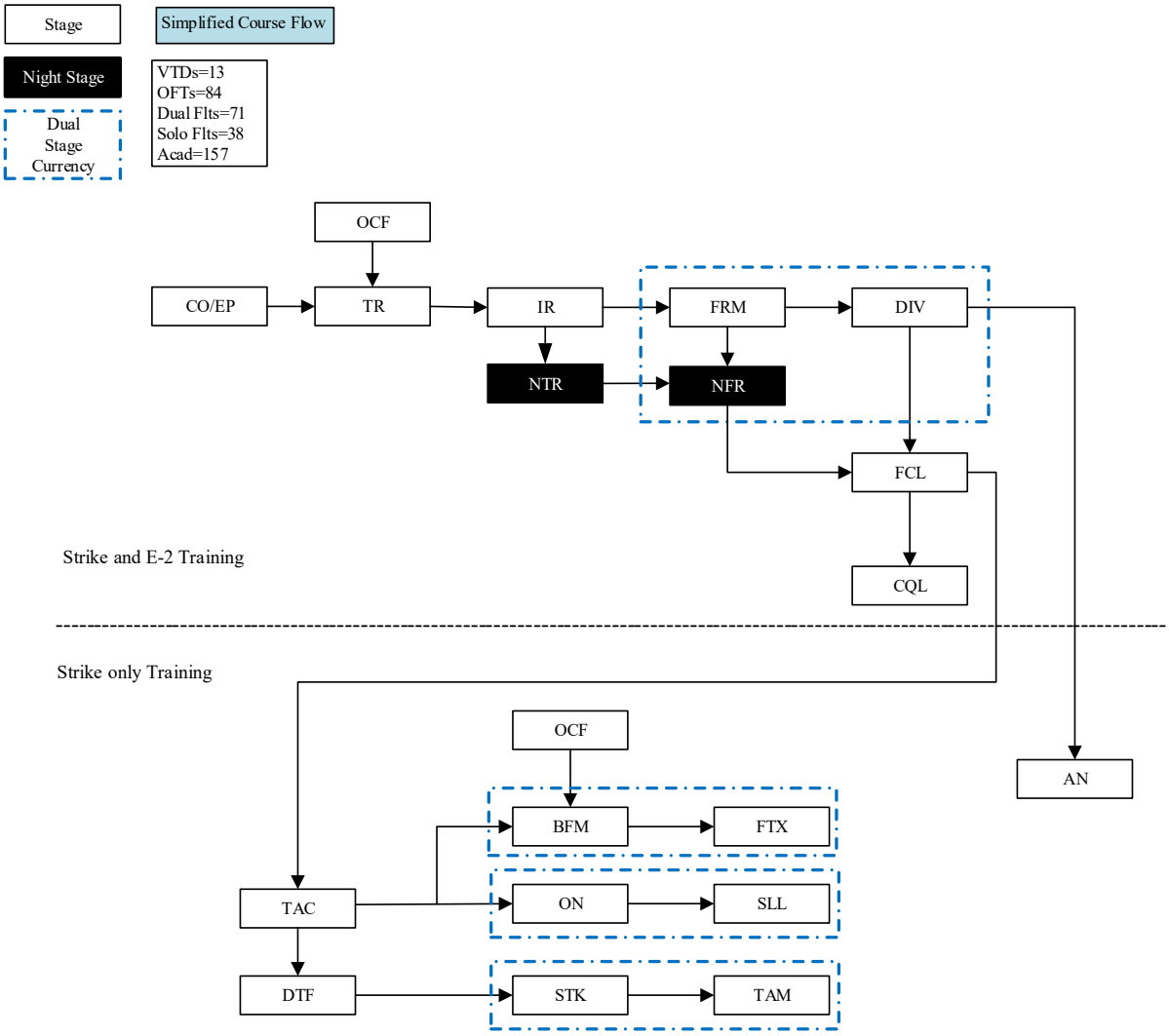
(c) BFM and FTX Notes. NFS shall not participate in any other stage of training until complete, unless pausing BFM/FTX training to execute Carrier Qualification (CQ) stage. Exception: NFS may also participate in AN or DTF in conjunction with BFM/FTX.

(d) CQ Notes. NFS may stop training in CQ due to unforeseen Carrier Qualification Detachment date slides. While training in CQ, NFS shall not execute events in other stages with the exception of AN events. With all prerequisites met, Carrier Qualification Pilot Program (CQPP) NFSs may start CQ at any point regardless of detachment dates. All NFSs shall receive a face-to-face debrief from a qualified LSO after a CQ/FCL event, and prior to the next CQ/FCL event.

(e) AN Notes. NFS may execute AN events to or from detachment sites when enrolled in other stages. AN events have no multiple stage enrollment restrictions.

d. Non-NFS Execution. Winged Aviators executing Pipeline Transitions (e.g., E-2 to VFA/VAQ transition) shall execute T-45C Undergraduate Strike training stages that were not previously completed during initial undergraduate training. Stages previously completed shall be waived and annotated on a supplemental ATF. TRAWING Commodores shall determine current stage equivalents from previous MCGs and note them on the supplemental ATF (ex: BI, RI, and FAM translate to TR stage).

SIMPLIFIED COURSE FLOW



Note:
This is a simplified Stage Flow Chart, additional prerequisites are listed within the syllabus.

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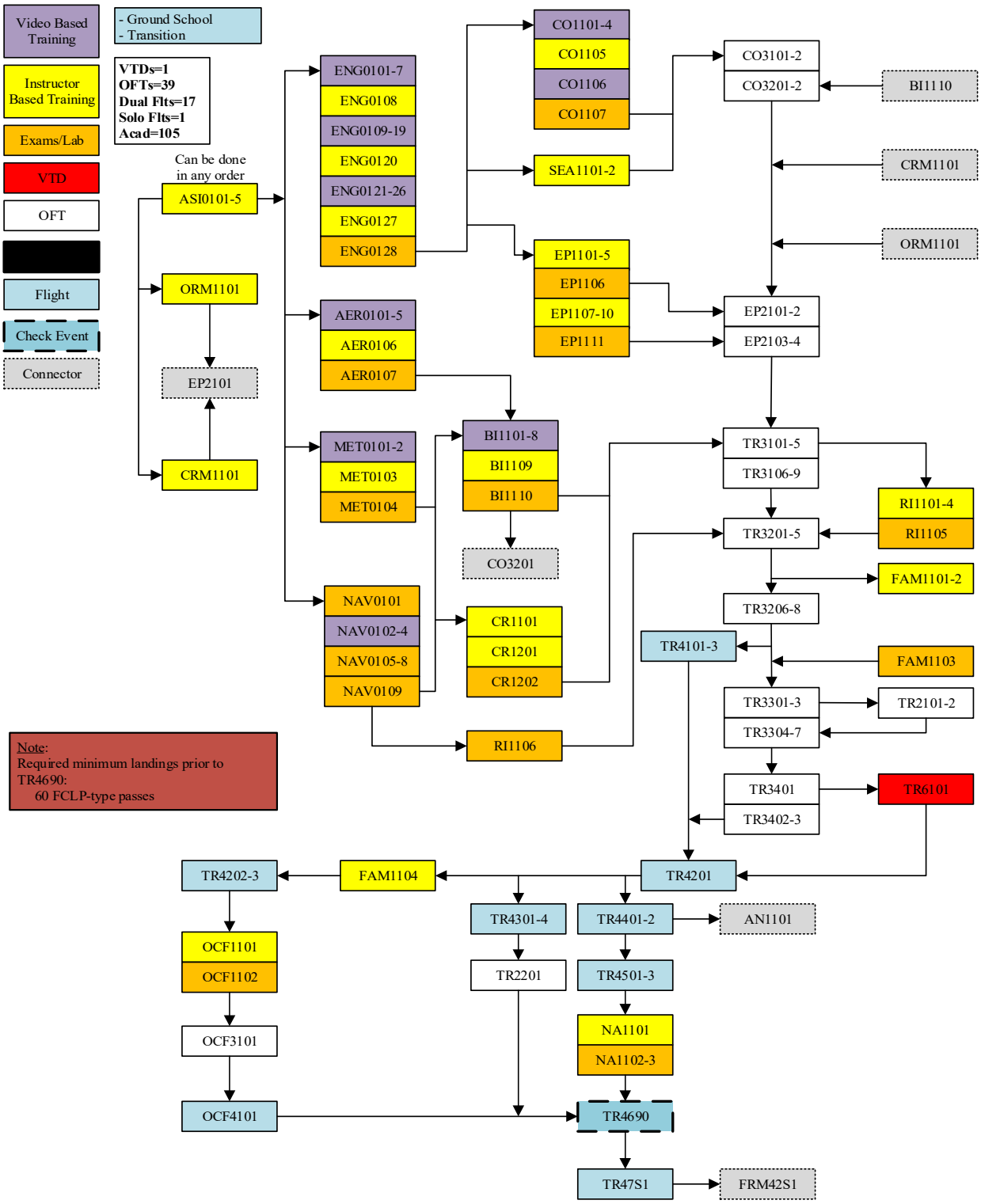
GROUND TRAINING AND TRANSITION FLOW



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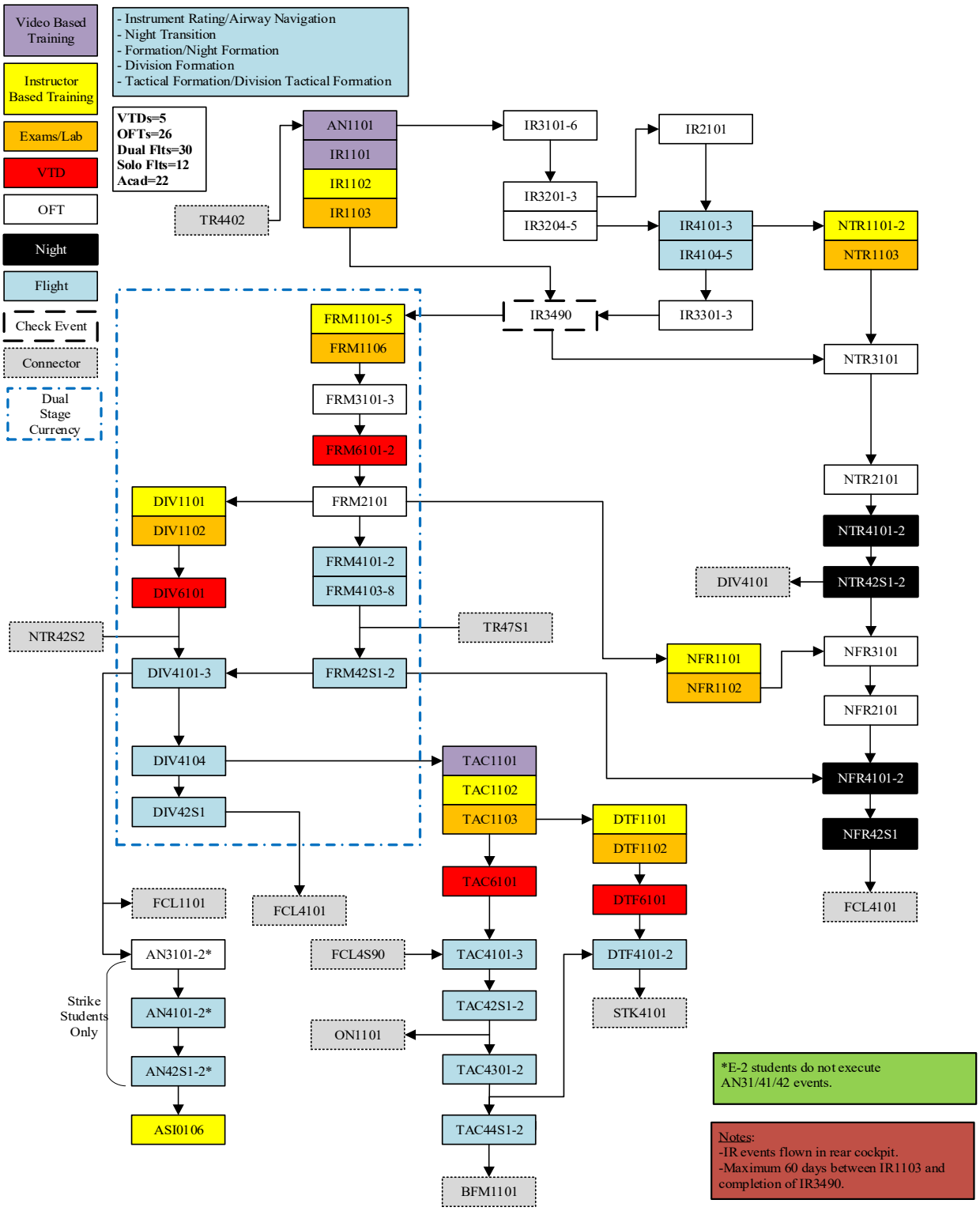
INSTRUMENT, NIGHT TRANSITION, AND FORMATION FLOW



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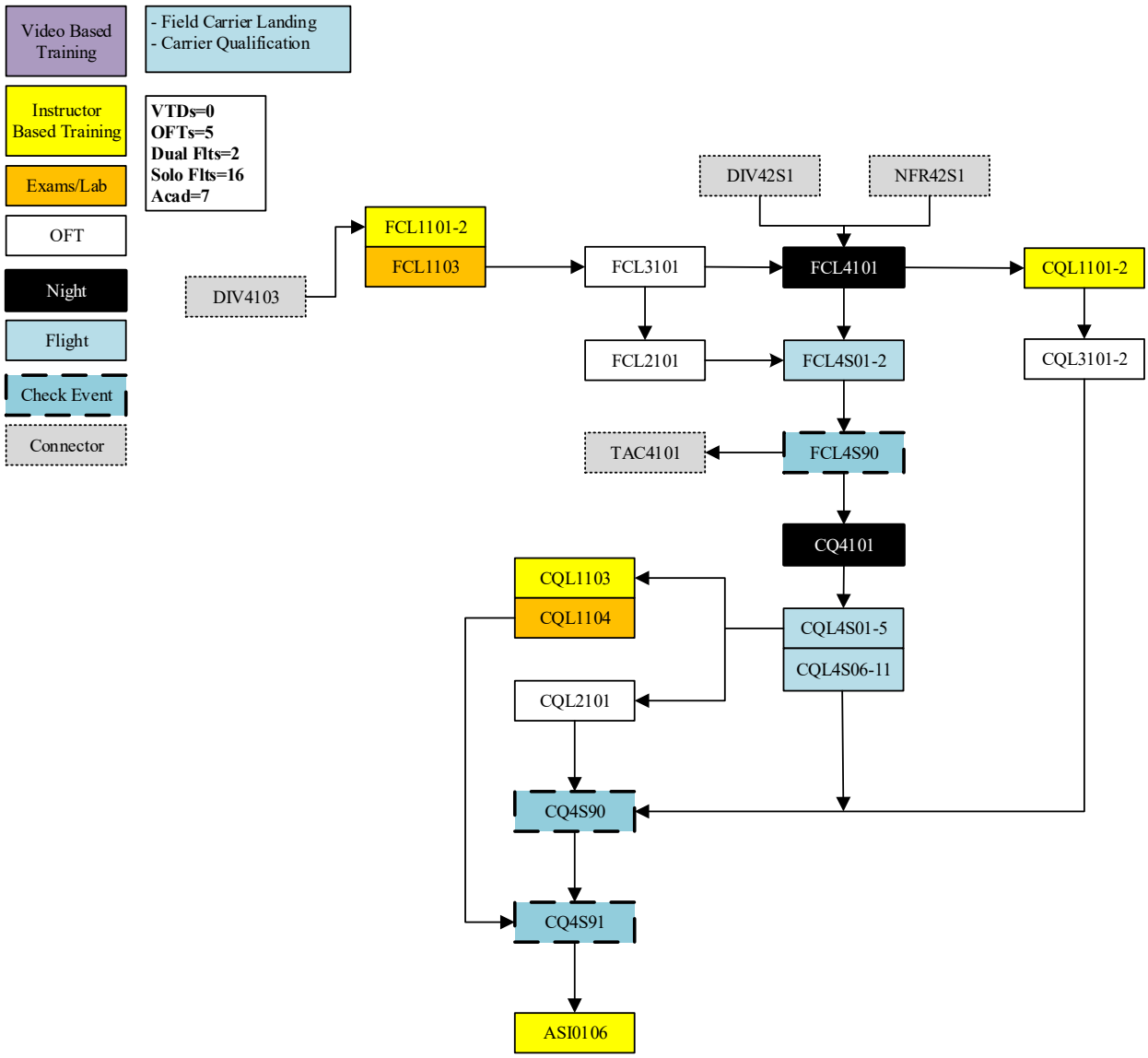
FIELD CARRIER LANDING AND CARRIER QUALIFICATION FLOW



Notes:
-180 FCLP-type landings required prior to FCL4S01.

-250 FCLP-type landings required prior to CQL4S01.

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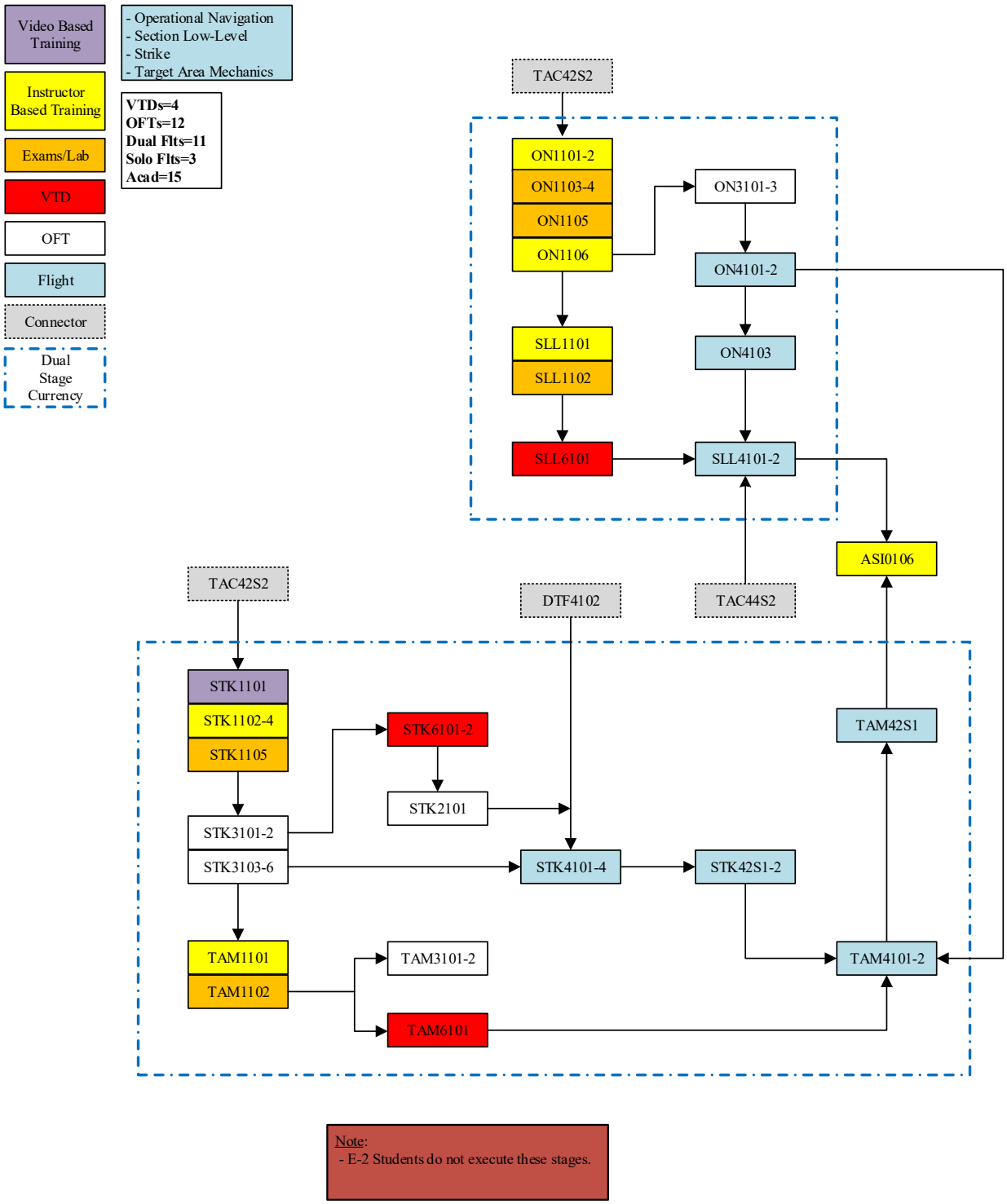
ON, SECTION LOW-LEVEL, STRIKE, AND TAM FLOW



Note:
- E-2 Students do not execute these stages.

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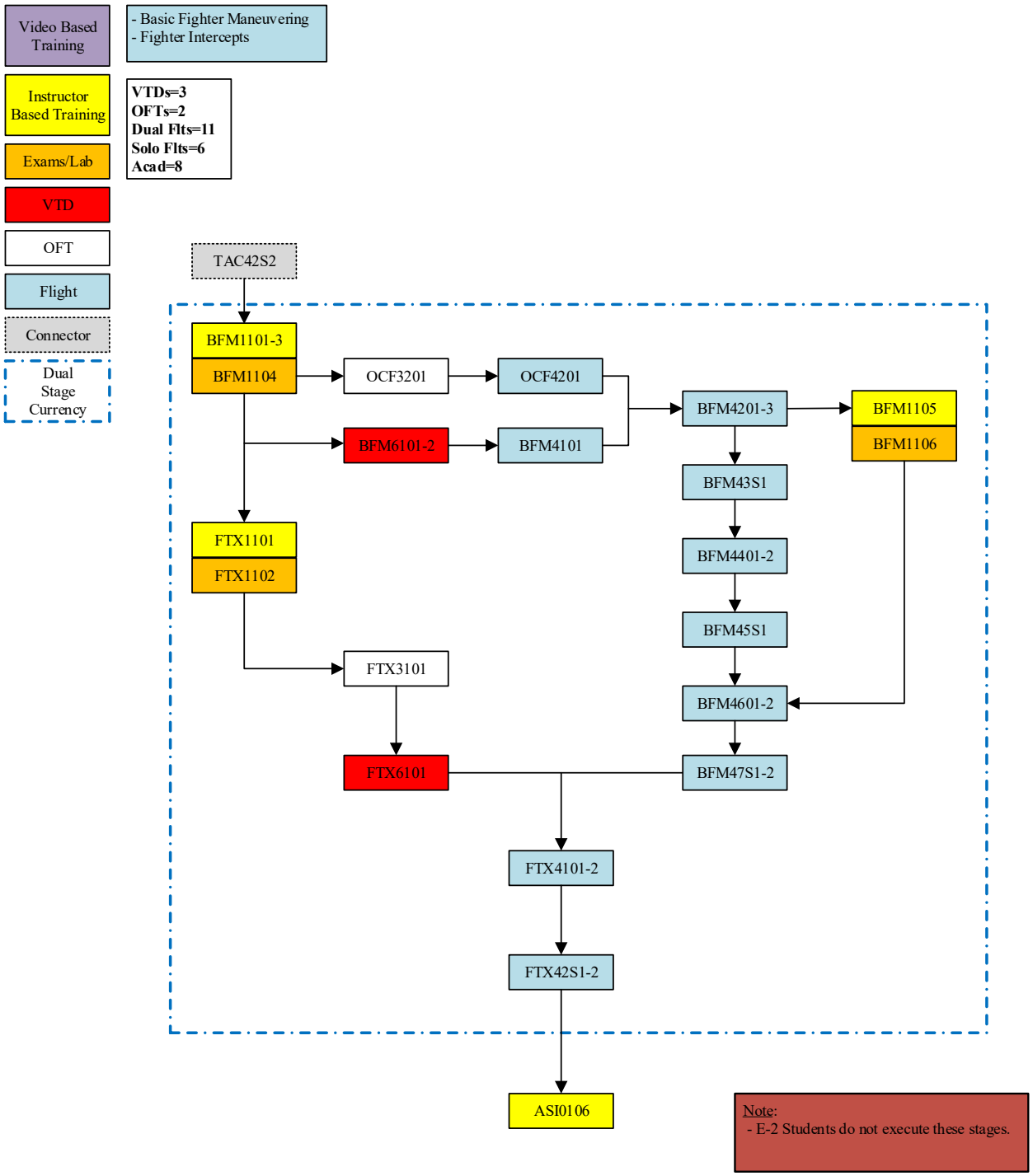
BASIC FIGHTER MANEUVERING AND FIGHTER INTERCEPT FLOW



3. Break in Training Warmup Events (XX86). See CNATRAINST 1500.4L for requirements. NFS currency for dual, solo, front-seat landing, and dynamic events are outlined in the tables below and take priority over CNATRAINST 1500.4L requirements if explicitly stated. A break equals current Julian Date minus the Julian Date of the last event executed. Individual TRAWING SOP may have additional aircraft flight currency requirements.

a. Warm-up Currency Requirements

Warm-up Currency Requirements	
Break	**Warm-up**
All Media in same stage	
Break Greater than 30 Days	CO-directed Warm-up Plan
All Media between stages	
Break 14 to 30 Days	1 Mandatory
Aircraft to Aircraft in same stage	
Break Greater than 3 Days within FCL and CQ Stage	1 Optional*
Break 7 to 13 Days	1 Optional
Break 14 to 30 Days	2 Optional**
OFT or VTD to Aircraft in same stage	
Break 7 to 30 Days	1 Mandatory OFT or VTD (same media)***
Lecture to first stage event	
Break greater than 30 Days	Execute refresher lecture

* May be flown solo at the discretion of the respective Landing Signal Officer (LSO). (See CNATRAINST 3740.9G for additional guidance during CQ training).
** If the next event is a Safe-for-Solo check event then first Warm-Up is mandatory.
***TR2101 shall be executed as a warmup event for breaks greater than 7 days between TR3101 and TR4101.
Note: Airway Navigation (AN) events do not require warmups as long as the NFS has logged First Pilot flight time within the preceding 14 days.

b. Solo Currency Requirements

Solo Currency Requirements	
Break	**Warm-up**
TR4690 Check Ride to TR47S1 Solo	
Break Greater than 2 Days	1 Front-Seat Landing Event
Break Greater than 3 Days	1 Mandatory (Re-Check)
Prior to First Night Solo (NTR42S1)	
Break Greater than 2 Days	1 Front-Seat Landing Event (day or night)
Prior to Subsequent Night Solo	
Break Greater than 3 Days	1 Front-Seat Landing Event (day or night)
Prior to subsequent Day or Night Solo	
Break Greater than 6 Days	1 Mandatory (Re-Check)

Notes:
- Executing a Solo event, complete or incomplete, resets Solo currency unless event was UNSAT.
- Front-seat landing events (XX86) should be executed as dedicated single-plane sorties to the maximum extent practical, but may be flown in the lead of a multi-plane flight with an IP in rear cockpit qualified to complete all training objectives for all NFSs in the flight. Front-Seat Landing flights **do not** reset safe-for-solo status.

c. Dynamic Currency Requirements

Dynamic Currency Requirements
In addition to CNAF-M 3710.7 requirements:
Flights shall be Syllabus Events (ex: BFM41XX, TAC41XX)
Warm-up Flights (XX86) shall be flown if CNAF-M 3710.7 currency requirements are not met
NFSs shall log First Pilot Time in the front-seat for events to count
If a dynamic flight is required, NFSs shall be granted a dedicated 1v0 event (BFM4X86)
OCF4201 updates OCF and Dynamic Flight Currency

4. Additional Flights and Simulators

a. Extra Training Events (XX87). In addition to CNATRAINST 1500.4L requirements, FCL or CQ may be solo as determined by the LSO.

b. Practice Simulators (XX85). Scheduling and instructor requirements shall be per CNATRAINST 1500.4L.

5. Ground Training and Briefing Requirements

a. Mission Preparation, Briefings, and Debriefings

(1) EOB Events. See CNATRAINST 1500.4L paragraph 601 for EOB MIF requirements. The IP shall carefully review the ATFs in planning the EOB event to ensure the profile includes opportunities to reach MIF on all items attempted in the block.

(2) Preparation. NFSs shall arrive for each flight with:

(a) Thorough knowledge of:

1. The flight's Discuss Items, as listed in Chapters III-VIII.

2. Procedural knowledge of all items for the event's training block.

(b) A flight profile tailored to training requirements, weak areas, and continuity.

(3) Briefing

(a) The instructor shall review the NFS's previous block ATFs before each event. Thoroughly cover the current mission's:

1. Discuss Items, as listed in Chapters III-VIII.

2. Specific objectives.

3. Techniques and required procedures for accomplishing those objectives.

4. Planned profile and contingencies.

(b) Duty officers shall provide a safety of flight brief to each solo NFS. In the case of multiplane events or Field Carrier Landing Practice (FCLP), this brief shall be given by the flight lead or LSO, as appropriate.

(4) Debriefing

(a) After each event, the instructor shall critique the NFS's performance using cause and effect analysis, particularly with respect to CTS.

(b) Mission complexity and NFS's progress shall govern the time required for debrief.

(c) Debriefing must be detailed and comprehensive. The ATF shall be completed prior to the NFS's next event. Exceptions may be made for out-and-ins, cross-country flights, and during FCL and CQ. In such instances, the NFS shall be provided feedback on performance as soon as possible following the event.

b. EP Briefing and Training

(1) EP training builds the NFS's confidence in the aircraft. The IP shall conduct EP training on all dual aircraft events, either on the ground or in the aircraft. Correct procedural deficiencies through additional instruction and study assignments.

(2) Incorporate EP training into non-EP simulator events when practical, however, instructional block objectives take precedence.

(3) Grade the NFS's overall EP knowledge and performance under EP situations.

6. Mission Grading Procedures and Evaluation Policies

a. General Grading and Evaluation Policy. Maneuver Item Files listed in MPTS are minimum stage or phase completion standards per maneuver. NFSs who consistently perform at the absolute minimum standard through multiple stages may not possess the skills required to complete follow-on training. MIF is designed to allow for minimum performance in a specific area with the understanding that performance above the minimum MIF will offset the weak area.

b. Grading Procedures (Aircraft and Training Devices). See CNATRAINST 1500.4L. ATFs for events are divided into critical and general items in order to draw attention to those maneuvers or skills that are seen as critical to follow-on training in other blocks and stages of undergraduate training. In addition to the absolute grading scale utilized for individual graded items, the instructor will use the relative grading criteria to assess overall sortie performance with grades of Excellent (E), Good (G), Fair (F) or Unsatisfactory (U). NFSs may receive grades below MIF on individual graded items new to them, but still receive a grade of G or E for overall sortie performance. Overall performance assessment is primarily a tool to assist in tracking overall NFS performance trends. The MIF for overall sortie performance on every event is Fair (F).

c. Pattern or Landing Failure. If an NFS triggers a Progress Check in a stage other than Transition because of an UNSAT pattern or landing, any subsequent re-fly events shall be flown in the same stage. If UNSAT occurred during a multiplane event, a lead is not required for re-fly events, and only items related to landing pattern shall be graded. IP need not be qualified in stage and shall add a comment to General Comments (ATF) with above information included. Example: BFM4402 may be flown with a non-BFM qualified LSO. BFM4588 would be flown with designated IPC Check Pilot, but need not be BFM-qualified.

7. Special Instructions and Restrictions

a. Programmed Hours and Events. Programmed syllabus flight hours are listed on page xii. Event lengths and XX86-XX89 events will cause variation.

b. Maximum Daily NFS Activities (Aircraft, Simulator, or Academic)

(1) NFSs shall not be scheduled for, or participate in, more than two aircraft or simulator events during one duty day with the following exceptions:

(a) Any combination of three simulator and dual flight cross-country or round robin events, but no academic training shall take place.

(b) Three CQ events in preparation for Carrier Qualification detachment. See CNATRAINST 3740.9G for CQ detachment restrictions.

(2) Academic training may be scheduled in addition to two aircraft or simulator events in a single day. The total time (MCG duration) of all training and required minimum-turn-time breaks between events shall not exceed 12 hours.

(a) Academic and flight support training must be kept within the 12-hour crew day (maximum of eight hours of academic training).

(b) NFSs are limited to three aircraft man-ups per day (four for cross-country and CQ events). A man-up is defined as entering the cockpit.

(c) Only one event per day shall be executed for the following events or blocks (excluding lectures, see specific events for exceptions):

1. CO31 and CO32 blocks.

2. TR4101.

3. TR3101-3104.

4. TR4201.

5. TAC4101.

Note: TRAWING Commander may waive requirements associated with maximum daily NFS activities on a case-by-case basis.

(3) Crew Day. See CNATRAINST 1500.4L. Official Government Travel (NALO, Commercial Air, or Ground Transportation) with no scheduled events does not count as part of the official work week (T-45C man-ups are scheduled events).

(4) Crew Rest. See CNATRAINST 1500.4L. Crew Rest begins upon completion of Official Government Travel for follow on day events.

(5) All lectures and exams that have online courseware shall utilize online courseware except during detachments. The detachment coordinator must contact the Curriculum Coordinator (CC) and request the exams for detachment.

c. Weather Requirements. Forecast weather shall be used for solo minimums. All dual flight events require CNAF M-3710 minima to launch.

STAGE[Notes]	SOLO	REMARKS AND WORKING AREA NOTES
TR[1,3,5]	1500/3	
NTR[2,3,4]	1500/3	
OCF	------	BFM Wx requirements. Max cloud tops 5000-ft AGL.
IR, AN[1,2]	500/2	
FRM, DIV, TAC, DTF[1]	500/2	
NFR[2]	500/2	
ON, SLL	-------	3000/5 on route.
STK[1]	500/2	10500/5 30-degree pattern. 5000/5 10-degree pattern.
TAM[1]	500/2	5000/5 (level) 8000/5 (displacement maneuver) in target area.
BFM, FTX[1]	500/2	Engagement Wx directed by CNATRA.
FCL, CQ[1,2]	500/2	CQ4S91 requires weather as outlined in CNATRAINST 3790.9G

Notes:

(1) All day NFS solo flights shall take off no earlier than 30 minutes after official sunrise and land no later than 30 minutes prior to official sunset.

(2) All night syllabus flights shall take off no earlier than 30 minutes after official sunset and land no later than 30 minutes prior to official sunrise.

(3) NFS solo flights shall maintain VMC prior to receiving an instrument rating.

(4) NTR flights shall execute either high work (area familiarization) or instrument work in addition to the landing pattern. NTR solo conduct **SHALL** be the same as the safe-for-solo event.

(5) Minimum of two flights with visual ground reference are required prior to TR4690. TR4690 shall maintain the ability to complete all required critical maneuvers VMC.

(6) TRAWING Commanders may waive the solo weather requirements for designated Naval Aviators enrolled in this syllabus, CNAF restrictions apply.

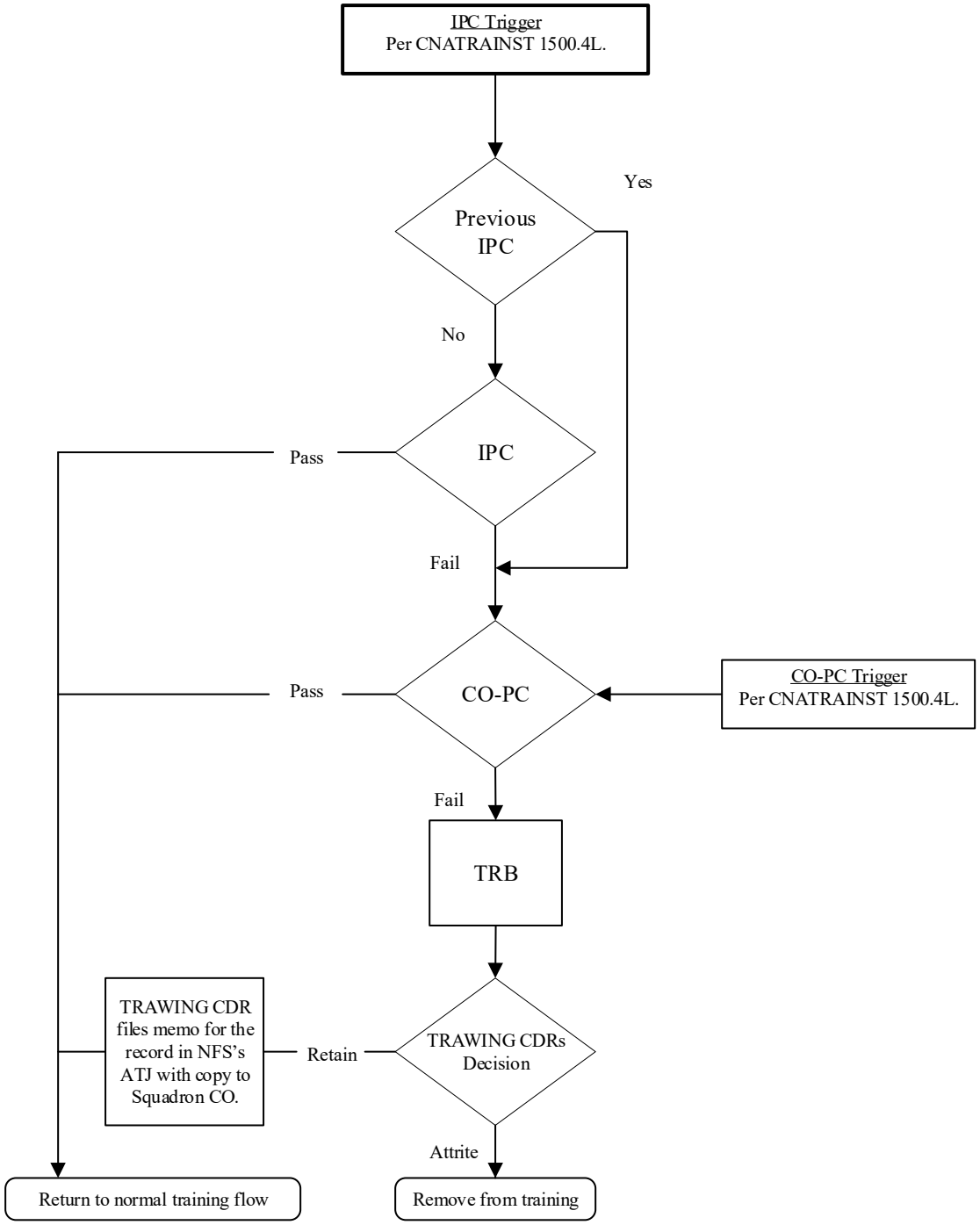
d. Aircraft and Simulator Interchangeability. Simulator events may be conducted in the aircraft when the simulator is unavailable for extended periods of time (excluding 2XXX (EP) events).

e. VTD Event Progression. If a scheduled VTD event cannot be executed or completed due to device issues (maintenance, software, etc.), instructor will conduct an in-depth event walk through with the NFS to act as an advancing event. The intent is to not unnecessarily delay syllabus progression while emerging technology is being integrated into the curriculum.

f. Gradesheets and ATF. FCLP-type landing comments shall be included on ATFs (using the CNATRA approved Training and Learning Management System (T/LMS)) for every FCLP-type pass flown on an IFLOLS lens. All Precautionary Approaches will be annotated in the ATF with the type and number. Example: 2 x Overhead PA, 1 x Straight in PA. For example:

	180	90	45	X	IM	IC	AR
B	650 ft TWA	450 ft	-200	TMP.HX	HIM	OC.NEP.CDIC	LOBAR

MPTS PROGRESS CHECK TRAINING REVIEW PROCESS



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Chapter II

Ground Training

GROUND TRAINING					
Block	**Title**		**Prerequisites**		**Total Hours**
ASI0101-06	Administration/Indoctrination (Any order)				**4.7**
	ASI0101	Wing or Squadron Check-in	Lect	1.3	
	ASI0102	Ground Rules	Lect	0.3	
	ASI0103	Intro to Learning Management System	Lect	0.5	
	ASI0104	Intro to CAI, OFT, VTD	Lect	2.0	
	ASI0105	Flight Record Training	Lect	0.5	
	ASI0106	Check-out (When syllabus complete)	Lect	0.1	
ENG0101-28	Engineering/Systems (In any order)		ASI0101-05		**18.5**
	ENG0101	Introduction to T-45C Configuration	VBT	0.5	
	ENG0102	Electrical System	VBT	0.5	
	ENG0103	Electrical System Malfunctions	VBT	0.5	
	ENG0104	Engine and Related Systems	VBT	0.5	
	ENG0105	Engine and Related Systems Malfunctions	VBT	0.5	
	ENG0106	Engine System Malfunctions	VBT	0.5	
	ENG0107	Aircraft Fuel System	VBT	0.5	
	ENG0108	Electrical, Engine, Fuel Review	MIL	1.5	
	ENG0109	Hydraulic System	VBT	0.5	
	ENG0110	Hydraulic System Malfunctions	VBT	0.5	
	ENG0111	Hydraulic Subsystems	VBT	0.5	
	ENG0112	Hydraulic Subsystems Malfunctions	VBT	0.5	
	ENG0113	Flight Control System	VBT	0.5	
	ENG0114	Flight Control System Malfunctions	VBT	0.5	
	ENG0115	Egress System	VBT	0.5	
	ENG0116	Egress System Malfunctions	VBT	0.5	
	ENG0117	ECS/Pressurization and OBOGS	VBT	0.5	
	ENG0118	ECS/Press. and OBOGS Malfunctions	VBT	0.5	
	ENG0119	Flight Instruments	VBT	0.5	
	ENG0120	Flight Control, Egress, ECS, Inst Review	MIL	1.5	
	ENG0121	CNI System	VBT	0.5	
	ENG0122	CNI System Malfunctions	VBT	0.5	
	ENG0123	Other T-45C Systems	VBT	0.5	
	ENG0124	INS/GPS Operation and Concepts	VBT	0.5	
	ENG0125	Display System and Malfunctions	VBT	0.5	
	ENG0126	Engine Start Procedures	VBT	0.5	
	ENG0127	Engineering Review	MIL	1.5	
	ENG0128	Engineering Block Exam	CAI	2.0	

GROUND TRAINING (CONT.)					
Block	**Title**		**Prerequisite**		**Total Hours**
AER0101-07	Aerodynamics		ASI0101-5		**5.0**
	AER0101	General Aerodynamics Review	VBT	0.5	
	AER0102	High Speed Flight	VBT	0.5	
	AER0103	Slow Speed Flight, Stall and Spin, and AOA	VBT	0.5	
	AER0104	Stability	VBT	0.5	
	AER0105	Engine Thrust and & Performance Charts	VBT	0.5	
	AER0106	Aeronautics Review	MIL	1.5	
	AER0107	Aeronautics Block Exam	CAI	1.0	
MET0101-04	Meteorological Principles		ASI0101-5		**3.0**
	MET0101	Review of Basic Meteorological Principles	VBT	0.5	
	MET0102	Meteorology and Flight Planning	VBT	0.5	
	MET0103	Meteorology Review	MIL	1.0	
	MET0104	Meteorology Exam	CAI	1.0	
NAV0101-09	Instrument Navigation		ASI0101-05		**8.8**
	NAV0101	FLIP/Electronic Flight Bag (EFB) Operation	Lab	1.8	
	NAV0102	Introduction to INAV and Voice Procedures	VBT	0.5	
	NAV0103	Departure and Terminal Procedures	VBT	0.5	
	NAV0104	Interpretation of High-Altitude Instrument Approach Plates	VBT	0.5	
	NAV0105	JMPS Introduction	Lab	1.2	
	NAV0106	JMPS Flight Planning - Departure/Arrival	Lab	0.8	
	NAV0107	JMPS Flight Planning - En route	Lab	1.0	
	NAV0108	JMPS Practical Problem	Lab	1.0	
	NAV0109	Instrument Navigation Exam	Lab/ CAI	1.5	
CRM1101	Crew Resource Management		ASI0101-05		**3.0**
	CRM1101	Crew Resource Management	MIL	3.0	
ORM1101	Operational Risk Management		ASI0101-05		**1.0**
	ORM1101	Operational Risk Management	Offline MIL	1.0	
SEA1101-02	NACES Flight Physiology		ENG0128		**3.0**
	SEA1101	NACES Flight Physiology	MIL	2.0	
	SEA1102	Ejection Seat Lecture/NACES Preflight	Lect	1.0	

Chapter III

NATOPS Training

This chapter does not apply to T-45C Undergraduate Strike or Advanced E-2 NFSs.

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Chapter IV

Transition Training

1. Matrices. Individual matrices following event block descriptions list the MIF requirements for the Transition training category. This category includes Cockpit Orientation, Emergency Procedure, Basic Instrument, Radio Instrument, Transition, Out-of-Control Flight, and Night Transition, Field Carrier Landing Practice, and Carrier Qualification stages. The purpose of these matrices is to provide the NFS and IP the easiest way to track progress and overall status in relation to MIF.

2. Instructor Continuity. On-wing instructors are not required, however, instructor continuity during the TR stage of training is encouraged. At a minimum, three flight events across TR42, TR43, and TR45 blocks shall be flown with TR "S" qualified IPs.

3. Scheduling. Only one event per day shall be flown for the following blocks and events (lectures may be executed in addition to the specified events):

a. CO31 and CO32 blocks.

b. TR3101-TR3104.

c. TR4101 and TR4201

4. FCLP-type Landing Notes. In order to ensure success in follow-on Carrier Qualification training, refer to the following guidance regarding the FCLP-type passes.

a. 180 FCLP-type landings on the Improved Fresnel Lens Optical Landing System (IFLOLS) are required to prior to FCL4S01. At least 10 of these landings shall be at night.

Recommended FCLP-type Landings per Block/Event	
TR42 and TR45	Six per event (36 total)
TR43	Nine per event (36 total)
OCF41	Four (4 total)
TR46	Five (5 total)
TR47	Six (6 total)
NTR41 and NTR42	Six per event (24 total)
FRM41	Four per event (32 total)
FRM42	Five per event (10 total)
NFR41	Four per event (8 total)
NFR42	Five (5 total)
DIV41 and DIV42	Two per event (10 total)

b. 250 FCLP-type landings are required prior to CQ4S01. At least 35 of these landings shall be at night.

Recommended FCLP-type Landings per Block/Event	
TAC	Two per event (18 total)
DTF	Two per event (4 total)
ON	Two per event (6 total)
SLL	Two per event (4 total)
STK and TAM	Two per event (18 total)
BFM and OCF42	Two per event (26 total)
FTX	Two per event (8 total)

c. TRAWING Commanders may waive these requirements on a case-by-case basis based on NFS performance.

d. NFSs shall not fly more than 12 FCLP-type passes during a single event. The intent is to avoid diminishing training value during an event.

5. Transition Training Flight Support

TRANSITION TRAINING FLIGHT SUPPORT						
Block	**Title**		**Prerequisite**			**Total Hours**
CO1101-07	Cockpit Orientation		ENG0128			**5.0**
	CO1101	Engine Start and Post-start		VBT	0.5	
	CO1102	Multi-function Display and Navigation		VBT	0.5	
	CO1103	Display System (HUD)		VBT	0.5	
	CO1104	Waypoint Navigation Procedures		VBT	0.5	
	CO1105	Velocity Vector		MIL	1.0	
	CO1106	Exterior Preflight Checks		VBT	0.5	
	CO1107	Aircraft Preflight/Strap-in Procedures		Lab	1.5	
EP1101-11	Emergency Flight Procedures		ENG0128			**14.5**
	EP1101	Start, Ground, and Takeoff EPs I		MIL	1.5	
	EP1102	Start, Ground, and Takeoff EPs II		MIL	1.5	
	EP1103	Operational and Ejection EPs		MIL	1.0	
	EP1104	Engine and Hydraulic EPs I		MIL	1.5	
	EP1105	Engine and Hydraulic EPs II		MIL	1.5	
	EP1106	Emergency Flight Procedures Exam I		CAI	1.0	
	EP1107	Canopy and Flight Control EPs		MIL	1.0	
	EP1108	Electrical and Indicator EPs I		MIL	1.5	
	EP1109	Electrical and Indicator EPs II		MIL	1.5	
	EP1110	Operational and Landing EPs		MIL	1.5	
	EP1111	Emergency Flight Procedures Exam II		CAI	1.0	

TRANSITION TRAINING FLIGHT SUPPORT (CONT.)					
Block	**Title**		**Prerequisite**		**Total Hours**
CR1101/ CR1201-02	Course Rules (Training Wing Specific)		NAV0109		**4.0**
	CR1101	In-Flight Guide/SOP	Lect	1.0	
	CR1201	Course Rules	Lect	2.0	
	CR1202	Course Rules Exam	CAI	1.0	
BI1101-10	Basic Instrument Flight Procedures		AER0107, MET0104, NAV0109		**8.0**
	BI1101	Instrument Takeoff and Climb with DP	VBT	1.0	
	BI1102	Introduction to Basic Instruments	VBT	0.5	
	BI1103	Instrument Turns	VBT	0.5	
	BI1104	Basic Flight Maneuvers and Transitions	VBT	0.5	
	BI1105	"S" Patterns	VBT	0.5	
	BI1106	Stalls, Unusual Attitudes, and Aerobatics	VBT	1.0	
	BI1107	TACAN/VOR/GCA/ILS Procedures	VBT	1.0	
	BI1108	Instrument Failures and GPS/INS Failures	VBT	1.0	
	BI1109	Basic Instrument Review	MIL	1.0	
	BI1110	Basic Instrument Stage Exam	CAI	1.0	
RI1101-06	Radio Instrument Flight Procedures		TR3105 (NAV0109 for RI1106)		**8.5**
	RI1101	Introduction to Radio Instruments	MIL	2.5	
	RI1102	TACAN and VOR Procedures	CAI	0.5	
	RI1103	TACAN and VOR Holding Procedures	CAI	0.5	
	RI1104	TACAN/VOR/ILS/PAR/ASR Procedures	CAI	1.0	
	RI1105	Radio Instrument Stage Exam	CAI	1.0	
	RI1106	JMPS Smart Packet Development	Lect	3.0	
FAM1101-04	Familiarization Flight Procedures		TR3205 (TR4201 for FAM1104)		**8.6**
	FAM1101	Familiarization Flight Procedures I	MIL	3.3	
	FAM1102	Familiarization Flight Procedures II	MIL	3.3	
	FAM1103	Familiarization Flight Procedures Exam	CAI	1.0	
	FAM1104	LSO Ball Flying Brief (Given by LSO)	Lect	1.0	
NTR1101-03	Night Transition Flight Procedures		IR4103		**3.5**
	NTR1101	Night Transition Flight Procedures I	MIL	1.3	
	NTR1102	Night EPs II	MIL	1.2	
	NTR1103	Night Transition Flight Procedures Exam	CAI	1.0	
OCF1101-02	Out-of-Control Flight Procedures		TR4203		**3.0**
	OCF1101	Out-of-Control Flight	MIL	2.0	
	OCF1102	OCF Exam	CAI	1.0	
NA1101-03	NATOPS/NATOPS Examination		TR4503		**6.0**
	NA1101	NATOPS Review	MIL	2.0	
	NA1102	NATOPS Open-Book Exam	P/P	2.0	
	NA1103	NATOPS Closed-Book Exam and SOP Exam	P/P	2.0	

TRANSITION TRAINING FLIGHT SUPPORT (CONT.)					
Block	**Title**		**Prerequisite**		**Total Hours**
FCL1101-03	Field Carrier Landing Flight Procedures		DIV4103		**2.5**
	FCL1101	Carrier Qualification Landing (FCLP) Procedures	MIL	1.0	
	FCL1102	Night FCLP Procedures	MIL	0.5	
	FCL1103	FCLP Exam	CAI	1.0	
CQ1101-04	Carrier Qualification Flight Procedures		FCL4101 (CQ4S05 for CQ1103)		**6.0**
	CQ1101	Day/Night FCLP Refresher	MIL	1.0	
	CQ1102	CQ Shipboard Procedures	MIL	1.0	
	CQ1103	Ship's Brief	MIL	3.0	
	CQ1104	Ship's Brief Exam	CAI	1.0	
Total					**69.6**

Blk#	Media	Title	Events	Hrs	H/X
CO31	OFT	Cockpit Orientation Simulators	2	3.0	1.5

1. Prerequisites

a. SEA1102 (Ejection Seat Lecture/NACES Preflight).

b. CO1107 (Aircraft Preflight/Strap-in procedures).

2. Syllabus Notes

a. Practice all checklists, applicable FTI briefings, radio calls, and basic aircraft control. Ensure NFS's checklist proficiency is adequate to proceed to flight operations.

b. Multiple items are listed as discuss items; however, due to time constraints, it may not be possible to discuss all items prior to the simulator event (SIM); therefore, a ***Discuss Item*** may be addressed during or after the SIM.

c. Only one event per day shall be flown in block.

d. The NFS shall perform the following procedures per FTI, NATOPS, and SOP on the indicated event:

(1) CO3101. Canopy and ejection seat preflight, cockpit preflight checklist, pre-start checklist, aircraft start, post-start checklist, pre-taxi checklist, ground communications, taxi checklist, flight instrument checks, takeoff clearance, takeoff checklist, engine checks, takeoff, departure communications, 10,000-feet checks and 15-minute report, descent and penetration checklist, landing checklist, after landing checklist, shutdown checklist, and normal egress procedures. Enter mission data into display system.

(2) CO3102. Canopy and ejection seat preflight, blindfold cockpit check, cockpit preflight checklist, pre-start checklist, aircraft start, post-start checklist, ground communications, taxi checklist, aircraft taxi, flight instrument checks, takeoff clearance, takeoff checklist, engine checks, takeoff, departure communications, 10,000-feet checks and 15-minute report, en route communications, approach control communications, descent and penetration checklist, VFR approach-to-pattern initial, communications to tower, landing checklist, after landing checklist, after landing communications, shutdown checklist, and normal egress procedures.

3. Special Syllabus Requirements. None.

4. Discuss Items

CO3101
QOD, IFT operation (if applicable), ground signals, final checker, and shutdown signals.

CO3102
QOD, OFT operation, ground signals, and final checker.

5. Block MIF

COCKPIT ORIENTATION		CO3102
CTS Ref	**Critical Items**	
0	Overall Sortie Performance	3+
1	Mission Planning/Brief/Debrief	3+
1	Headwork/Situational Awareness	3+
2	Communications	2+
2	Ground Operations	2+
CTS Ref	**General Items**	
2	Flight Management	3+
2	Departure	2+
2	Arrival	2+

Blk #	Media	Title	Events	Hrs	H/X
CO32	OFT	Cockpit Orientation Simulators	2	3.0	1.5

1. Prerequisites

   a. CO3102 (Cockpit Orientation).

   b. BI1110 (Basic Instrument Stage Exam).

2. Syllabus Notes

   a. CO3201 and CO3202 should be flown with different instructors.

   b. CO32 block meets stage currency requirements for TR31 block (lectures and simulators).

   c. Only one event per day shall be flown in block.

3. Special Syllabus Requirements. None.

4. Discuss Items. QOD, control, performance, and position instruments, instrument scan, and scan technique.

5. Block MIF

COCKPIT ORIENTATION		CO3202
CTS Ref	**Critical Items**	
0	Overall Sortie Performance	3+
1	Mission Planning/Brief/Debrief	3+
1	Headwork/Situational Awareness	3+
2	Communications	2+
2	Ground Operations	3+
CTS Ref	**General Items**	
1	Basic Air Work	2+
2	Flight Management	3+
5	Turn Pattern	2+
5	One-Half Standard Rate Turn (SRT)	2+
5	Level Speed Change	2+
5	Slow Flight Maneuver	2+

MIF continued on next page.

COCKPIT ORIENTATION		CO3202
CTS Ref	**General Items**	
5	S-1 Pattern	2+
4	High-Altitude Penetration	2+
4	TACAN/VOR DME Approach	2+
4	PAR Approach	2+

Blk #	Media	Title	Events	Hrs	H/X
EP21	OFT	Emergency Procedures	4	6.0	1.5

1. Prerequisites

a. CO3202 (Cockpit Procedures).

b. ORM1101 (Operational Risk Management).

c. CRM1101 (Crew Resource Management).

d. EP1106 (Emergency Flight Procedures Exam I).

e. EP1111 (Emergency Flight Procedures Exam II) prior to EP2103.

2. Syllabus Notes. The NFS shall perform the following procedures per FTI, NATOPS, and SOP on the indicated event:

a. EP2101. No READY light, wet start, low oil pressure on start, hot start, ground emergency communications, unsafe gear (up), fuel leak, LP fuel pump failure, boost pump failure, initial shot failure, engine fire no secondary indications, GINA failure, OXYGEN light, engine fire on shutdown, and emergency egress.

b. EP2102. Engine fire on start, hung start, GTS fire, trim malfunctions, engine fire with secondary indications, engine over-speed, engine flameout, air-start (high altitude), ECA failure (full trim), engine vibration, engine stalls, engine failure (seizure), oil pressure failure, ejection, and ground emergency communications.

c. EP2103. Hot start, bleed valve failure, engine failure on takeoff, generator failure, inverter failure, total electrical failure, uncommanded RAT extension, HYD 1 EDP failure, HYD 2 EDP failure, HYD 1 and 2 failure RAT OK, total HYD failure, accumulator failure, CONTR AUG failure, emergency communications, and MFD failure.

d. EP2104. takeoff, runaway rudder trim, rudder hard-over, runaway stabilator trim, runaway aileron trim, aileron trim failure, speed brake fails to retract, split flaps, pitot static malfunctions, main/nose gear unsafe down, gear emergency extend failure, brake accumulator failure, brake failure after touchdown, on rollout, anti-skid failure after touchdown and on rollout, OXYGEN light, ECS failure, GINA failure, emergency communications, and ejection.

3. Special Syllabus Requirements. None.

4. Discuss Items

EP2101
QOD, canopy malfunctions, engine fire on deck, OBOGS malfunctions and air-start.

EP2102
QOD, ground ejection situations, engine stalls, short-field arrested landing, and go-around.

EP2103
QOD, gear door malfunctions, and long-field arrested landing, aileron PCU checklist procedures and failure modes, BASH mitigation profile procedures.

EP2104
QOD, smoke or fumes in cockpit, rudder trim failure, stabilator trim failure, flaps fail to retract, slats fail to retract, flaps fail to extend, slats fail to extend, split slats, gear unsafe after extension, gear door malfunctions after extension, and go-around.

5. Block MIF

EMERGENCY PROCEDURES		**EP2104**
CTS Ref	**Critical Items**	
0	Overall Sortie Performance	3+
1	Mission Planning/Brief/Debrief	3+
1	Headwork/Situational Awareness	3+
2	Communications	2+
2	Ground Operations	3+
CTS Ref	**General Items**	
2	EPs	3+
2	Flight Management	3+
2	Start Malfunctions	3+
2	Ground EPs	3+
2	Aborted Takeoff	3+
2	Takeoff EPs	3+
2	Engine EPs	3+
2	OBOGS EPs	3+
2	Flight Control EPs	3+
2	Landing Gear EPs	3+
2	Electrical EPs	3+
2	Hydraulic EPs	3+
2	ECS EPs	3+
2	Fuel System EPs	3+
2	Ejection	3+

Blk #	Media	Title	Events	Hrs	H/X
TR31	OFT	Basic Instrument Simulators	9	13.5	1.5

1. Prerequisites

   a. BI1110 (Basic Instrument Stage Exam).

   b. CR1202 (Course Rules Exam).

   c. EP2104 (Emergency Procedures).

2. Syllabus Notes

   a. Introduce partial panel during TR3104.

   b. During this block, NFSs must fly at least the approaches listed below (approaches may be combined, e.g., a low oil PAR may be logged as a PAR and a low oil approach):

MANEUVERS	
High Altitude Penetration	3
TACAN/VOR DME	5 full panel and 1 partial panel
VOR	1
ASR	2
ILS	3 full panel and 1 partial panel
PAR	5 full panel and 1 partial panel
No-Gyro GCA	1

   c. CO32 block meets currency requirements for TR31 block (lectures and simulators).

3. Special Syllabus Requirements. None.

4. Discuss Items

   TR3101-2
   QOD, instrument scan, control instruments, performance instruments, and position instruments.

   TR3103
   QOD, main ADI failure, GINA malfunctions, turn-and-slip failure, HSI failure, and MFD failure.

TR3104-9
QOD and partial panel.

5. Block MIF

BASIC INSTRUMENT OFT		TR3109
CTS Ref	**Critical Items**	
0	Overall Sortie Performance	3+
1	Partial Panel Air Work	3+
2	Communications	3+
2	Ground Operations	3+
2	Departure	3+
2	Intercept/Maintain Course	3+
4	Missed Approach	3+
CTS Ref	**General Items**	
1	Mission Planning/Brief/Debrief	4+
1	Headwork/Situational Awareness	3+
2	EPs	3+
1	Basic Air Work	3+
2	Takeoff	3+
2	En route Navigation/Course Rules	3+
5	Climbs/Descents	3+
5	Turn Pattern	4+
5	One-Half Standard Rate Turn (SRT)	4+
5	Standard Rate Turn (SRT)	4+
5	Level Speed Change	4+
5	Level Speed Change in One-Half SRT	4+
5	Slow Flight Maneuver	3+
5	S-1 Pattern	4+
5	S-3 Pattern	4+
2	Arcing	3+
5	Stall Series	4+
5	Wingover	3+
5	Barrel Roll	3+
5	Unusual Attitude	4+
5	Partial Panel Unusual Attitudes	4+
4	High-Altitude Penetration	3+
4	TACAN/VOR DME Approach	3+
4	Partial Panel TACAN/VOR DME Approach	3+
4	VOR Approach	3+
4	ASR Approach	3+
4	ILS Approach	3+
4	Partial Panel ILS Approach	3+

Blk #	Media	Title	Events	Hrs	H/X
TR32	OFT	Radio Instrument Simulators	8	12.0	1.5

1. Prerequisites

a. TR3109 (Basic Instrument).

b. RI1105 (Radio Instrument Stage Exam).

c. RI1106 (JMPS En route Flight Planning).

2. Syllabus Notes

a. Handouts listing the route of flight to plan and study for each simulator event shall be obtained by NFSs from book issue at the ground training building.

b. NFSs shall bring a **copy** of their DD-1801 and single-engine jet log to all simulator events for instructor use.

c. TR3205 shall introduce radar altimeter failure, VOR holding, localizer approach partial panel, and emergency oil instrument approach.

d. TR3206 shall introduce visual takeoff, low ceiling ITO, localizer back course approach (if able), and instrument-to-visual scan.

e. TR3207 shall introduce direct routing and circling approach-to-land.

f. During this block, NFSs must fly at least the approaches listed below (approaches may be combined, e.g., a low oil PAR may be logged as a PAR and a low oil Approach):

MANEUVERS	
High Altitude Penetration	2 full panel and 1 partial panel
TACAN or VOR DME	2 full panel and 2 partial panel
ASR	3
ILS	4 full panel and 2 partial panel
PAR	2 full panel and 2 partial panel
Localizer Approach	1 full panel and 1 partial panel
Low Oil Approach	2
Min or Emergency Fuel Approach	1
No-Gyro GCA	2
Circle-to-Land Approach	1

3. Special Syllabus Requirements. None.

4. Discuss Items

TR3201
QOD, radial intercepts, and point-to-point navigation.

TR3202
QOD and MFD failure.

TR3203
QOD and minimum or emergency fuel GCA.

TR3204
QOD, lost communication, and en route descent.

TR3205
QOD, marker beacon failure, and oil pressure warning.

TR3206-8
QOD.

5. Block MIF

RADIO INSTRUMENT OFT		TR3208
CTS Ref	**Critical Items**	
0	Overall Sortie Performance	3+
1	Partial Panel Air Work	3+
2	Communications	3+
2	Ground Operations	3+
2	Departure	4+
2	Intercept/Maintain Course	3+
4	Missed Approach	4+
CTS Ref	**General Items**	
1	Mission Planning/Brief/Debrief	4+
1	Headwork/Situational Awareness	3+
1	Basic Air Work	3+
2	EPs	3+
2	Flight Management	3+
2	Takeoff	4+
2	En route Navigation/Course Rules	4+

MIF continued on next page.

RADIO INSTRUMENT OFT		TR3208
CTS Ref	**General Items**	
5	Climbs/Descents	3+
2	Non-system Point-to-Point Navigation	3+
2	System Point-to-Point Navigation	3+
2	Arcing	3+
2	Holding	4+
4	High-Altitude Penetration	4+
4	TACAN/VOR DME Approach	4+
4	Partial Panel TACAN/VOR DME Approach	3+
4	ASR Approach	4+
4	Localizer Approach	3+
4	Partial Panel Localizer Approach	3+
4	ILS Approach	4+
4	Partial Panel ILS Approach	3+

Blk #	Media	Title	Events	Hrs	H/X
TR33	OFT	Familiarization Simulators	7	10.5	1.5

1. Prerequisites

a. TR3208 (Radio Instruments).

b. FAM1103A or FAM1103B (applicable Familiarization Flight Procedures Exam).

c. TR2102 prior to TR3304.

2. Syllabus Notes

a. Instructor shall demonstrate area familiarization.

b. The NFS shall perform the following maneuvers per FTI, NATOPS, and SOP on the indicated event:

(1) TR3301. Area familiarization, level flight accelerate and decelerate, overhead pattern entry (break), wave-off, taxi-to-line, and shutdown.

(2) TR3302. NWS failure, wheel brake failure, generator malfunction and EPs, overhead pattern entry (break), one-half flap arrested landing (roll-in, fly in), taxi-to-line, and shutdown.

(3) TR3303. Engine fire during takeoff, abort, lost communications situation, electrical EPs, overhead pattern entry (break), taxi-to-line, and shutdown.

(4) TR3304. OLF operations, blown tire on takeoff, abort, waypoint navigation, and swerve on landing rollout.

(5) TR3305. Failure to reach line speed, abort, crosswind takeoff, engine EPs, CNI failure, inadvertent IMC, blown tire during field landing, and short-field arrested landing with blown tire.

(6) TR3306. IFR recovery to visual pattern and engine surge and compressor stall. Practice BASH mitigation profile procedures per local SOP.

(7) TR3307. Suspend GINA alignment on power-up, and rejected landing and go-around.

3. Special Syllabus Requirements. None.

4. Discuss Items

TR3301
QOD, parking brake failure, lost aircraft, lost communications, loss of ECS temperature control, and OBOGS malfunction.

TR3302
QOD, flap indicator failure, landing gear indicator failure, trim indicator failure, and swerve on touchdown.

TR3303
QOD, fuel flow indicator failure, IFF failure, and long-field arrestment.

TR3304
QOD and warning and caution tones.

TR3305
QOD, lost canopy, and aircraft configurations for field arrestments.

TR3306
QOD, aileron PCU checklist procedures, PCU failure modes, BASH mitigation profile procedures.

TR3307
QOD and aircraft systems.

5. Block MIF

FAMILIARIZATION OFT		TR3307
CTS Ref	**Critical Items**	
0	Overall Sortie Performance	3+
1	Mission Planning/Brief/Debrief	4+
1	Headwork/Situational Awareness	3+
1	Basic Air Work	3+
2	EPs	3+
2	Communications	3+

MIF continued on next page.

FAMILIARIZATION OFT		TR3307
CTS Ref	**General Items**	
2	Ground Operations	3+
2	Flight Management	3+
2	Takeoff	3+
2	Departure	3+
2	En route Navigation/Course Rules	3+
2	Arrival	3+
3	Crosswind Landings	2+
3	Wave-off	3+
3	Full-Stop Landing	3+
5	Turn Pattern	3+
5	Accelerated Stall	3+
5	Break Turn Stall	3+
5	Power Off Stall	3+
5	Landing Attitude Maneuver	3+
5	Landing Attitude Stall	2+
5	Approach Turn Stall	2+
5	Stall Series	3+
5	Pattern Stall and Recovery	3+
5	Vertical Recovery	3+
7	G-Warm	3+
5	Minimum Radius Turn	3+
5	Aileron Roll	3+
5	Wingover	3+
5	Barrel Roll	3+
5	Aerobatics	3+
5	Squirrel Cage	3+
5	Unusual Attitude Recovery	3+
3	Straight-in Approach	3+
3	IFR Recovery to VFR Pattern	3+
3	VFR Landing Pattern	3+
3	Field Carrier Landing	2+
3	No-Flap Roll-and-Go	3+
3	Half-Flap Roll-and-Go	3+
3	Full-Flap Roll-and-Go	3+
3	No-HUD Landings	3+
3	Precautionary Approach	3+
3	PA to Full-Stop	3+
2	Start Malfunctions	3+

MIF continued on next page.

FAMILIARIZATION OFT		TR3307
CTS Ref	**General Items**	
2	Ground EPs	3+
2	Aborted Takeoff	3+
2	Takeoff EPs	3+
2	Engine EPs	3+
2	Electrical EPs	3+
2	Hydraulic EPs	3+
2	Swerve/Blown Tire on Landing	3+
3	Short-Field Arrestment	3+
3	Half-Flap Arrestment	3+
3	Rejected Landing/Go-Around	3+
2	Lost Communications	3+

Blk #	Media	Title	Events	Hrs	H/X
TR21	OFT	Familiarization EPs	2	3.0	1.5

1. Prerequisites. TR3303 (Familiarization).

2. Syllabus Notes. The NFS shall perform the following procedures per FTI, NATOPS, and SOP on the indicated event:

a. TR2101. Takeoff with direct entry into landing pattern, pattern depart and reenter (break), downwind entry, VFR straight-in approach (FF, NF), VFR landing pattern, field carrier landings, no-flap touch-and-go, precautionary approaches (straight-in, abeam, overhead, pattern), rejected landing and go-around, short-field arrestment (roll-in, fly-in), long field arrestment, full-stop with blown tire (non-arrested), lost communications, taxi-to-line, and shutdown.

b. TR2102. Start malfunction/EP (any), failure to reach line speed, abort situation (any), takeoff EP (any), one gear unsafe down, NWS caution light illumination airborne, tailpipe overheat, cabin pressurization failure, trim malfunctions, tail hook malfunctions, anti-skid failure, blown tire during field landing, locked-in low-altitude compressor stall, bird strike/dirty precautionary approach, ejection (low altitude), fuel system EPs (any), electrical EPs (any), hydraulic EPs (any), flight control EPs (any), one-half flap field-arrested landing (roll-in, fly-in), and landing with NWS failure.

3. Special Syllabus Requirements. None.

4. Discuss Items

TR2101
QOD, brake pressure caution light illuminated airborne, weight-on-wheels proximity switch failure (AOA indexers with aircraft on landing rollout), and arrested landing.

TR2102
QOD and OBOGS malfunctions.

5. Block MIF

FAMILIARIZATION EP OFT		TR2102
CTS Ref	**Critical Items**	
0	Overall Sortie Performance	3+
1	Mission Planning/Brief/Debrief	4+
1	Headwork/Situational Awareness	3+
1	Basic Air Work	3+
2	EPs	3+
2	Communications	3+
CTS Ref	**General Items**	
2	Ground Operations	3+
2	Flight Management	3+
2	Takeoff	3+
2	Departure	3+
2	Arrival	3+
3	Straight-in Approach	3+
3	Downwind Entry	3+
3	VFR Landing Pattern	3+
3	No-Flap Touch-and-Go	3+
3	No-HUD Landings	3+
3	Precautionary Approach	2+
3	Simulated Bird Strike PA	3+
2	Start Malfunctions	3+
2	Ground EPs	3+
2	Aborted Takeoff	3+
2	Takeoff EPs	3+
2	Engine EPs	3+
2	Flight Control EPs	3+
2	Landing Gear EPs	3+
2	Electrical EPs	3+
2	Hydraulic EPs	3+
2	ECS EPs	3+
2	Fuel System EPs	3+
2	Ejection	3+
2	Swerve/Blown Tire on Landing	3+
3	Short-Field Arrestment	3+
3	Long-Field Arrestment	3+
3	Rejected Landing/Go-Around	3+
2	Full-Stop with Blown Tire (Non-Arrested)	3+
3	No-Flaps Landings	3+

Blk #	Media	Title	Events	Hrs	H/X
TR34	OFT	Transition Simulators	3	4.5	1.5

1. Prerequisites. TR3307 (Familiarization).

2. Syllabus Notes

a. The intent of this block is to refresh instrument training and to prepare NFSs for a typical TR41 event.

b. The NFS shall perform the following maneuvers per FTI, NATOPS, and SOP on the indicated event:

(1) TR3401. OLF operations, blown tire on takeoff, abort, and waypoint navigation. Practice BASH mitigation profile procedures per local SOP.

(2) TR3402. Failure to reach line speed, abort, crosswind takeoff, inadvertent IMC, blown tire during field landing, and short-field arrested landing with blown tire.

(3) TR3403. IFR departure to working area, IFR recovery to visual pattern and engine surge or compressor stall.

3. Special Syllabus Requirements. None.

4. Discuss Items

TR3401
QOD, warning and caution tones, and BASH mitigation profile procedures.

TR3402
QOD, lost canopy, and aircraft configurations for field arrestments.

TR3403
QOD, aileron PCU checklist procedures, and PCU failure modes.

5. Block MIF

TRANSITION OFT		TR3403
CTS Ref	**Critical Items**	
0	Overall Sortie Performance	3+
1	Headwork/Situational Awareness	3+
2	EPs	4+
5	Stalls/Stall Series	4+
5	Unusual Attitude Recovery	4+
3	Full-Stop Landing	4+
CTS Ref	**General Items**	
1	Mission Planning/Brief/Debrief	4+
1	Basic Air Work	4+
1	Partial Panel Air Work	3+
2	Communications	4+
2	Ground Operations	4+
2	Takeoff	4+
2	Departure	4+
2	Flight Management	4+
2	Arrival	4+
2	Course Rules	4+
3	Wave-off	3+
3	FCLP	3+
3	No-HUD Landings	3+
3	Precautionary Approach	4+
3	Roll-and-Go	3+
5	Vertical Recovery	4+
5	S-3 Pattern	4+
5	Minimum Radius Turn	3+
5	Aerobatics	3+
3	Visual Straight-in	3+
4	High-Altitude Penetration	3+
4	Non-Precision Approach	3+
4	Precision Approach	3+
4	Partial Panel Approach	3+
4	No-Gyro GCA	3+
4	Missed Approach	3+
3	PA to Full-Stop	3+

Blk#	Media	Title	Events	Hrs	H/X
TR61	VTD	Transition Exposure	1	1.0	1.0

1. Prerequisites. TR3401 (Transition)

2. Syllabus Notes

a. Mission objective is to execute VFR pattern operations in order to rehearse and familiarize NFSs prior to further execution in aircraft.

b. One CI may link and instruct up to six NFSs in separate devices simultaneously.

c. TR61 block events are considered academic in nature, and are graded on a pass or fail standard.

3. Special Syllabus Requirements. None.

4. Discuss Items. QOD, VFR pattern interval sight picture, pattern communications, pattern EPs, and delta procedures.

5. Block MIF

	TRANSITION VTD	**TR6101**
3	Pattern Entry	1
3	VFR Pattern Operations	1
2	Communications	1

Blk #	Media	Title	Events	Hrs	H/X
TR41	T-45C	Instrument	3	3.6	1.2

1. Prerequisites. TR3208 (Radio Instruments).

2. Syllabus Notes

a. Fly events from the rear cockpit with hood.

b. TR4101 shall be only flight event flown that day, and the brief shall be 2+00 prior to scheduled takeoff.

c. TR4101 is basic instrument training and shall be conducted within the local working area and include S-3 pattern, timed turns, stalls, unusual attitudes, and partial panel.

d. TR4102/3 are Radio Instrument flights and should prioritize instrument basic air work and approaches.

e. The NFS shall perform the following maneuvers per FTI, NATOPS, and SOP within the block:

MANEUVERS	
Departure/SID	1
High Altitude Penetration	1
Non-Precision Approaches (TACAN, VOR ASR)	2 full panel
Precision Approaches (ILS,PAR)	4 full panel, 1 partial panel
No-Gyro GCA	1

3. Special Syllabus Requirements. Instructor must demonstrate aircraft exterior preflight, man-up, interior preflight and post flight aircraft inspection on TR4101.

4. Discuss Items

TR4101
QOD, RADALT usage, approach configurations, and compressor stall.

TR4102-3
QOD, engine surge or compressor stall, and inadvertent engine shutdown (finger lifts).

5. Block MIF

INSTRUMENT FLIGHT		TR4103
CTS Ref	**Critical Items**	
0	Overall Sortie Performance	3+
1	Headwork/Situational Awareness	3+
2	EPs	4+
5	Stalls/Stall Series	4+
5	Unusual Attitude Recovery	4+
3	Full-Stop Landing	4+
CTS Ref	**General Items**	
1	Mission Planning/Brief/Debrief	4+
1	Basic Air Work	4+
1	Partial Panel Air Work	3+
2	Communications	4+
2	Departure	4+
2	Flight Management	4+
2	Arrival	4+
2	Course Rules	4+
5	S-3 Pattern	4+
2	Holding	4+
4	High-Altitude Penetration	3+

MIF continued on next page.

INSTRUMENT FLIGHT		TR4103
CTS Ref	**General Items**	
4	Non-Precision Approach	3+
4	Precision Approach	3+
4	Partial Panel Approach/Missed Approach	3+
4	No-Gyro GCA	3+
4	Missed Approach	3+
3	Low Oil Approach	3+
3	Minimum/Emergency Fuel Approach	3+
	Special Syllabus Requirements	
	Pre and Post-flight Inspection	1
	Man-up and seat preflight	1

Blk #	Media	Title	Events	Hrs	H/X
TR42	T-45C	Familiarization	3	3.6	1.2

1. Prerequisites

a. TR6101 (Transition Exposure).

b. TR4103 (Instrument).

c. TR3403 (Transition).

2. Syllabus Notes

a. TR4201 shall be only flight event flown that day, and the brief shall be 2+00 prior to scheduled takeoff.

b. TR4202/3 events, TR43 block and TR44 block can be executed concurrently. Based on weather, decision on which event to execute shall be made by IP no later than starting the brief.

c. NFS shall have a minimum of FCLP-type landings prior to TR4690. If requirement not met, TR4587 (ET) flights shall be awarded as necessary.

d. NFS shall perform two full-stop landings with roll-out to the end of the runway on separate TR flights prior to TR4690. RTB without TACAN or waypoint on two events prior to TR4690.

e. The NFS shall perform the following maneuvers per FTI, NATOPS, and SOP within the block:

MANEUVERS	
Overhead recovery (break)	1
Precision Approach	3 (1 to full stop)
Simulated short-field arrestment at full-flaps	2 (1 without directional control)

3. Special Syllabus Requirements. Instructor must demonstrate area familiarization, abeam PA, and simulated short-field arrestment (directional control in question) in full-flap configuration to avoid hard landing.

4. Discuss Items

TR4201
QOD, G-Warm maneuver, PA configuration management, course rules execution.

TR4202-3
QOD, short-field arrestment procedures, swerve on touchdown.

5. Block MIF

FAMILIARIZATION FLIGHT		TR4203
CTS Ref	**Critical Items**	
0	Overall Sortie Performance	3+
1	Headwork/Situational Awareness	3+
2	EPs	4+
5	Stalls/Stall Series	4+
5	Unusual Attitude Recovery	4+
3	Full-Stop Landing	4+
CTS Ref	**General Items**	
1	Mission Planning/Brief/Debrief	4+
1	Basic Air Work	4+
2	Communications	4+
2	Ground Operations	4+
2	Takeoff	4+
2	Departure	4+
2	Flight Management	4+
2	Arrival	4+
2	Course Rules	4+
3	Wave-off	3+
3	FCLP	3+
3	No-HUD Landings	3+
3	Half-Flap Landing	3+
3	No-Flap Landing(s)	3+
3	Precautionary Approach	4+
3	Roll-and-Go	3+
5	Turn Pattern	4+
5	Vertical Recovery	4+
7	G-Warm	3+
5	Minimum Radius Turn	3+
5	Aerobatics	3+
3	Visual Straight-in	3+
3	Downwind Entry	3+
3	PA to Full-Stop	3+
	Special Syllabus Requirements	
	Area Familiarization	1
	Abeam PA	1
	Simulated Short-field Arrestment	1

Blk #	Media	Title	Events	Hrs	H/X
TR43	T-45C	Landing Pattern	4	4.8	1.2

1. Prerequisites

a. TR4201 (Familiarization).

b. FAM1104 (LSO Ball Flying Brief).

2. Syllabus Notes

a. NFS shall have a minimum of 60 FCLP-type landings prior to TR4690. If requirement not met, TR4587 (ET) flights shall be awarded as necessary.

b. TR4202/3 events, TR43 block and TR44 block can be executed concurrently. Based on weather, decision on which event to execute shall be made by IP no later than starting the brief.

c. Two of the three following maneuvers are desired on each flight in block (weather permitting): straight-in PA, overhead PA, or abeam PA.

d. RTB without TACAN or waypoint on two events prior to TR4690.

3. Special Syllabus Requirements. None.

4. Discuss Items

TR4301-4
QOD, swerve on touchdown, go-around procedure, ground ejection, and ejection envelope.

5. Block MIF

LANDING PATTERN FLIGHT		TR4304
CTS Ref	**Critical Items**	
0	Overall Sortie Performance	3+
1	Headwork/Situational Awareness	3+
2	EPs	4+
3	Full-Stop Landing	4+
CTS Ref	**General Items**	
1	Mission Planning/Brief/Debrief	4+

MIF continued on next page.

LANDING PATTERN FLIGHT		TR4304
CTS Ref	**General Items**	
1	Basic Air Work	4+
2	Communication	4+
2	Ground Operations	4+
2	Takeoff	4+
2	Departure	4+
2	Flight Management	4+
2	Arrival	4+
2	Course Rules	4+
3	Wave-off	3+
3	FCLP	3+
3	No-HUD Landings	3+
3	Half-Flap Landing	3+
3	No-Flap Landing(s)	3+
3	Precautionary Approach	4+
3	Roll-and-Go	3+
3	PA to Full-Stop	3+

Blk #	Media	Title	Events	Hrs	H/X
TR44	T-45C	Instrument	2	2.4	1.2

1. Prerequisites. TR4201 (Familiarization).

2. Syllabus Notes

a. NFS shall contact their instructor the day prior to brief to determine the route of flight to plan. A copy of a completed DD-1801 and flight planning documents shall be provided for the instructor during the brief.

b. TR4202/3 events, TR43 block and TR44 block can be executed concurrently. Based on weather, decision on which event to execute shall be made by IP no later than starting the brief.

c. Should be flown outside the local area, if able.

d. During this block, NFS must fly at least the approaches listed below.

MANEUVERS	
Non-Precision Approach	1
Precision Approach	2
Partial Panel Approach	1
Emergency Instrument Approach	1

3. Special Syllabus Requirements. None.

4. Discuss Items

TR4401-2
QOD, and in-flight emergencies.

5. Block MIF

INSTRUMENT FLIGHT		TR4402
CTS Ref	**Critical Items**	
0	Overall Sortie Performance	3+
1	Headwork/Situational Awareness	3+
2	EPs	4+
3	Full-Stop Landing	4+

MIF continued on next page.

INSTRUMENT FLIGHT		TR4402
CTS Ref	**General Items**	
1	Mission Planning/Brief/Debrief	4+
1	Basic Air Work	4+
1	Partial Panel Air Work	3+
2	Communications	4+
2	Ground Operations	4+
2	Takeoff	4+
2	Departure	4+
2	Flight Management	4+
2	Arrival	4+
2	Holding	1
4	High-Altitude Penetration	1
4	Non-Precision Approach	3+
4	Precision Approach	3+
4	Partial Panel Approach/Missed Approach	3+
4	No-Gyro GCA	1
4	Missed Approach	3+
3	Low Oil Approach	1
3	Minimum/Emergency Fuel Approach	1

Blk #	Media	Title	Events	Hrs	H/X
TR45	T-45C	Familiarization	3	3.6	1.2

1. Prerequisites. TR4402 (Instrument).

2. Syllabus Notes

a. NFS shall have a minimum of 60 FCLP-type landings prior to TR4690. If requirement not met, TR4587 (ET) flights shall be awarded as necessary.

b. NFS shall perform the break turn stall, landing attitude stall, and approach turn stall on every flight.

c. Two full-stop landings with roll-out to the end of the runway on separate TR stage flights prior to TR4690. RTB without TACAN or waypoint on two TR stage flights prior to TR4690.

d. The NFS shall perform the following maneuvers per FTI, NATOPS, and SOP within the block:

MANEUVERS	
Overhead recovery (break)	1
Precautionary Approach	3 (1 to full stop)
Simulated short-field arrestment at full-flaps	2 (1 without directional control)

3. Special Syllabus Requirements. Instructor must demonstrate area familiarization, abeam PA, and simulated short-field arrestment (directional control in question) in full-flap configuration to avoid hard landing.

4. Discuss Items

TR4501
QOD and electrical system.

TR4502
QOD and hydraulic system.

TR4503
QOD, engine/accessory gear box, PA to full-stop (approach versus ground idle on rollout), and fuel system.

5. Block MIF

FAMILIARIZATION FLIGHT		TR4503
CTS Ref	**Critical Items**	
0	Overall Sortie Performance	3+
1	Headwork/Situational Awareness	3+
2	EPs	4+
5	Stalls/Stall Series	4+
5	Unusual Attitude Recovery	4+
3	Full-Stop Landing	4+
CTS Ref	**General Items**	
1	Mission Planning/Brief/Debrief	4+
1	Basic Air Work	4+
2	Communications	4+
2	Ground Operations	4+
2	Takeoff	4+
2	Departure	4+
2	Flight Management	4+
2	Arrival	4+
2	Course Rules	4+
3	Wave-off	3+
3	FCLP	3+
3	No-HUD Landings	3+
3	Half-Flap Landing	3+
3	No-Flap Landing(s)	3+
3	Precautionary Approach	4+
3	Roll-and-Go	3+
5	Turn Pattern	4+
5	Vertical Recovery	4+
7	G-Warm	4+
5	Minimum Radius Turn	3+
5	Aerobatics	3+
3	Visual Straight-in	3+
3	Downwind Entry	3+
3	PA to Full-Stop	3+

Blk #	Media	Title	Events	Hrs	H/X
TR22	OFT	Transition EPs	1	1.5	1.5

1. Prerequisites. TR4304 (Landing Pattern).

2. Syllabus Notes. The NFS shall perform the following procedures per FTI, NATOPS, and SOP on this event: start malfunction or EP (any), taxi EPs, takeoff EP (any), engine EPs, fuel system EPs (any), electrical EPs (any), OBOGS malfunctions, ECS malfunction or EP, hydraulic system malfunction or EPs (any), flight control EPs (any), GINA malfunction, ejection (low altitude), approach or landing EPs, and post landing malfunctions or EPs.

3. Special Syllabus Requirements. None.

4. Discuss Items. QOD, lost aircraft situations, and start sequence.

5. Block MIF

TRANSITION EP OFT		**TR2201**
CTS Ref	**Critical Items**	
0	Overall Sortie Performance	3+
1	Mission Planning/Brief/Debrief	4+
1	Headwork/Situational Awareness	3+
2	EPs	3+
1	Basic Air Work	3+
2	Communications	3+
CTS Ref	**General Items**	
2	Ground Operations	3+
2	Flight Management	3+
2	Takeoff	3+
2	Departure	3+
2	Arrival	3+
3	Straight-in Approach	3+
3	VFR Landing Pattern	3+
3	No-Flap Touch-and-Go	3+
3	Precautionary Approach	2+
3	Simulated Bird Strike PA	3+
2	Start Malfunctions	3+
2	Ground EPs	3+
2	Aborted Takeoff	3+
2	Landing Gear EPs	3+
2	Electrical EPs	3+

MIF continued on next page.

TRANSITION EP OFT		TR2201
CTS Ref	**General Items**	
2	Hydraulic EPs	3+
2	Ejection	3+
2	Swerve/Blown Tire on Landing	3+
3	Short-Field Arrestment	3+
3	Long-Field Arrestment	3+
3	Rejected Landing/Go-Around	3+
2	Full-Stop with Blown Tire (Non-Arrested)	3+
3	No-Flaps Landings	3+

Blk #	Media	Title	Events	Hrs	H/X
OCF31	OFT	Out-of-Control Flight Simulator	1	1.5	1.5

1. Prerequisites. OCF1102 (OCF Exam).

2. Syllabus Notes. The NFS shall perform air-start, blown tire during field landing, and field-arrested landing with blown tire. Two stuck throttle approaches are required.

3. Special Syllabus Requirements. None.

4. Discuss Items. QOD, runaway trim, engine flameout, ejection situations, locked-in compressor stall, air-start, and NATOPS, Chapter 11.

5. Block MIF

OUT-OF-CONTROL FLIGHT OFT		**OCF3101**
CTS Ref	**Critical Items**	
0	Overall Sortie Performance	3+
2	Ground Operations	4+
2	Takeoff	4+
2	Departure	4+
3	Precautionary Approach	3+
CTS Ref	**General Items**	
1	Mission Planning/Brief/Debrief	3+
1	Headwork/Situational Awareness	3+
1	Basic Air Work	3+
2	EPs	3+
2	Communications	4+
2	Flight Management	3+
3	FCLP	2+
3	Full-Stop Landing	4+
5	Pattern Stall and Recovery	3+
5	High AOA/Deep Stall Investigation/Rudder-Induced Dep.	3+
5	70-Degree Nose-High Departure	3+
5	90-Degree Nose-High Departure	3+
5	110-Degree Nose-High Departure	3+
5	Lateral Stick Adverse Yaw Departure	3+
5	Spin/Spin Recovery	3+
2	Stuck Throttle Approach	3+
3	No-Flap Touch-and-Go	3+
3	No-HUD Landings	3+

Blk #	Media	Title	Events	Hrs	H/X
OCF41	T-45C	Out-of-Control Flight	1	0.7	0.7

1. Prerequisites. OCF3101 (Out-of-Control Flight Sim).

2. Syllabus Notes. None.

3. Special Syllabus Requirements. None.

4. Discuss Items. QOD and NATOPS, Chapter 11.

5. Block MIF

OUT-OF-CONTROL FLIGHT		OCF4101
CTS Ref	**Critical Items**	
0	Overall Sortie Performance	3+
2	Ground Operations	4+
2	Takeoff	4+
2	Departure	4+
CTS Ref	**General Items**	
1	Mission Planning/Brief/Debrief	4+
1	Headwork/Situational Awareness	3+
1	Basic Air Work	3+
2	EPs	4+
2	Communications	3+
2	Flight Management	2+
3	Full-Stop Landing	3+
5	High AOA/Deep Stall Investigation/Rudder-induced Dep.	3+
5	70-Degree Nose-High Departure	3+
5	110-Degree Nose-High Departure	3+
5	Lateral Stick Adverse Yaw Departure	3+

Blk #	Media	Title	Events	Hrs	H/X
TR4690	T-45C	Safe-for-Solo Check Flight	1	1.2	1.2

1. Prerequisites

a. TR4503 (Familiarization).

b. TR2201 (Transition EPs).

c. OCF4101 (Out-of-Control Flight).

d. NA1103 (NATOPS Closed-Book Exam and SOP Exam).

2. Syllabus Notes

a. Bring Transition QA card to brief to ensure all FCLP-type landing, PA, and approach requirements complete.

b. The NFS shall, at a minimum, fly the following maneuvers in the stall series: break turn stall, landing attitude stall, and approach turn stall.

c. NFS shall have a minimum of 60 FCLP-type landings prior to TR4690. If this requirement is not met, TR4587 (ET) flights shall be awarded as necessary. Reference FCLP-type landing notes at the beginning of this chapter for additional guidance.

d. NFSs shall complete the NATOPS open- and closed-book exams prior to TR4690. NFSs shall present the graded open-and closed-book exams to their instructor prior to the brief.

3. Special Syllabus Requirements. None.

4. Discuss Items. QOD and aircraft systems.

5. Block MIF

SAFE-FOR-SOLO CHECK FLIGHT		TR4690
CTS Ref	**Critical Items**	
0	Overall Sortie Performance	3+
1	Headwork/Situational Awareness	3+
2	EPs	4+
5	Stalls/Stall Series	4+
5	Unusual Attitude Recovery	4+
3	Full-Stop Landing	4+
CTS Ref	**General Items**	
1	Mission Planning/Brief/Debrief	4+
1	Basic Air Work	4+
2	Communications	4+
2	Ground Operations	4+
2	Takeoff	4+
2	Departure	4+
2	Flight Management	4+
2	Arrival	4+
2	Course Rules	4+
3	Wave-off	3+
3	FCLP	3+
3	No-HUD Landings	3+
3	No-Flap Landing(s)	3+
3	Precautionary Approach	4+
3	Roll-and-Go	3+
5	Vertical Recovery	4+
5	Minimum Radius Turn	3+
5	Aerobatics	3+

Blk #	Media	Title	Events	Hrs	H/X
TR47	T-45C	Initial Solo	1	1.2	1.2

1. Prerequisites. TR4690 (Safe-for-Solo Check Flight).

2. Syllabus Notes

a. Intentional spins, stalls, unusual attitudes, and vertical recoveries are prohibited maneuvers for solo NFSs.

b. Event shall maintain VMC.

3. Special Syllabus Requirements. None.

4. Discuss Items. QOD, solo brief, and lost aircraft situations.

5. Block MIF

INITIAL SOLO		**TR47S1**
CTS Ref	**Critical Items**	
0	Overall Sortie Performance	3+
1	Headwork/Situational Awareness	3+
CTS Ref	**General Items**	
1	Mission Planning/Brief/Debrief	4+

Blk #	Media	Title	Events	Hrs	H/X
NTR31	OFT	Night Transition	1	1.5	1.5

1. Prerequisites

   a. NTR1103A or NTR1103B (applicable Night Transition Procedures Exam).

   b. IR3490 (NATOPS Instrument Check Flight).

2. Syllabus Notes

   a. NFS shall execute inadvertent IMC, and a touch-and-go landing without IFLOLS.

   b. Entire route is not required to be flown.

3. Special Syllabus Requirements. None.

4. Discuss Items. QOD, lost aircraft situations, night preflight and night cockpit lighting.

5. Block MIF

NIGHT TRANSITION OFT		NTR3101
CTS Ref	**Critical Items**	
0	Overall Sortie Performance	3+
1	Mission Planning/Brief/Debrief	3+
1	Headwork/Situational Awareness	3+
1	Basic Air Work	3+
2	Communications	3+
2	Flight Management	3+
CTS Ref	**General Items**	
2	EPs	4+
2	Ground Operations	3+
2	Takeoff	3+
2	Departure	3+
2	Visual Navigation	3+
2	Dead Reckoning	3+
2	Arrival	3+
4	Precision or Non-precision Approach	4+

MIF continued on next page.

NIGHT TRANSITION OFT		NTR3101
CTS Ref	**General Items**	
2	Course Rules	3+
3	Wave-off	3+
3	FCLP	2+
3	Full-Flap Roll-and-Go	3+
3	Full-Stop Landing	3+

Blk #	Media	Title	Events	Hrs	H/X
NTR21	OFT	Night Transition EPs	1	0.9	0.9

1. Prerequisites. NTR3101 (Night Transition).

2. Syllabus Notes. The NFS shall perform the following maneuvers per FTI, NATOPS, and SOP on this event: total electrical failure, trim malfunction in landing pattern, swerve on touchdown, night abort, pattern stall and recovery, and lost comm.

3. Special Syllabus Requirements. None.

4. Discuss Items. QOD, NORDO light signals, night Bingo considerations, airfield lighting, and cockpit fogging.

5. Block MIF

NIGHT TRANSITION EP OFT		NTR2101
CTS Ref	**Critical Items**	
0	Overall Sortie Performance	3+
1	Mission Planning/Brief/Debrief	3+
1	Headwork/Situational Awareness	3+
1	Basic Air Work	3+
2	Communications	3+
2	Flight Management	3+
CTS Ref	**General Items**	
2	EPs	4+
2	Ground Operations	3+
2	Takeoff	3+
2	Course Rules	3+
3	No-Flap Touch-and-Go	3+
3	Full-Stop Landing	2+
5	Pattern Stall and Recovery	4+
2	Lost Communications	3+
2	Aborted Takeoff	3+
2	Takeoff EPs	3+
2	In-Flight EPs	3+
2	Approach/Landing EPs	3+
2	Electrical EPs	3+
3	No-Flap Landing(s)s	3+
3	Short-Field Arrestment	4+

Blk #	Media	Title	Events	Hrs	H/X
NTR41	T-45C	Night Transition	2	2.2	1.1

1. Prerequisites. NTR2101 (Night Transition EPs).

2. Syllabus Notes

a. Events shall take off no earlier than 30 minutes after official sunset.

b. NFS shall perform at least one night break at the field in block.

c. NTR flights shall execute either high work (area familiarization) or instrument work in addition to the landing pattern. NTR solo conduct **SHALL** be the same as the safe-for-solo event.

3. Special Syllabus Requirements. None.

4. Discuss Items. QOD, Night preflight and Cockpit Lighting.

5. Block MIF

NIGHT TRANSITION		**NTR4102**
CTS Ref	**Critical Items**	
0	Overall Sortie Performance	3+
1	Mission Planning/Brief/Debrief	4+
1	Headwork/Situational Awareness	3+
1	Basic Air Work	4+
2	Communications	4+
2	Flight Management	4+
CTS Ref	**General Items**	
2	EPs	4+
2	Ground Operations	4+
2	Takeoff	3+
2	Departure	3+
2	Arrival	4+
4	Precision or Non-precision Approach	4+
2	Course Rules	4+

MIF continued on next page.

NIGHT TRANSITION		NTR4102
CTS Ref	**General Items**	
3	Wave-off	4+
3	FCLP	3+
3	No-Flap Touch-and-Go	3+
3	Full-Flap Roll-and-Go	3+
3	Full-Stop Landing	4+

Blk #	Media	Title	Events	Hrs	H/X
NTR42	T-45C	Night Transition Solo	2	2.2	1.1

1. Prerequisites. NTR4102 (Night Transition).

2. Syllabus Notes

a. Event shall take off no earlier than 30 minutes after official sunset.

b. NTR flights shall execute either high work or instrument work in addition to the landing pattern. NTR solo conduct **SHALL** be the same as the safe-for-solo event.

3. Special Syllabus Requirements. None.

4. Discuss Items. QOD, solo brief, and waypoint navigation.

5. Block MIF

NIGHT TRANSITION SOLO		**NTR42S2**
CTS Ref	**Critical Items**	
0	Overall Sortie Performance	3+
1	Headwork/Situational Awareness	3+
1	Mission Planning/Brief/Debrief	4+

Blk #	Media	Title	Events	Hrs	H/X
FCL31	OFT	Field Carrier Landing Practice	1	1.3	1.3

1. Prerequisites. FCL1103A or FCL1103B (applicable FCLP Exam).

2. Syllabus Notes

a. 180 FCLP-type landings on the Improved Fresnel Lens Optical Landing System (IFLOLS) are required to prior to FCL4S01. At least 10 of these landings shall be at night. Reference FCLP-type landing notes at the beginning of this chapter for additional guidance.

b. Demonstrate CVN flight operations.

3. Special Syllabus Requirements. None.

4. Discuss Items. QOD, scan technique, field Delta pattern, and "Dirty Bingo."

5. Block MIF

FIELD CARRIER LANDING PATTERN		FCL3101
CTS Ref	**Critical Items**	
0	Overall Sortie Performance	3+
3	FCLP Pattern	2+
3	Start Position	2+
3	Response to LSO Calls	2+
CTS Ref	**General Items**	
1	Mission Analysis/Plan/Brief/Debrief/Brief/Debrief	4+
1	Headwork/Situational Awareness	4+
1	Basic Air Work	3+
2	Admin	3+
2	EPs	4+
3	Field Carrier Landing	2+
3	Full-Stop Landing	3+
3	Wave-off/Wave-off Technique	3+
3	AOA Control	2+
3	Glideslope Control	2+
3	Power Control	2+
3	Lineup Control	2+
3	Error Detection/Correction	2+
3	Bolter/Touch-and-Go Technique	2+

MIF continued on next page.

FIELD CARRIER LANDING PATTERN		FCL3101
CTS Ref	**General Items**	
3	Rejected Landing/Go-Around Scenario	3+
3	Short-Field Arrestment	3+
2	Swerve/Blown Tire on Landing	3+
2	Pattern Stall/ Recovery	3+
2	Divert	3+

Blk #	Media	Title	Events	Hrs	H/X
FCL21	OFT	FCLP EPs	1	1.5	1.5

1. Prerequisites. FCL3101 (Field Carrier Landing Practice).

2. Syllabus Notes. The NFS shall perform the following procedures per FTI, NATOPS, and SOP on this event: in-flight EPs (any), lost communications in pattern, NWS failure on deck, brake failure on deck, GINA failures, Bingo profile, swerve after touchdown, blown tire on landing, short-field arrestment with blown tire(s), and ejection.

3. Special Syllabus Requirements. None.

4. Discuss Items. QOD and ditching situations, BINGO profile.

5. Block MIF

FCLP EPs		**FCL2101**
CTS Ref	**Critical Items**	
0	Overall Sortie Performance	3+
3	FCLP Pattern	2+
3	Start Position	2+
CTS Ref	**General Items**	
1	Mission Analysis/Plan/Brief/Debrief	4+
1	Headwork/Situational Awareness	4+
1	Basic Air Work	4+
2	Admin	4+
2	EPs	4+
3	Field Carrier Landing	2+
3	Wave-off/Wave-off Technique	3+
2	Ground EPs	3+
2	OBOGS EPs	4+
2	Gear EPs	3+
3	Short-Field Arrestment	3+
2	Swerve/Blown Tire on Landing	3+
2	Ejection	3+
2	Divert	3+

Blk #	Media	Title	Events	Hrs	H/X
FCL41	T-45C	Night FCLP Safe-for-Solo	1	0.6	0.6

1. Prerequisites

a. FCL3101 (FCLP Practice).

b. NFR42S1 (Night Formation Solo).

c. DIV42S1 (Division Formation Solo).

2. Syllabus Notes

a. 180 FCLP-type landings on the Improved Fresnel Lens Optical Landing System (IFLOLS) are required to prior to FCL4S01. At least 10 of these landings shall be at night. Reference FCLP-type landing notes at the beginning of this chapter for additional guidance.

b. A total of three solo FCLP periods shall be flown at night prior to CQ4S90. TRAWING Commanders may waive the night requirement for FCL4101 and CQ4101. If this safe for solo night requirement is waived, all solo events in the applicable stage shall be executed during daytime until a night safe for solo is completed. One event in FCL4S or CQ4S0 may be flown dual at night to provide night safe-for-solo training and count as an advancing event in that block.

c. IP shall demonstrate proper wave-off technique and lineup adjustments.

d. LSO required on station. CQ Demo IP required.

3. Special Syllabus Requirements. None.

4. Discuss Items. QOD.

5. Block MIF

FCLP SAFE-FOR-SOLO		FCL4101
CTS Ref	**Critical Items**	
0	Overall Sortie Performance	3+
3	FCLP Pattern	3+
3	Start Position	3+
3	Response to LSO Calls	2+
CTS Ref	**General Items**	
1	Mission Analysis/Planning/Briefing/Debriefing	4+
1	Headwork/Situational Awareness	4+
1	Basic Air Work	4+
2	Admin	4+
2	EPs	4+
3	Field Carrier Landing	3+
3	Full-Stop Landing	4+
3	Full-Flap Roll-and-Go	3+
3	AOA Control	3+
3	Glideslope Control	3+
3	Power Control	3+
3	Lineup Control	3+
3	Error Detection/Correction	3+
3	Bolter/Touch-and-Go Technique	3+

Blk #	Media	Title	Events	Hrs	H/X
FCL4S	T-45C	FCLP Solo	2	1.2	0.6

1. Prerequisites

a. FCL4101 (Night FCLP Safe for Solo).

b. FCL2101 (FCLP EPs).

2. Syllabus Notes

a. 180 FCLP-type landings on the Improved Fresnel Lens Optical Landing System (IFLOLS) are required to prior to FCL4S01. At least 10 of these landings shall be at night. Reference FCLP-type landing notes at the beginning of this chapter for additional guidance.

b. An SOP Exam is required to be completed prior to beginning FCL4S.

c. LSOs shall evaluate and critique each individual pass as well as landing trends; landing grades are at the sole discretion of the LSOs. A maximum of two LSOs shall instruct during the FCL stage.

d. One night period is desired in order to have a minimum of three night SOLO FCLP periods prior to the CQ4S90. TRAWING Commanders may waive this restriction when unforeseen circumstances preclude completion (BASH, weather, etc.).

e. A minimum of six FCLP-type passes are required on each flight; eight are desired.

f. These events shall not be shot-gunned for any reason.

3. Special Syllabus Requirements. None.

4. Discuss Items. QOD, pattern procedures, arrestment procedures, scan techniques, glideslope and lineup corrections, and trend analysis.

5. Block MIF

FCLP SOLO		FCL4S02
CTS Ref	**Critical Items**	
0	Overall Sortie Performance	3+
3	FCLP Pattern	4+
3	Start Position	4+
3	Response to LSO Calls	4+
CTS Ref	**General Items**	
1	Mission Analysis/Planning/Briefing/Debriefing	4+
1	Headwork/Situational Awareness	4+
1	Basic Air Work	4+
2	Admin	4+
2	EPs	4+
3	Field Carrier Landing	3+
3	Full-Stop Landing	4+
3	Wave-off/Wave-off Technique	4+
3	AOA Control	3+
3	Glideslope Control	3+
3	Power Control	3+
3	Lineup Control	3+
3	Error Detection/Correction	3+
3	Bolter/Touch-and-Go Technique	4+

Blk #	Media	Title	Events	Hrs	H/X
FCL4S90	T-45C	FCLP Check Flight Solo	1	0.6	0.6

1. Prerequisites. FCL4S02 (FCLP Solo).

2. Syllabus Notes

a. LSOs shall evaluate and critique each individual pass as well as landing trends; landing grades are at the sole discretion of the LSO.

b. FCL4S90 shall not be shot-gunned for any reason.

3. Special Syllabus Requirements. None.

4. Discuss Items. QOD.

5. Block MIF

FIELD CARRIER LANDING PATTERN		**FCL4S90**
CTS Ref	**Critical Items**	
0	Overall Sortie Performance	3+
3	FCLP Pattern	4+
3	Start Position	4+
3	Response to LSO Calls	4+
CTS Ref	**General Items**	
1	Mission Analysis/Planning/Briefing/Debriefing	4+
1	Headwork/Situational Awareness	4+
1	Basic Air Work	4+
2	Admin	4+
2	EPs	4+
3	Field Carrier Landing	3+
3	Full-Stop Landing	4+
3	Wave-off/Wave-off Technique	4+
3	AOA Control	3+
3	Glideslope Control	3+
3	Power Control	3+
3	Lineup Control	3+
3	Error Detection/Correction	3+
3	Bolter/Touch-and-Go Technique	4+

Blk #	Media	Title	Events	Hrs	H/X
CQ41	T-45C	Night CQ Safe-for-Solo	1	0.6	0.6

1. Prerequisites. FCL4S90 (FCLP Check Flight Solo).

2. Syllabus Notes

a. 250 FCLP-type landings are required prior to CQ4S01. At least 35 of these landings shall be at night. TRAWING Commanders may waive these requirements on a case-by-case basis based on NFS performance.

b. A total of three solo FCLP periods shall be flown at night prior to CQ4S90. TRAWING Commanders may waive the night requirement for FCL4101 and CQ4101. If this safe-for-solo night requirement is waived, all solo events in the applicable stage shall be executed during daytime until a night safe-for-solo is completed. One event in FCL4S or CQ4S0 may be flown dual at night to provide night safe-for-solo training and count as an advancing event in that block.

c. CQ Demo IP required.

3. Special Syllabus Requirements. None.

4. Discuss Items. QOD, Delta pattern, preflight and ground operations, pattern entry, and communications.

5. Block MIF

CARRIER QUALIFICATION		**CQ4101**
CTS Ref	**Critical Items**	
0	Overall Sortie Performance	3+
2	Admin	4+
3	FCLP/CV Pattern	4+
CTS Ref	**General Items**	
1	Mission Analysis/Planning/Briefing/Debriefing	4+
1	Headwork/Situational Awareness	4+
1	Basic Air Work	4+
2	EPs	4+
3	Full-Stop Landing	4+
3	Full-Flap Roll-and-Go	4+

Blk #	Media	Title	Events	Hrs	H/X
CQ31	OFT	Carrier Qualification Landing Simulators	2	2.8	1.4

1. Prerequisites. CQ1102 (CQ Shipboard Procedures).

2. Syllabus Notes

a. During CQ3101, demonstrate CVN flight operations with emphasis on field departure to shipboard recovery.

b. CQ3102 shall continue shipboard procedures with emphasis on EPs.

3. Special Syllabus Requirements. None.

4. Discuss Items

CQ3101
QOD, ship-to-shore checklist, Delta pattern (CV versus field), Case I arrival, shipboard alignment, Case I departure, use of IFLOLS in NORDO, Bingo, and wave-off situations.

CQ3102
QOD, preflight and ground operations, communications, CV terms and comm brevity, pattern entry, Case II arrival and Case II departure, ship-to-shore checklist, and Bingo card data.

5. Block MIF

CARRIER QUALIFICATION		CQ3102
CTS Ref	**Critical Items**	
0	Overall Sortie Performance	3+
2	Admin	4+
3	FCLP/CV Pattern	3+
3	Start Position	3+
3	Response to LSO Calls	4+
3	Error Detection/Correction	2+
CTS Ref	**General Items**	
1	Mission Analysis/Planning/Briefing/Debriefing	4+
1	Headwork/Situational Awareness	4+
1	Basic Air Work	4+

MIF continued on next page.

CARRIER QUALIFICATION		CQ3102
CTS Ref	**General Items**	
2	EPs	4+
3	Wave-off/Wave-off Technique	4+
3	AOA Control	2+
3	Glideslope Control	2+
3	Power Control	2+
3	Lineup Control	2+
3	Bolter/Touch-and-Go Technique	2+
3	CV Arrival (Case I/II)	3+
3	CVN Flight Deck Procedures	3+
3	CVN Arrestment Procedures	3+
3	Catapult Launch Procedures	3+
3	Bingo	4+
3	CV EPs	3+

Blk #	Media	Title	Events	Hrs	H/X
CQ4S0	T-45C	Carrier Qualification Landing Solo	11	6.6	0.6

1. Prerequisites. CQ4101 (Night CQ Safe-for-Solo).

2. Syllabus Notes

a. 250 FCLP-type landings are required prior to CQ4S01. At least 35 of these landings shall be at night. TRAWING Commanders may waive these requirements on a case-by-case basis based on NFS performance.

b. Two night FCLP periods under LSO control are required during CQ. A total of three night SOLO FCLP periods (FCL and CQ stages) under LSO control must be flown prior to CQ4S90. TRAWING Commanders may waive this restriction when unforeseen circumstances preclude completion (BASH, weather, etc.).

c. A minimum of six FCLP-type passes are required on each event (eight are desired).

d. All night CQ flights shall take off no earlier than 30 minutes after official sunset.

e. Up to three CQ events may be flown per day.

f. Landing grades are at the sole discretion of the LSOs.

g. Only CQ events shall be scheduled from CQ4S01 through the completion of CQ4S91.

h. Blown Tire Exercise (BTX) shall be flown at half flaps between CQ4S09-11.

i. These events shall not be shot-gunned for any reason.

3. Special Syllabus Requirements. None.

4. Discuss Items

CQ4S01
QOD, Delta pattern, preflight and ground operations, pattern entry, and communications.

CQ4S02
QOD, pattern procedures, arrestment procedures, Case I procedures, and trend analysis.

CQ4S03
QOD and deck procedures.

CQ4S04
QOD, Bingo and divert procedures, and GINA failure.

CQ4S05
QOD, departure procedures, and return-to-base procedures.

CQ4S06
QOD, carrier-related EPs, and Case II arrival procedures.

CQ4S07
QOD and carrier pattern.

CQ4S08
QOD and carrier procedures.

CQ4S09
QOD, carrier pattern, and BTX.

CQ4S10-11
QOD.

5. Block MIF

CARRIER QUALIFICATION		CQ4S11
CTS Ref	**Critical Items**	
0	Overall Sortie Performance	3+
2	Admin	4+
3	FCLP/CV Pattern	4+
3	Start Position	4+
3	Response to LSO Calls	4+
3	Error Detection/Correction	4+
CTS Ref	**General Items**	
1	Mission Analysis/Planning/Briefing/Debriefing	4+
1	Headwork/Situational Awareness	4+
1	Basic Air Work	4+
2	EPs	4+
3	Full-Stop Landing	4+
3	Wave-off/Wave-off Technique	4+
3	AOA Control	4+
3	Glideslope Control	4+
3	Power Control	4+
3	Lineup Control	4+
3	Bolter/Touch-and-Go Technique	4+

Blk #	Media	Title	Events	Hrs	H/X
CQ21	OFT	CQ EPs	1	1.5	1.5

1. Prerequisites. CQ4S05 (CQ Landing Solo).

2. Syllabus Notes. None.

3. Special Syllabus Requirements. None.

4. Discuss Items. QOD, carrier-related EPs, ditching situations, and short-field arrestments.

5. Block MIF

CARRIER QUALIFICATION		CQ2101
CTS Ref	**Critical Items**	
0	Overall Sortie Performance	3+
2	Admin	4+
3	FCLP/CV Pattern	4+
3	Start Position	3+
3	Response to LSO Calls	4+
3	Error Detection/Correction	2+
CTS Ref	**General Items**	
1	Mission Analysis/Planning/Briefing/Debriefing	4+
1	Headwork/Situational Awareness	4+
1	Basic Air Work	4+
2	EPs	4+
3	Wave-off/Wave-off Technique	4+
3	AOA Control	2+
3	Glideslope Control	2+
3	Power Control	2+
3	Lineup Control	2+
3	Bolter/Touch-and-Go Technique	2+
3	CV Arrival (Case I/II)	4+
3	CVN Flight Deck Procedures	4+
3	CVN Arrestment Procedures	4+
3	Catapult Launch Procedures	4+
2	Ground EPs	4+
3	Bingo	4+

MIF continued on next page.

CARRIER QUALIFICATION		CQ2101
CTS Ref	**General Items**	
3	CV EPs	4+
3	CV Arrestment with Blown Tire(s)	4+
3	Bolter with Blown Tire(s)	4+
3	Field Arrestment with Blown Tire(s)	4+
3	Suspend Procedures	4+
3	Brake Failure on Deck	4+
3	Lost Communications at CVN	4+
2	NWS Failure	4+
2	Launch Bar Malfunction	4+
2	Catapult Malfunctions	4+
2	GINA Failure	3+
2	OBOGS EP	3+
2	Swerve on Touchdown	3+
2	Ejection	3+

Blk #	Media	Title	Events	Hrs	H/X
CQ4S90	T-45C	Carrier Qualification Landing Solo Check Flight (Field)	1	0.6	0.6

1. Prerequisites

   a. CQ4S11 (CQ Landing Solo).

   b. CQ2101 (CQ EPs).

   c. CQ3102 (CQ Landing Sim).

2. Syllabus Notes

   a. 250 FCLP-type landings are required prior to CQ4S01. At least 35 of these landings shall be at night. TRAWING Commanders may waive these requirements on a case-by-case basis based on NFS performance.

   b. CQ4S90 shall not be shot-gunned for any reason.

   c. Landing grades are at the sole discretion of the LSOs.

3. Special Syllabus Requirements. None.

4. Discuss Items. QOD.

5. Block MIF

CARRIER QUALIFICATION		**CQ4S90**
CTS Ref	**Critical Items**	
0	Overall Sortie Performance	3+
2	Admin	4+
3	FCLP/CV Pattern	4+
3	Start Position	4+
3	Response to LSO Calls	4+
3	Error Detection/Correction	4+

MIF continued on next page.

CARRIER QUALIFICATION		CQ4S90
CTS Ref	**General Items**	
1	Mission Analysis/Planning/Briefing/Debriefing	4+
1	Headwork/Situational Awareness	4+
1	Basic Air Work	4+
2	EPs	4+
3	Full-Stop Landing	4+
3	Wave-off/Wave-off Technique	4+
3	AOA Control	4+
3	Glideslope Control	4+
3	Power Control	4+
3	Lineup Control	4+
3	Bolter/Touch-and-Go Technique	4+

Blk #	Media	Title	Events	Hrs	H/X
CQ4S91	T-45C	Carrier Qualification Landing Solo Check Flight (Ship)	1	3.6	3.6

1. Prerequisites

a. CQ4S90 (CQ Landing Solo Check Flight [Field]).

b. CQ1104 (Ship's Brief Exam).

2. Syllabus Notes (Reference CNATRAINST 3740.9G for amplifying information)

a. CQ4S91 shall not be shot-gunned for any reason.

b. Four carrier touch-and-go landings and ten carrier-arrested landings required for completion.

c. An NFS shall have a warmup CQ4S86 if more than two days have elapsed since CQ4S90 or a day touch-and-go and arrestment at the ship.

d. A maximum of six carrier arrestments are permitted for CNATRA NFSs per day. This is waiverable by the CNATRA LSO.

e. A maximum of 3.5 flight hours are permitted per one Carrier Qualification Landing flight for NFSs, to commence at takeoff and terminate with engine shutdown.

f. A maximum of 5 total flight hours per day is permitted for NFSs.

g. NFSs shall be designated as qualified with a GPA of 2.50 or better and a 60-percent boarding rate or better, provided MIF for the block has been met. Additionally, the TRAWING LSO, with CNATRA LSO approval, may qualify NFSs with less than a 2.50 GPA based on improving trends. Conversely, LSO may disqualify an NFS with a GPA above 2.5 due to a decreasing trend or unsafe tendencies. Grading criteria is provided in Landing Signal Officer NATOPS Manual (NAVAIR 00-80T-104) and Carrier Qualification Flight Training Instruction (CNATRA P-1211).

h. Landing grades are at the sole discretion of the LSOs.

3. Special Syllabus Requirements. None.

4. Discuss Items. QOD and Carrier Qualification Landing procedures.

5. Block MIF

CARRIER QUALIFICATION		CQ4S91
CTS Ref	**Critical Items**	
0	Overall Sortie Performance	3+
2	Admin	4+
3	FCLP/CV Pattern	4+
3	Start Position	4+
3	Response to LSO Calls	4+
3	Error Detection/Correction	4+
CTS Ref	**General Items**	
1	Mission Analysis/Planning/Briefing/Debriefing	4+
1	Headwork/Situational Awareness	4+
1	Basic Air Work	4+
2	EPs	4+
3	AOA Control	4+
3	Glideslope Control	4+
3	Power Control	4+
3	Lineup Control	4+
3	Bolter/Touch-and-Go Technique	4+
3	CV Arrival (Case I/II)	4+
3	CVN Flight Deck Procedures	4+
3	CVN Arrestment Procedures	4+
3	Catapult Launch Procedures	4+

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Chapter V

Instrument Training

1. Matrices. Individual matrices following event block descriptions list the MIF requirements for the Instrument training category. This category expands upon Basic Instrument and Radio Instrument skills acquired during Transition training, and includes Instrument Rating and Airways Navigation stages. The purpose of these matrices is to provide the NFS and IP the easiest way to track progress and overall status in relation to MIF.

2. Scheduling

a. Advanced E-2 NFSs execute all instrument training with the exception of:

(1) AN31 block.

(2) AN41 and AN42 blocks.

b. The following blocks have no warm-up or simulator to flight limitations and do not factor into multiple stage enrollment:

(1) AN31 block.

(2) AN41 and AN42 blocks (AN solo events must remain within AN safe for solo currency).

3. Training Notes

a. Heads-Up Display (HUD) shall be incorporated into the instrument scan for all instrument events. Partial panel training shall be utilized throughout the syllabus, particularly in instrument stages of training, in order to ensure NFSs are capable of maintaining Basic Air Work standards with degraded cockpit display functionality.

b. A minimum of one event in both the IR31 and IR32 blocks shall be executed in night conditions.

c. At least one flight event within IR41 shall be flown at night.

d. Any specific approach requirements listed for a flight event may be replaced with another type of precision or non-precision approach if unavailable. For example, an ASR approach requirement may be replaced with a TACAN or VOR approach if ASR is unavailable.

e. All approach requirements listed for flight events may be combined. For example, a Low-Oil PAR will meet the requirements for a PAR or precision approach as well as a Low-Oil approach.

f. NFSs shall bring a **copy** of all completed flight planning documentation and a jet log to the brief for instructor use during the event. For OFT events, NFSs shall retrieve planning handouts from book issue at the ground training building for event specific routes.

g. Electronic Kneeboard (EKB) navigation functions shall only be utilized on instrument events (simulator or aircraft) executed outside the local area.

h. The EKB is primarily a reference tool and shall be properly stowed during simulator and flight events unless specific instances require its temporary usage.

4. IR Notes

a. No more than 60 days shall elapse between completion of the IR1103 exam and successful completion of IR3490 or IR1101-3 shall be retaken.

b. Successful completion of IR3490 shall warrant issuance of a USN standard NATOPS instrument rating. If this NATOPS instrument rating will expire within 180 days of completion of the T-45C undergraduate training, the instrument rating process shall be updated prior to detaching.

c. One out-and-in flight (outside the local area) is required in the IR syllabus prior to IR3490. A cross-country may be substituted for this requirement.

5. Instrument Training Flight Support

INSTRUMENT TRAINING FLIGHT SUPPORT					
Block	**Title**		**Prerequisite**		**Total Hours**
AN1101	Airways Navigation Flight Procedures		TR4402		**2.0**
	AN1101	Airways Navigation Flight Procedures	MIL	2.0	
IR1101-03	Instrument Rating Flight Procedures		AN1101		**3.3**
	IR1101	Meteorology Review	CAI	0.3	
	IR1102	Instrument Rules and CRM Review	MIL	2.0	
	IR1103	Instrument Rating Exam	CAI	1.0	
Total					**5.3**

Blk #	Media	Title	Events	Hrs	H/X
IR31	OFT	Instruments	6	9.0	1.5

1. Prerequisites. AN1101 (Airways Navigation Flight Procedures).

2. Syllabus Notes

a. Handouts listing the route of flight to plan and study for each simulator event shall be obtained by NFSs from book issue at the ground training building.

b. NFSs shall bring a **copy** of their DD-1801 and single-engine jet log to all simulator events for instructor use.

c. During this block, NFSs must fly at least the approaches listed below (approaches may be combined, e.g., a low oil PAR may be logged as a PAR and a low oil approach):

MANEUVERS	
High Altitude Penetration	3
TACAN/VOR DME	2 full panel and 2 partial panel
ASR	2 partial panel
ILS	2 full panel and 2 partial panel
PAR	2 full panel and 2 partial panel
Low Oil Approach	2
Min Fuel/Emer Fuel Approach	2
No-Gyro GCA	2

d. Discuss and introduce a circle-to-land approach on one event in this block. Emphasis shall be placed on instrument-to-visual scan procedures.

e. One event in block shall be executed in night conditions.

3. Special Syllabus Requirements. None.

4. Discuss Items. QOD, circle-to-land, and instrument-to-visual scan.

5. Block MIF

INSTRUMENT RATING		IR3106
CTS Ref	**Critical Items**	
0	Overall Sortie Performance	3+
1	Mission Planning/Brief/Debrief	4+
2	Communications	3+
4	Non-Precision Approach	4+
4	Precision Approach	4+
CTS Ref	**General Items**	
2	EPs	4+
1	Headwork/Situational Awareness	3+
1	Basic Air Work	4+
1	Partial Panel Air Work	4+
2	Ground Operations	4+
2	Takeoff	4+
2	Flight Management	4+
2	Departure	4+
2	En route Navigation	4+
2	Lost Communications	4+
2	Route/Destination Change	3+
2	STAR	3+
2	Arrival	4+
2	Holding	4+
4	Partial Panel Approach/Missed Approach	4+
4	No-Gyro GCA	4+
4	Missed Approach	4+
3	Low Oil Approach	4+
3	Minimum/Emergency Fuel Approach	4+
4	Circling Approach	3+
3	Instrument-to-Visual Scan	3+

Blk #	Media	Title	Events	Hrs	H/X
IR32	OFT	Airways Navigation	5	7.5	1.5

1. Prerequisites. IR3106 (Instruments).

2. Syllabus Notes

a. Handouts listing the route of flight to plan and study for each simulator event shall be obtained by NFSs from book issue at the ground training building.

b. NFSs shall bring a **copy** of their DD-1801 and single-engine jet log to all simulator events for instructor use.

c. IR3203 shall introduce unfamiliar field ground operations and erroneous GINA data.

d. IR3204 shall introduce a STAR.

e. One event in block shall be executed in night conditions.

f. During this block, NFSs shall fly at least the approaches listed below (approaches may be combined, e.g., a low oil PAR may be logged as a PAR and a low oil approach):

MANEUVERS	
TACAN/VOR DEM	2 full panel and 1 partial panel
VOR	1 full panel and 1 partial panel
ASR	2 full panel
ILS	4 full panel and 1 partial panel
PAR	1 full panel an 1 partial panel
No-Gyro GCA	2
STAR	1
Low Oil Approach	1
Min Fuel or Emergency Fuel Approach	1
Circle-to-Lan	1

3. Special Syllabus Requirements. None.

4. Discuss Items

IR3201
QOD, weather criteria, lost communications, and en route descent.

IR3202-5
QOD, en route weather updates, and in-flight fuel calculations.

5. Block MIF

INSTRUMENT RATING		IR3205
CTS Ref	**Critical Items**	
0	Overall Sortie Performance	3+
1	Mission Planning/Brief/Debrief	4+
2	Communications	3+
4	Non-Precision Approach	4+
4	Precision Approach	4+
CTS Ref	**General Items**	
2	EPs	4+
1	Headwork/Situational Awareness	3+
1	Basic Air Work	4+
1	Partial Panel Air Work	4+
2	Ground Operations	4+
2	Takeoff	4+
2	Flight Management	4+
2	Departure	4+
2	En route Navigation	4+
2	Lost Communications	4+
2	Route/Destination Change	3+
2	STAR	3+
2	Arrival	4+
2	Holding	4+
4	Partial Panel Approach/Missed Approach	4+
4	No-Gyro GCA	4+
4	Missed Approach	4+
3	Low Oil Approach	4+
3	Minimum/Emergency Fuel Approach	4+
4	Circling Approach	3+
3	Instrument-to-Visual Scan	3+

Blk #	Media	Title	Events	Hrs	H/X
IR21	OFT	Airways Navigation EP	1	1.3	1.3

1. Prerequisites. IR3203 (Airways Navigation).

2. Syllabus Notes. The NFS shall perform the following procedures per FTI, NATOPS, and SOP on this event: start malfunction or emergency (any), takeoff emergency (any), engine flameout, electrical EPs (any), HYD 2 EDP failure, CONTR AUG failure, runaway stabilator trim, engine fire (secondary indications), lost communications, ejection, main or nose gear unsafe down, brake accumulator failure, and post landing EPs.

3. Special Syllabus Requirements. None.

4. Discuss Items. QOD.

5. Block MIF

INSTRUMENT RATING		**IR2101**
CTS Ref	**Critical Items**	
0	Overall Sortie Performance	3+
1	Mission Planning/Brief/Debrief	4+
2	Communications	3+
CTS Ref	**General Items**	
2	EPs	4+
1	Headwork/Situational Awareness	3+
1	Basic Air Work	4+
1	Partial Panel Air Work	3+
2	Ground Operations	4+
2	Takeoff	4+
2	Flight Management	4+
2	Departure	4+
2	Lost Communications	4+
2	Arrival	4+
2	Start Malfunctions	3+
2	Ground EPs	3+
2	Aborted Takeoff	3+
2	Takeoff EPs	3+
2	Engine EPs	3+
2	Landing Gear EPs	3+
2	Electrical EPs	3+
2	Hydraulic EPs	3+

Blk #	Media	Title	Events	Hrs	H/X
IR41	T-45C	Instrument Rating	5	7.5	1.5

1. Prerequisites

a. IR2101 (Airways Navigation EP).

b. IR3205 (Airways Navigation).

2. Syllabus Notes

a. Fly events from the rear cockpit with hood

b. It is highly recommended that Navigation flights be conducted outside the local flying areas to the maximum extent possible.

c. One out-and-in flight (outside the local area) is required in the IR stage prior to IR3490. A cross-country may be substituted for this requirement.

d. NFSs shall contact their instructor the day prior to determine the route of flight to plan.

e. NFSs shall bring a **copy** of a completed DD-1801 and jet log to the brief for instructor use during the flight.

f. During this block, NFSs shall fly at least the approaches listed below (approaches may be combined where appropriate, e.g., a low oil PAR may be logged as a low oil and a precision approach):

MANEUVERS	
Precision Approach	4 full panel and 2 partial panel
Non-Precision Approach	4 full panel and 2 partial panel
Low Oil Approach	1
Min or Emer Fuel Approach	1
No-Gyro GCA	1

3. Special Syllabus Requirements. None.

4. Discuss Items

IR4101, IR4103, IR4104
QOD, in-flight EPs, fuel planning, and en route descents.

IR4102, IR4105
QOD, in-flight EPs, lost communications, and route or destination change.

5. Block MIF

INSTRUMENT RATING		IR4105
CTS Ref	**Critical Items**	
0	Overall Sortie Performance	3+
1	Mission Planning/Brief/Debrief	4+
2	Communications	4+
4	Non-Precision Approach	4+
4	Precision Approach	4+
CTS Ref	**General Items**	
2	EPs	4+
1	Headwork/Situational Awareness	4+
1	Basic Air Work	4+
1	Partial Panel Air Work	4+
2	Ground Operations	4+
2	Takeoff	4+
2	Flight Management	4+
2	Departure	4+
2	En route Navigation	4+
2	Route/Destination Change	4+
2	STAR	1
2	Arrival	4+
2	Holding	1
4	Partial Panel Approach/Missed Approach	4+
4	No-Gyro GCA	4+
4	Missed Approach	4+
3	Low Oil Approach	4+
3	Minimum/Emergency Fuel Approach	4+
4	Circling Approach	1
3	Instrument-to-Visual Scan	1

Blk #	Media	Title	Events	Hrs	H/X
IR33	OFT	Instrument Rating	3	4.5	1.5

1. Prerequisite. IR4105.

2. Syllabus Notes

a. Handouts listing the route of flight to plan and study for each simulator event shall be obtained by NFSs from book issue at the ground training building.

b. NFSs shall bring a **copy** of their DD-1801 and single-engine jet log to all simulator events for instructor use.

c. During this block, NFSs must fly at least the approaches and maneuvers listed below (approaches may be combined, e.g., a low oil PAR may be logged as a PAR and a low oil approach):

MANEUVERS	
TACAN/VOR DME	1 full panel and 2 partial panel
VOR	1 full panel
ILS	2 full panel and 1 partial panel
PAR	1 full panel
Min or Emer Fuel Approach	1
Low Oil Approach	1
No-Gyro GCA	1
Route or Destination Change	3
STAR	1

3. Special Syllabus Requirements. None.

4. Discuss Items

IR3301
QOD and weather minimums required per CNAF-M 3710.7.

IR3302
QOD and PIREP.

IR3303
QOD and en route descent.

5. Block MIF

INSTRUMENT RATING		IR3303
CTS Ref	**Critical Items**	
0	Overall Sortie Performance	3+
1	Mission Planning/Brief/Debrief	4+
2	Communications	4+
4	Non-Precision Approach	4+
4	Precision Approach	4+
CTS Ref	**General Items**	
2	EPs	4+
1	Headwork/Situational Awareness	4+
1	Basic Air Work	4+
1	Partial Panel Air Work	4+
2	Ground Operations	4+
2	Flight Management	4+
2	Departure	4+
2	En route Navigation	4+
2	Arrival	4+
4	Partial Panel Approach/Missed Approach	4+
4	No-Gyro GCA	4+
4	Missed Approach	4+
3	Low Oil Approach	4+
3	Minimum/Emergency Fuel Approach	4+
4	Circling Approach	3+
3	Instrument-to-Visual Scan	3+

Blk #	Media	Title	Events	Hrs	H/X
IR3490	OFT	NATOPS Instrument Rating Check	1	1.5	1.5

1. Prerequisites

a. IR1103 (Instrument Rating Exam).

b. IR3303 (Instrument Rating).

2. Syllabus Notes

a. Event shall be flown in the local area, but must have at least one approach not at home field.

b. No more than 60 days shall elapse between completion of the IR1103 exam and successful completion of IR3490 or IR1101-3 shall be retaken.

c. Successful completion of this block shall warrant issuance of a USN standard NATOPS instrument rating. If this NATOPS instrument rating will expire within 180 days of completion of the T-45C Undergraduate Strike and Advanced E-2 Curriculum, the instrument rating process shall be updated prior to detaching.

d. One out-and-in flight (outside the local area) is required in the IR syllabus prior to IR3490. A cross-country may be substituted for this requirement.

e. NFSs shall contact their instructor the day prior to determine the route of flight to plan.

f. NFSs shall bring a **copy** of a completed DD-1801 and jet log to the brief for instructor use during the flight.

g. NFSs shall be prepared to discuss in detail any and all aspects of instrument flight in the brief. These include (but are not limited to): procedures; rules governing instrument flight from FARs, NATOPS, or the AIM; information contained in DOD FLIP publications; and EPs.

h. During this block, NFSs must fly at least the approaches listed below:

MANEUVERS	
Precision Approach	1 full panel and 1 partial panel
Non-Precision Approach	1

3. Special Syllabus Requirements. None.

4. Discuss Items. QOD, lost communications, in-flight EPs, and general instrument procedures and knowledge.

5. Block MIF

INSTRUMENT RATING		IR3490
CTS Ref	**Critical Items**	
0	Overall Sortie Performance	3+
1	Mission Planning/Brief/Debrief	4+
2	Communications	4+
4	Non-precision Approach	4+
4	Precision Approach	4+
CTS Ref	**General Items**	
2	EPs	4+
1	Headwork/Situational Awareness	4+
1	Basic Air Work	4+
1	Partial Panel Air Work	4+
2	Ground Operations	4+
2	Flight Management	4+
2	Departure	4+
2	En route Navigation	4+
2	Arrival	4+
4	Partial Panel Approach/Missed Approach	4+
4	Missed Approach	4+
3	Emergency Instrument Approach	4+
3	Instrument-to-Visual Scan	4+

Blk #	Media	Title	Events	Hrs	H/X
AN31	OFT	Advanced Airways Navigation	2	3.0	1.5

1. Prerequisites. DIV4103 (Division Formation).

2. Syllabus Notes

a. Advanced E-2 NFSs do not execute.

b. Handouts listing the route of flight to plan and study for each simulator event shall be obtained by NFSs from book issue at the ground training building.

c. NFSs shall bring a **copy** of their DD-1801 and single-engine jet log to all simulator events for instructor use.

d. AN3101 event shall be flown in night conditions.

e. Lost communications procedures shall be conducted at some point during AN3102.

f. NFSs shall fly the following approaches at a minimum (approaches may be combined, e.g., a low oil PAR may be logged as a PAR and a low oil approach):

MANEUVERS	
Precision Approach	2
Non-Precision Approach	1
Partial Panel Approach	1

3. Special Syllabus Requirements. None.

4. Discuss Item. QOD.

5. Block MIF

AIRWAY NAVIGATION		AN3102
CTS Ref	**Critical Items**	
0	Overall Sortie Performance	3+
1	Mission Planning/Brief/Debrief	4+
2	Communications	4+
4	Non-Precision Approach	4+
4	Precision Approach	4+
CTS Ref	**General Items**	
1	Headwork/Situational Awareness	4+
2	EPs	4+
1	Basic Air Work	4+
1	Partial Panel Air Work	4+
2	Ground Operations	4+
2	Takeoff	4+
2	Departure	4+
2	En route Navigation	4+
2	Arrival	4+
4	High Altitude Penetration	4+
4	Partial Panel Approach	4+
4	Missed Approach	4+
4	Transition to Full-Flap Off Instrument Approach	3+
3	Emergency Instrument Approach	4+
3	Instrument-to-Visual Scan	4+
3	Landings	3+

Blk #	Media	Title	Events	Hrs	H/X
AN41	T-45C	Airways Navigation	2	2.8	1.4

1. Prerequisites. AN3102 (Advanced Airways Navigation).

2. Syllabus Notes

a. Advanced E-2 NFSs do not execute.

b. NFSs shall contact their instructor prior to brief to determine the route of flight to plan.

c. NFSs shall bring a **copy** of a completed DD-1801 and jet log to the brief for instructor use during the flight.

d. Should be flown outside local area, if able.

e. During this block, NFSs must fly a minimum of four total approaches, to include the items listed below (approaches may be combined, e.g., a low oil PAR may be logged as a PAR and a low oil approach):

MANEUVERS	
High Altitude Penetration	1
Non-Precision	1 full panel
Precision	1 full panel
PAR	1
Partial Panel Approach	1
Emergency Instrument Approach	1

3. Special Syllabus Requirements. None.

4. Discuss Items

AN4101-2
QOD, night landing at a field without an IFLOLS, transition from one-half to full flaps for full-stop landing, circle-to-land procedures, uncontrolled airport procedures, UNICOM voice procedures, and in-flight EPs.

5. Block MIF

AIRWAY NAVIGATION		AN4102
CTS Ref	**Critical Items**	
0	Overall Sortie Performance	3+
1	Mission Planning/Brief/Debrief	4+
2	Communications	4+
4	Non-Precision Approach	4+
4	Precision Approach	4+
CTS Ref	**General Items**	
1	Headwork/Situational Awareness	4+
2	EPs	4+
1	Basic Air Work	4+
1	Partial Panel Air Work	4+
2	Ground Operations	4+
2	Takeoff	4+
2	Departure	4+
2	En route Navigation	4+
2	Arrival	4+
4	High Altitude Penetration	4+
4	Partial Panel Approach	4+
4	Missed Approach	4+
4	Transition to Full-Flap Off Instrument Approach	3+
3	Emergency Instrument Approach	4+
3	Instrument-to-Visual Scan	4+
3	Landings	3+

Blk #	Media	Title	Events	Hrs	H/X
AN42	T-45C	Airways Navigation Solo	2	2.4	1.2

1. Prerequisites. AN4102 (Airways Navigation).

2. Syllabus Notes

a. Advanced E-2 NFSs do not execute.

b. AN42 block is a two-leg cross-country or out-and-in solo flight to build confidence in unfamiliar field operations. AN42S1 and shall be scheduled consecutively.

c. NFSs shall brief with the Wing or Squadron Duty Officer (WDO or SDO) and shall have a completed DD-1801 and jet log for the route of flight as well as all pertinent weather and NOTAM information.

d. During this block, NFSs must fly at least one precision approach and one non-precision approach; however, additional approaches are desired if able.

3. Special Syllabus Requirements. None.

4. Discuss Items

AN42S1-2
QOD and in-flight EPs.

5. Block MIF

AIRWAY NAVIGATION		**AN42S2**
CTS Ref	**Critical Items**	
0	Overall Sortie Performance	3+
1	Mission Planning/Brief/Debrief	4+
CTS Ref	**General Items**	
1	Headwork/Situational Awareness	4+

Chapter VI

Navigation Training

This chapter does not apply to T-45C Undergraduate Strike or Advanced E-2 NFSs.

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Chapter VII

Formation Training

1. Matrices. Individual matrices following event block descriptions list the MIF requirements for the Formation training category. This category includes Formation, Night Formation, and Division Formation stages. The purpose of these matrices is to provide the NFS and IP the easiest way to track progress, and overall status in relation to MIF.

2. Formation Training Flight Support

FORMATION TRAINING FLIGHT SUPPORT					
Block	**Title and Objectives**		**Prerequisites**		**Total Hours**
FRM1101-06	Formation Flight Procedures		IR3490		**5.5**
	FRM1101	Marshal, Takeoff, Rendezvous, Departure	MIL	1.0	
	FRM1102	Section Parade Formation	MIL	1.0	
	FRM1103	Formation Recovery, Approaches, Landing Configuration	MIL	0.7	
	FRM1104	Formation Section Cruise	MIL	0.8	
	FRM1105	Formation EPs	MIL	1.0	
	FRM1106	Formation Exam I	CAI	1.0	
NFR1101-02	Night Formation Flight Procedures		FRM2101		**2.2**
	NFR1101	Night Formation Flight Procedures	MIL	1.2	
	NFR1102	Night Formation Exam	CAI	1.0	
DIV1101-02	Division Parade Formation Flight Procedures		FRM2101		**2.5**
	DIV1101	Division Parade Formation	MIL	1.5	
	DIV1102	Formation Exam II	CAI	1.0	
Total					**10.2**

Blk #	Media	Title	Events	Hrs	H/X
FRM31	OFT	Formation Simulators	3	4.5	1.5

1. Prerequisites. FRM1106 (Formation Exam I).

2. Syllabus Notes

a. FRM3101 shall introduce interval takeoff, parade, echelon (VFR parade turns away), cross-unders, breakup and rendezvous, and section break.

b. FRM3102 shall introduce interval takeoff abort, TACAN rendezvous, section approach, and section missed approach.

c. FRM3103 shall introduce section takeoff, cruise position, and section touch-and-go and rejoin.

d. NFS shall demonstrate lost sight procedures in block to include climbs and descents with turns into and away.

3. Special Syllabus Requirements. None.

4. Discuss Items

FRM3101
QOD, lost sight, and underrun.

FRM3102
QOD and section approach minima.

FRM3103
QOD and midair collision.

5. Block MIF

FORMATION		FRM3103
CTS Ref	**Critical Items**	
0	Overall Sortie Performance	3+
2	Admin	3+
2	Initial Rendezvous	3+
6	Parade	3+
6	Breakup and Rendezvous	3+
CTS Ref	**General Items**	
1	Mission Analysis/Plan/Brief/Debrief	3+
2	EPs	4+
1	Headwork/Situational Awareness	3+
1	Basic Air Work	3+
2	Section Takeoff	3+
6	Cross-under/Shuffle	3+
6	TACAN Rendezvous	3+
6	Running Rendezvous	3+
6	Cruise Maneuvering	3+
6	Section Approach	3+
6	Break	3+
3	Precautionary Approach	3+
3	Landing Pattern	3+
3	Full-Stop Landing	3+

Blk#	Media	Title	Events	Hrs	H/X
FRM61	VTD	Formation Exposure	2	2.0	1.0

1. Prerequisites. FRM3103 (Formation).

2. Syllabus Notes

a. Mission objective is formation procedure rehearsal and familiarization prior to execution in aircraft.

b. One CI may link and instruct multiple devices simultaneously.

c. FRM61 block events are considered academic in nature, and are graded on a pass/fail standard.

3. Special Syllabus Requirements. None.

4. Discuss Items. QOD, lost sight procedures, in-flight emergency considerations.

5. Block MIF

FORMATION		**FRM6102**
CTS Ref		
2	Admin	1
2	Initial Rendezvous	1
6	Parade	1
6	Breakup and Rendezvous	1
6	Running Rendezvous	1

Blk #	Media	Title	Events	Hrs	H/X
FRM21	OFT	Formation EPs	1	1.3	1.3

1. Prerequisites. FRM6102 (Formation Exposure).

2. Syllabus Notes. The NFS shall perform the following maneuvers per FTI, NATOPS, and SOP on this event: formation abort, ECS EPs, structural failure or damage, NWS failure, anti-skid failure, pattern stall and recovery, and ejection.

3. Special Syllabus Requirements. None.

4. Discuss Items. QOD and SAR situations.

5. Block MIF

FORMATION EMERGENCY PROCEDURES		FRM2101
CTS Ref	**Critical Items**	
0	Overall Sortie Performance	3+
2	Admin	3+
2	Initial Rendezvous	3+
6	Parade	3+
CTS Ref	**General Items**	
1	Mission Analysis/Plan/Brief/Debrief	3+
2	EPs	4+
1	Headwork/Situational Awareness	3+
1	Basic Air Work	3+
3	Precautionary Approach	3+
2	Ground EPs	4+
2	Aborted Takeoff	4+
2	Takeoff EPs	4+
2	Engine EPs	3+
2	Flight Control EPs	3+
2	Electrical EPs	3+
2	ECS EPs	3+

Blk #	Media	Title	Events	Hrs	H/X
FRM41	T-45C	Formation	8	10.4	1.3

1. Prerequisites. FRM2101 (Formation EPs).

2. Syllabus Notes

a. Lead will brief join position.

b. Brief 2+00 prior to takeoff for FRM4101.

c. FRM4101 shall provide opportunity for IP demonstration, however, NFS may accomplish flight at IP's discretion.

d. Cruise maneuvering, 300 KIAS rendezvous, and Tail Chase should not be introduced until FRM4104.

e. NFSs shall complete a minimum amount of the following maneuvers during the block:

MANEUVERS	
Interval Takeoff	3
TACAN Rendezvous	3
Running Rendezvous	3
Section Takeoff	2
Breakup and Rendezvous (250)	30
Breakup and Rendezvous (300)	5
Join to Inside of Turn	6
4-, 6- Second Break Intervals	4 of each
Underrun	5
Lost Sight Drill	2
Cruise Maneuvering	2
Section Break	2 (as wing) and 1 (as lead)
Section Approach to Missed (May be simulated at altitude)	2 (as wing) and 1 (as lead)
Section Approach to Flight Separation on Final	1
Section Approach to Touch-and-Go and Rejoin	1
Precautionary Approach	1 (2 desired)

3. Special Syllabus Requirements. None.

4. Discuss Items. QOD, lost sight procedures, weather requirements for formation departures, formation launch contingencies, BASH mitigation procedures, trimming techniques, and formation EPs.

5. Block MIF

FORMATION		FRM4108
CTS Ref	**Critical Items**	
0	Overall Sortie Performance	3+
2	Admin	4+
2	Initial Rendezvous	4+
6	Parade	4+
6	Breakup and Rendezvous	4+
CTS Ref	**General Items**	
1	Mission Analysis/Plan/Brief/Debrief	4+
2	EPs	4+
1	Headwork/Situational Awareness	4+
1	Basic Air Work	4+
2	Section Takeoff	3+
6	Cross-under/Shuffle	4+
6	Lead Change	4+
6	TACAN Rendezvous	4+
6	Running Rendezvous	4+
6	Underrun	4+
6	Cruise Position	4+
6	Tail Chase	3+
6	Section Approach	3+
6	Break	4+
6	Formation Lead	3+
2	Lost Sight Drill	4+
3	Precautionary Approach	4+
3	Landing Pattern	4+
3	Full-Stop Landing	3+

Blk #	Media	Title	Events	Hrs	H/X
FRM42	T-45C	Basic Formation Solo	2	2.6	1.3

1. Prerequisites

   a. FRM4108 (FRM4107 if safe-for-solo granted).

   b. TR47S1 (Initial Solo).

2. Syllabus Notes

   a. Running rendezvous may be performed at altitude.

   b. Minimum of 7 Breakup and Rendezvous in block.

   c. Minimum of 1 TACAN rendezvous in block.

   d. In order to execute a section takeoff, one must have been executed on safe-for-solo event.

3. Special Syllabus Requirements. None.

4. Discuss Items. QOD.

5. Block MIF

FORMATION		**FRM42S2**
CTS Ref	**Critical Items**	
0	Overall Sortie Performance	3+
2	Admin	4+
2	Initial Rendezvous	4+
6	Parade	4+
6	Breakup and Rendezvous	4+
CTS Ref	**General Items**	
1	Mission Analysis/Plan/Brief/Debrief	4+
2	EPs	4+
1	Headwork/Situational Awareness	4+
1	Basic Air Work	4+
2	Section Takeoff	1

MIF continued on next page.

FORMATION		FRM42S2
CTS Ref	**General Items**	
6	Cross-under/Shuffle	4+
6	Lead Change	4+
6	TACAN Rendezvous	4+
6	Running Rendezvous	4+
6	Underrun	1
6	Cruise Position	4+
6	Tail Chase	1
6	Section Approach	3+
6	Break	4+
6	Formation Lead	1

Blk#	Media	Title	Events	Hrs	H/X
DIV61	VTD	Division Formation Exposure	1	1.0	1.0

1. Prerequisites. DIV1102 (Formation Exam II).

2. Syllabus Notes

a. Mission objective is division formation procedure rehearsal and familiarization prior to execution in aircraft.

b. One CI will act as flight lead and link and instruct multiple devices simultaneously.

c. DIV61 block events are considered academic in nature, and are graded on a pass or fail standard.

3. Special Syllabus Requirements. None.

4. Discuss Items. QOD, lost sight procedures, in-flight emergency considerations.

5. Block MIF

CTS REF	**DIVISION**	**DIV6101**
2	Admin	1
2	Initial Rendezvous	1
6	Parade	1
6	Cross-under/Shuffle	1
6	Breakup and Rendezvous	1

Blk #	Media	Title	Events	Hrs	H/X
DIV41	T-45C	Division Formation	4	5.6	1.4

1. Prerequisites

   a. DIV6101 (Division Formation Exposure).

   b. FRM42S2 (Basic Formation Solo).

   c. NTR42S2 (Night Transition Solo).

2. Syllabus Notes

   a. A maximum of two flights in this block may be completed as three-plane flights.

   b. NFS must have one division takeoff running or CV rendezvous and one division break prior to DIV4103 (a three-plane rendezvous and break shall meet this requirement).

   c. The following maneuvers shall be flown on every flight:

MANEUVERS	
Section Cross-under	
Balanced Parade and Turns	
Breakup and Rendezvous	6 required (2 per position in a 4-plane, 3 per position in a 3-plane)
Division Cruise	At least twice in block
Precautionary Approaches	1 in block

3. Special Syllabus Requirements. None.

4. Discuss Items. QOD.

5. Block MIF

DIVISION		DIV4104
CTS Ref	**Critical Items**	
0	Overall Sortie Performance	3+
2	Admin	4+
2	Initial Rendezvous	3+
6	Parade	3+
6	Breakup and Rendezvous	4+
CTS Ref	**General Items**	
1	Mission Analysis/Plan/Brief/Debrief	4+
1	Headwork/Situational Awareness	4+
1	Basic Air Work	4+
6	Cross-under/Shuffle	4+
6	Cruise Maneuvering	4+
6	Break	4+
3	Precautionary Approach	4+
3	Landing Pattern	4+
3	Full-Stop Landing	3+

Blk #	Media	Title	Events	Hrs	H/X
DIV42	T-45C	Division Formation Solo	1	1.4	1.4

1. Prerequisites. DIV4104 (Division Formation).

2. Syllabus Notes

a. All maneuvers except landings shall be graded by the flight lead.

b. NFSs must fly six (6) breakup and rendezvous, two per position in a four-plane or three per position in a three-plane.

3. Special Syllabus Requirements. None.

4. Discuss Items. QOD.

5. Block MIF

DIVISION		**DIV42S1**
CTS Ref	**Critical Items**	
0	Overall Sortie Performance	3+
2	Admin	4+
2	Initial Rendezvous	3+
6	Parade	3+
6	Breakup and Rendezvous	4+
CTS Ref	**General Items**	
1	Mission Analysis/Plan/Brief/Debrief	4+
1	Headwork/Situational Awareness	4+
1	Basic Air Work	4+
6	Cross-under/Shuffle	4+
6	Cruise Position	4+
6	Break	4+

Blk #	Media	Title	Events	Hrs	H/X
NFR31	OFT	Night Formation	1	1.5	1.5

1. Prerequisites

   a. NFR1102 (Night Formation Exam).

   b. NTR42S2 (Night Transition solo).

2. Syllabus Notes. Conduct "blind" cockpit switch check.

3. Special Syllabus Requirements. None.

4. Discuss Items. QOD and NORDO light signals.

5. Block MIF

	NIGHT FORMATION	**NFR3101**
CTS Ref	**Critical Items**	
0	Overall Sortie Performance	3+
2	Admin	3+
6	TACAN Rendezvous	3+
6	Parade	2+
6	Breakup and Rendezvous	2+
CTS Ref	**General Items**	
1	Mission Analysis/Plan/Brief/Debrief	3+
2	EPs	4+
1	Headwork/Situational Awareness	3+
1	Basic Air work	3+
6	Cross-under/Shuffle	2+
6	Running Rendezvous	2+
6	Section Approach to Touch-and-Go/Rejoin	2+
6	Break	2+
3	Landing Pattern	3+
3	Full-Stop Landing	2+

Blk #	Media	Title	Events	Hrs	H/X
NFR21	OFT	Night Formation EPs	1	0.9	0.9

1. Prerequisites. NFR3101 (Night Formation).

2. Syllabus Notes. None.

3. Special Syllabus Requirements. None.

4. Discuss Items. QOD, NORDO light signals, night Bingo considerations, airfield lighting, cockpit fogging, and pattern stall.

5. Block MIF

NIGHT FORMATION		NFR2101
CTS Ref	**Critical Items**	
0	Overall Sortie Performance	3+
2	Admin	3+
CTS Ref	**General Items**	
1	Mission Analysis/Plan/Brief/Debrief	4+
2	EPs	4+
1	Headwork/Situational Awareness	4+
1	Basic Air Work	3+
3	Full-Stop Landing	2+
2	Takeoff EPs	4+
2	Aborted Takeoff	4+
2	Electrical EPs	3+
2	In-Flight EPs	3+
2	Landing EPs	4+
2	Lost Communications	4+
2	Field Arrestment	4+
5	Pattern Stall/Recovery	4+

Blk #	Media	Title	Events	Hrs	H/X
NFR41	T-45C	Night Formation	2	2.6	1.3

1. Prerequisites

   a. NFR2101 (Night Formation EPs).

   b. FRM42S2 (Basic Formation Solo).

2. Syllabus Notes

   a. These flights shall take off no earlier than 30 minutes after official sunset.

   b. At least one section break must be flown in this block.

   c. The following maneuvers at a minimum shall be flown on each flight:

MANEUVERS	
TACAN Rendezvous	2
Breakup and Rendezvous	4 on NFR4101 3 on NFR4102
Running Rendezvous at Altitude	
Cross-unders	
Parade	
Lead Change	
Section Approach (May be simulated at altitude)	
Touch-and-Go Rejoin or Section Missed Approach	
Section Break (WX permitting)	

3. Special Syllabus Requirements. None.

4. Discuss Items

   NFR4101
   QOD, landing pattern, formation safety, EPs, and night lead.

   NFR4102
   QOD, NORDO lead change, and total electrical failure.

5. Block MIF

NIGHT FORMATION		NFR4102
CTS Ref	**Critical Items**	
0	Overall Sortie Performance	3+
2	Admin	4+
6	TACAN Rendezvous	3+
6	Parade	3+
6	Breakup and Rendezvous	3+
CTS Ref	**General Items**	
1	Mission Analysis/Plan/Brief/Debrief	4+
2	EPs	4+
1	Headwork/Situational Awareness	4+
1	Basic Air Work	4+
6	Cross-under/Shuffle	3+
6	Lead Change	3+
6	Running Rendezvous	3+
6	Section Approach	4+
6	Touch-and-Go/Rejoin	1
6	Break	4+
3	Landing Pattern	4+
3	Full-Stop Landing	3+

Blk #	Media	Title	Events	Hrs	H/X
NFR42	T-45C	Night Formation Solo	1	1.2	1.2

1. Prerequisites. NFR4102 (Night Formation).

2. Syllabus Notes

a. This flight shall take off no earlier than 30 minutes after official sunset.

b. All maneuvers except landings shall be graded by the flight lead.

c. Running rendezvous shall be done at altitude.

d. Section approach may be simulated at altitude.

e. Section break is desired (weather permitting), but is not required.

f. A day or night front-seat landing within the previous three days is a prerequisite for a night solo flight.

3. Special Syllabus Requirements. None.

4. Discuss Items. QOD and night Bingo procedures.

5. Block MIF

NIGHT FORMATION		NFR42S1
CTS Ref	**Critical Items**	
0	Overall Sortie Performance	3+
2	Admin	4+
6	TACAN Rendezvous	3+
6	Parade	3+
6	Breakup and Rendezvous	3+

MIF continued on next page.

NIGHT FORMATION		NFR42S1
CTS Ref	**General Items**	
1	Mission Analysis/Plan/Brief/Debrief	4+
2	EPs	4+
1	Headwork/Situational Awareness	4+
1	Basic Air Work	4+
6	Cross-under/Shuffle	3+
6	Lead Change	3+
6	Running Rendezvous	3+
6	Section Approach	4+
6	Break	4+

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Chapter VIII

Tactical Training

1. Matrices. Individual matrices following event block descriptions list the MIF requirements for the Tactical training category. This category includes Tactical Formation (TAC), Division Tactical Formation (DTF), Operational Navigation (ON), Section Low-Level (SLL), Strike, Target Area Mechanics (TAM), Basic Fighter Maneuvering (BFM), and Fighter Intercepts (FTX) stages. The purpose of these matrices is to provide the NFS and IP the easiest way to track progress, and overall status in relation to MIF.

Note: Advanced E-2 NFSs do not execute stages within this chapter.

2. Strike Notes. A Strike "E" may be awarded when the Circular Error Probability (CEP) is 50 feet or less with at least four bombs dropped.

3. ON Notes

a. NFSs shall not be scheduled for other events while executing ON11 block.

b. ON1101-3 shall not be scheduled the same day as ON1104-6.

c. NFS is required to produce two copies of two original JMPS-generated ON charts for use during ON and SLL events.

4. Multiple Stage Enrollment Notes. See Chapter 1, paragraph 2.

5. Tactical Training Flight Support

TACTICAL TRAINING FLIGHT SUPPORT					
Block	**Title**		**Prerequisite**		**Total Hours**
TAC1101-03	Tactical Formation Flight Procedures		DIV4104		**4.0**
	TAC1101	TAC HUD/MFD Data Entry Procedures	VBT	0.5	
	TAC1102	Intro to Tactical Formation Procedures	MIL	2.5	
	TAC1103	Tactical Formation Exam	CAI	1.0	
DTF1101-02	Division Tactical Formation Flight Procedures		TAC1103		**2.5**
	DTF1101	Intro to Division Tactical Formation	MIL	1.5	
	DTF1102	Division Tactical Formation Exam	CAI	1.0	

TACTICAL TRAINING FLIGHT SUPPORT (CONT.)					
Block	**Title**		**Prerequisite**		**Total Hours**
ON1101-06	Operational Navigation Flight Procedures		TAC42S2		**15.0**
	ON1101	ON Flight Procedures	MIL	2.0	
	ON1102	Intro into JMPS	MIL	1.0	
	ON1103	JMPS Lab	MIL	4.5	
	ON1104	ONFP Exam	CAI	1.0	
	ON1105	JMPS Lab 2	Lab	5.5	
	ON1106	ON Review	MIL	1.0	
SLL1101-02	Section Low-Level Flight Procedures		ON1106		**2.5**
	SLL1101	Section Low-Level	Lect	2.0	
	SLL1102	Section Low-Level Exam	P/P	0.5	
STK1101-05	Strike Flight Procedures		TAC42S2		**5.0**
	STK1101	Weapons Data Entry Procedures	VBT	0.7	
	STK1102	Weapons Delivery I	MIL	0.9	
	STK1103	Weapons Delivery II	MIL	0.9	
	STK1104	Weapons Delivery III	MIL	1.5	
	STK1105	Weapons Exam	CAI	1.0	
TAM1101-02	Target Area Mechanics Flight Procedures		STK3106		**2.5**
	TAM1101	Target Area Mechanics	MIL	1.5	
	TAM1102	Target Area Mechanics Exam	CAI	1.0	
BFM1101-06	Basic Fighter Maneuvering Flight Procedures		TAC44S2 BFM4203 for BFM1105		**5.8**
	BFM1101	Introduction to BFM/OCF Refresher	MIL	0.8	
	BFM1102	BFM 1v1 Offensive Maneuvering	MIL	1.0	
	BFM1103	BFM 1v1 Defensive Maneuvering	MIL	1.0	
	BFM1104	BFM 1v1 Offensive/Defensive Exam	CAI	1.0	
	BFM1105	BFM 1v1 High Aspect	MIL	1.0	
	BFM1106	BFM 1v1 High Aspect Exam	CAI	1.0	
FTX1101-02	Fighter Intercept Flight Procedures		BFM1104		**2.5**
	FTX1101	Fighter Intercepts	MIL	1.5	
	FTX1102	Fighter Intercepts Exam	CAI	1.0	
Total					**39.8**

Blk#	Media	Title	Events	Hrs	H/X
TAC61	VTD	Tactical Formation Exposure	1	1.0	1.0

1. Prerequisites. TAC1103 (Tactical Formation Exam).

2. Syllabus Notes

a. Mission objective is TAC rehearsal and familiarization prior to execution in aircraft.

b. TAC61 block events are considered academic in nature, and are graded on a pass or fail standard.

3. Special Syllabus Requirements. None.

4. Discuss Items. QOD, lost sight game plan, G-warm maneuver, 500-ft safety bubble, shackles, tac turns, in-place turns, tactical wing position, off-heading turns, co-altitude considerations, gunsight tracking exercise, and rejoin from spread.

5. Block MIF

TACTICAL FORMATION		**TAC6101**
7	Tactical Admin	1
8	Combat Spread	1
8	Tac Wing Position	1
8	Timing/Sight Picture	1
8	Error Corrections	1
7	Visual Lookout	1
8	Gun-Sight Tracking	1

Blk #	Media	Title	Events	Hrs	H/X
TAC41	T-45C	Basic Tactical Formation	3	3.3	1.1

1. Prerequisites

   a. TAC6101 (Tactical Formation Exposure).

   b. FCL4S90 (FCLP Check Flight Solo).

2. Syllabus Notes

   a. Brief 2+00 prior to takeoff for TAC4101.

   b. TAC4101 shall be the only flight flown that day.

   c. The following maneuvers shall be executed on TAC4101:

MANEUVERS	
Push to ATC or Combat Spread (demo)	G-Warm
Rejoin from Combat Spread (demo)	Tac Wing Position (demo/do)
Push to Combat Spread	Shackles (demo/do)
Tac Turns (demo/do)	In-place Turns (demo/do)
Cross Turns (demo/do)	Rejoin(s) from Combat Spread (demo/do)
Battle Damage Checks (demo/do)	

   d. NFS shall execute one instrument approach in block.

3. Special Syllabus Requirements. None.

4. Discuss Items

   TAC4101
   QOD, lost sight game plan, G-warm maneuver, and 500-ft safety bubble.

   TAC4102
   QOD, lost sight game plan.

   TAC4103
   QOD, wingman de-confliction responsibilities, lost sight game plan.

5. Block MIF

TACTICAL FORMATION		TAC4103
CTS Ref	**Critical Items**	
0	Overall Sortie Performance	3+
2	Admin	4+
7	Tactical Admin	3+
2	Initial Rendezvous	4+
8	Combat Spread	3+
8	Tactical Air Work	3+
CTS Ref	**General Items**	
1	Mission Analysis/Plan/Brief/Debrief	4+
1	Headwork/Situational Awareness	4+
1	Basic Air Work	4+
8	Tac Wing Position	3+
8	Timing/Sight Picture	3+
8	Error Corrections	3+
7	Visual Lookout	3+
6	Recovery	4+
4	Precision or Non-Precision Approach	4+
3	Precautionary Approach	4+
3	Full-Stop Landing	4+

Blk #	Media	Title	Events	Hrs	H/X
TAC42	T-45C	Basic Tactical Formation Solo	2	2.2	1.1

1. Prerequisites. TAC4103 (Basic Tactical Formation).

2. Syllabus Notes. NFS shall execute one instrument approach in block.

3. Special Syllabus Requirements. None.

4. Discuss Items. QOD, wingman de-confliction responsibilities, lost sight game plan, G-warm maneuver, and 500-ft safety bubble.

5. Block MIF

TACTICAL FORMATION SOLO		**TAC42S2**
CTS Ref	**Critical Items**	
0	Overall Sortie Performance	3+
2	Admin	4+
7	Tactical Admin	3+
2	Initial Rendezvous	4+
8	Combat Spread	3+
8	Tactical Air Work	3+
CTS Ref	**General Items**	
1	Mission Analysis/Plan/Brief/Debrief	4+
1	Headwork/Situational Awareness	4+
1	Basic Air Work	4+
8	Tac Wing Position	3+
8	Timing/Sight Picture	3+
8	Error Corrections	3+
7	Visual Lookout	3+
4	Precision or Non-Precision Approach	1
6	Recovery	4+
3	Full-Stop Landing	4+

Blk #	Media	Title	Events	Hrs	H/X
TAC43	T-45C	Advanced Tactical Formation	2	2.2	1.1

1. Prerequisites. TAC42S2 (Basic Tactical Formation Solo).

2. Syllabus Notes

a. Advanced Tac Form Maneuvering and the Gunsight Tracking Exercise shall be introduced on TAC4301. T-45 Air-to-Air Training Rules shall be briefed if Gunsight Tracking will be executed.

b. NFS shall brief and lead the admin to and from the working area on one event in block.

3. Special Syllabus Requirements. None.

4. Discuss Items. QOD, wingman de-confliction responsibilities, lost sight game plan, G-warm maneuver, and 500-ft safety bubble.

5. Block MIF

TACTICAL FORMATION		TAC4302
CTS Ref	**Critical Items**	
0	Overall Sortie Performance	3+
2	Admin	4+
7	Tactical Admin	4+
2	Initial Rendezvous	4+
8	Combat Spread	4+
8	Tactical Air Work	4+
CTS Ref	**General Items**	
1	Mission Analysis/Plan/Brief/Debrief	4+
1	Headwork/Situational Awareness	4+
1	Basic Air Work	4+
6	Formation Lead	3+
8	Tac Wing Position	1
8	Timing/Sight Picture	4+
8	Error Corrections	4+
7	Visual Lookout	4+
8	Gun-Sight Tracking	4+
8	Advanced Tac-Form Maneuvering	3+
6	Recovery	4+

Blk #	Media	Title	Events	Hrs	H/X
TAC44	T-45C	Advanced Tactical Formation Solo	2	2.2	1.1

1. Prerequisites. TAC4302 (Advanced Tactical Formation).

2. Syllabus Notes

a. The number and type of maneuvers conducted during each event are at the IP's discretion. Focus should be on getting each maneuver to MIF and building NFS proficiency.

b. NFS shall brief and lead the admin to and from the working area on one event in block.

c. T-45 Air-to-Air Training Rules shall be briefed if Gunsight Tracking will be executed.

3. Special Syllabus Requirements. None.

4. Discuss Items. QOD, wingman de-confliction responsibilities, lost sight game plan, G-warm maneuver, and 500-ft safety bubble.

5. Block MIF

TACTICAL FORMATION		**TAC44S2**
CTS Ref	**Critical Items**	
0	Overall Sortie Performance	3+
2	Admin	4+
7	Tactical Admin	4+
2	Initial Rendezvous	4+
8	Combat Spread	4+
8	Tactical Air Work	4+
CTS Ref	**General Items**	
1	Mission Analysis/Plan/Brief/Debrief	4+
1	Headwork/Situational Awareness	4+
1	Basic Air Work	4+
6	Formation Lead	3+
8	Tac Wing Position	1
8	Timing/Sight Picture	4+
8	Error Corrections	4+
7	Visual Lookout	4+
8	Gun-Sight Tracking	4+
8	Advanced Tac-Form Maneuvering	3+

Blk#	Media	Title	Events	Hrs	H/X
DTF61	VTD	Division Tactical Formation Exposure	1	1.0	1.0

1. Prerequisites. DTF1102 (Division Tactical Formation Exam).

2. Syllabus Notes

a. Mission objective is DTF rehearsal and familiarization prior to execution in aircraft.

b. Four VTD devices should be linked. A minimum of three (3) linked is required.

c. Two CIs will link and instruct simultaneously as lead and -3.

d. DTF61 block events are considered academic in nature, and are graded on a pass or fail standard.

3. Special Syllabus Requirements. None.

4. Discuss Items. QOD, lost sight game plan, division G-warm, and division maneuvers.

5. Block MIF

DIVISION TACTICAL FORMATION		**DTF6101**
CTS REF		
7	Tactical Admin	1
8	Combat Spread	1
8	Tac Wing Position	1
8	Timing/Sight Picture	1
8	Error Corrections	1
7	Visual Lookout	1

Blk #	Media	Title	Events	Hrs	H/X
DTF41	T-45C	Division Tactical Formation	2	2.2	1.1

1. Prerequisites

   a. DTF6101 (Division Tactical Formation Exposure).

   b. TAC4302 (Advanced Tactical Formation).

2. Syllabus Notes

   a. IP lead required as -1 and -3.

   b. Events shall launch as sections and execute a TACAN rendezvous to form into division.

   c. Division tactical formation maneuvers shall be the following:

      (1) Division deploy to wall formation.

      (2) Division G-warm.

      (3) FENCE-in and set PADS (wall formation with altitude de-confliction).

         (a) Tac turns.

         (b) In-place turns.

      (4) Collapse to Parade.

      (5) Deploy to fluid four formation.

         (a) Tac turns.

         (b) In-place turns.

      (6) Battle damage checks.

      (7) Division rejoin.

   d. NFS shall execute at least one instrument approach in block.

3. Special Syllabus Requirements. None.

4. Discuss Items. QOD, lost sight game plan, division G-warm maneuver, altitude de-confliction, yardstick game plan and division maneuvers.

5. Block MIF

	DIVISION TACTICAL FORMATION	**DTF4102**
CTS Ref	**Critical Items**	
0	Overall Sortie Performance	3+
2	Admin	4+
7	Tactical Admin	3+
2	Initial Rendezvous	4+
8	Combat Spread	3+
8	Tactical Air Work	3+
CTS Ref	**General Items**	
1	Mission Analysis/Plan/Brief/Debrief	4+
1	Headwork/Situational Awareness	4+
1	Basic Air Work	4+
8	Tac Wing Position	3+
8	Timing/Sight Picture	3+
8	Error Corrections	3+
7	Visual Lookout	3+
6	Recovery	4+
4	Precision or Non-Precision Approach	4+
3	Full-Stop Landing	4+

Blk #	Media	Title	Events	Hrs	H/X
ON31	OFT	Operational Navigation	3	3.9	1.3

1. Prerequisites. ON1106 (ON Review).

2. Syllabus Notes

a. NFS may use previously made charts but must update the obstacles by using CHUM in JMPS or the *Obstacle* function on a current ForeFlight account.

b. NFS will return all ON charts in their possession to the squadron ON standardization officer when complete with ON stage.

c. NFSs must have a minimum 24-hour notice prior to each previously unplanned ON event for preflight planning.

d. All ON routes should be flown at 400 KGS.

e. T-45C Low Altitude Training (LAT) Training Rules Shall be briefed.

f. ON3101. Shall utilize dead reckoning.

g. ON3102. Introduce response to weather on route and execute Bingo profile. Shall utilize dead reckoning.

h. ON3103. Shall utilize system navigation and be flown on route with significant terrain (VR-1266 desired).

3. Special Syllabus Requirements. None.

4. Discuss Items

ON3101
QOD, low altitude hazards, T-45C dive recovery rules, and Small Descent ROT.

ON3102
QOD, MFD failure, and Small Descent ROT.

ON3103
QOD and Mission Crosscheck Time.

5. Block MIF

	OPERATIONAL NAVIGATION	ON3103
CTS Ref	**Critical Items**	
0	Overall Sortie Performance	3+
9	Low-Level Air Work	3+
9	Navigation	3+
CTS Ref	**General Items**	
1	Mission Planning/Brief/Debrief	4+
1	Headwork/Situational Awareness	3+
1	Basic Air Work	4+
2	Admin	3+
9	Route Entry	3+
9	Route Exit	3+
9	Terrain Awareness	3+
9	Timing	3+
9	Dead Reckoning	3+
3	Full-Stop Landing	3+
2	HUD Failure	3+
3	Minimum/Emergency Fuel Approach	3+
3	Bingo	3+

Blk #	Media	Title	Events	Hrs	H/X
ON41	T-45C	Operational Navigation	3	3.3	1.1

1. Prerequisites. ON3103 (Operational Navigation).

2. Syllabus Notes

a. T-45C LAT Rules shall be briefed.

b. Brief 2+00 prior to takeoff for ON4101.

c. NFS may use previously made charts but must update the obstacles by using CHUM in JMPS or the *Obstacle* function on a current ForeFlight account.

d. NFS will return all ON charts in their possession to the squadron ON standardization officer when complete with ON stage.

e. NFSs must have a minimum 24-hour notice prior to each previously unplanned ON event for preflight planning.

f. All ON routes should be flown at 400 KGS.

g. NFSs must fly two different routes in this block of training. Out and In desired to allow access to more significant terrain.

h. All ON routes may be flown as legs of a cross-country.

i. ON4101-3 are system navigation flights. This does not eliminate the requirement for chart preparation.

3. Special Syllabus Requirements. None.

4. Discuss Items

ON4101
QOD, emergency Bingo, maximum range profile, inadvertent low altitude IMC, and low altitude EPs.

ON4102
QOD, low altitude flight safety, sun angles, shadows, and mission task management.

ON4103
QOD, low altitude flight safety, ridgeline crossing, waypoint data entry, and auto-sequential steering.

5. Block MIF

OPERATIONAL NAVIGATION		ON4103
CTS Ref	**Critical Items**	
0	Overall Sortie Performance	3+
9	Low-Level Air Work	4+
9	Navigation	4+
CTS Ref	**General Items**	
1	Mission Planning/Brief/Debrief	4+
1	Headwork/Situational Awareness	4+
1	Basic Air Work	4+
2	Admin	4+
9	Route Entry	4+
9	Route Exit	4+
9	Terrain Awareness	4+
9	Timing	4+
4	Precision or Non-Precision Approach	4+
3	Full-Stop Landing	4+

Blk #	Media	Title	Events	Hrs	H/X
SLL61	VTD	Section Low-Level Exposure	1	1.0	1.0

1. Prerequisites. SLL1102 (Section Low-Level Exam).

2. Syllabus Notes. None.

a. Mission objective is Section Low-Level procedure rehearsal and familiarization prior to execution in aircraft.

b. One CI may link and instruct multiple devices simultaneously.

c. SLL61 block events are considered academic in nature, and are graded on a pass or fail standard.

3. Special Syllabus Requirements. None.

4. Discuss Items. QOD, lost sight procedures, in-flight emergency considerations.

5. Block MIF

SECTION LOW-LEVEL		**SLL6101**
CTS REF		
7	Tactical Admin	1
9	Route Entry	1
9	Route Exit	1
9	Terrain Awareness	1
8	Tactical Formation	1

Blk #	Media	Title	Events	Hrs	H/X
SLL41	T-45C	Section Low-Level	2	2.0	1.0

1. Prerequisites

a. SLL6101 (Section Low-Level Exposure). The currency window between SLL6101 to SLL4101 is 30 days.

b. ON4103 (Operational Navigation).

c. TAC44S2 (Advanced Tactical Formation Solo).

2. Syllabus Notes

a. T-45C Low Altitude Training Rules shall be briefed.

b. Level laydown tactics shall be conducted.

c. NFS may use previously made charts but they must have update the obstacles by using CHUM in JMPS or the *Obstacle* function on a current ForeFlight account.

d. NFS will return all ON charts in their possession to the squadron ON standardization officer when complete with ON stage.

e. NFS shall lead the administrative portion of SLL4102.

3. Special Syllabus Requirements. None.

4. Discuss Items

SLL4101
QOD, wingman de-confliction responsibilities, and target area de-confliction.

SLL4102
QOD and lead responsibilities.

5. Block MIF

SECTION LOW-LEVEL		SLL4102
CTS Ref	**Critical Items**	
0	Overall Sortie Performance	3+
9	Low-Level Air Work	4+
9	Navigation	3+
CTS Ref	**General Items**	
1	Mission Planning/Brief/Debrief	4+
1	Headwork/Situational Awareness	4+
1	Basic Air Work	4+
2	Admin	4+
7	Tactical Admin	4+
9	Route Entry	4+
9	Route Exit	4+
9	Terrain Awareness	4+
3	Full-Stop Landing	4+
8	Tactical Formation	4+
10	Strike Maneuvering	4+

Blk #	Media	Title	Events	Hrs	H/X
STK31	OFT	Strike	6	7.8	1.3

1. Prerequisites. STK1105 (Weapons Exam).

2. Syllabus Notes

a. NFS shall see a minimum of four passes in each pattern.

b. STK3101 shall introduce and concentrate on the 30-degree pattern. NFS is required to practice aborting a bombing run and dealing with HUD failure. Manual deliveries only.

c. STK3102 shall concentrate on the 30-degree pattern, manual deliveries only.

d. STK3103 shall introduce the 10-degree pattern, then concentrate on 30-degree pattern, manual deliveries only with wind correction (time permitting).

e. STK3104 shall concentrate on the 30-/10-degree pattern, manual deliveries with wind corrections.

f. STK3105 shall introduce 30-/10-degree pattern, CCIP deliveries.

g. With NFS proficiency, winds may be introduced earlier than the STK3104.

h. Winds should be between 10-30 knots with a variant of crosswinds and be consistent for at least four passes.

i. NFS is required to execute at least one instrument approach in block.

3. Special Syllabus Requirements. None.

4. Discuss Items

STK3101
QOD, compute offset aimpoint, and weapons pattern.

STK3102
QOD, abort criteria, and weapons pattern.

STK3103
QOD, hung ordnance approach, weapons pattern, wind corrections, compute CEP, weapons emergencies, and emergency jettison.

STK3104
QOD, delivery validation criteria, heavy wind corrections, and weapons pattern.

STK3105-6
QOD and CCIP target tracking.

5. Block MIF

STRIKE		STK3106
CTS Ref	**Critical Items**	
0	Overall Sortie Performance	3+
10	Pattern Communications	3+
10	Release Parameters	3+
10	Off-Target Rendezvous	3+
CTS Ref	**General Items**	
1	Mission Planning/Brief/Debrief	4+
1	Headwork/Situational Awareness	4+
1	Basic Air Work	4+
2	Admin	3+
7	Tac Admin	3+
10	Weapons Pattern Management	3+
10	Roll-in Mechanics	3+
10	Tracking/Dive Angle	3+
10	Error Corrections	3+
11	Off-Target Mechanics	3+
10	Accuracy	3+
2	Abort Run	4+
4	Precision or Non-Precision Approach	4+
2	HUD Failure	3+

Blk #	Media	Title	Events	Hrs	H/X
STK61	VTD	Strike Exposure	2	2.0	1.0

1. Prerequisites. STK3102 (Strike).

2. Syllabus Notes

a. Mission objective is Strike procedure rehearsal and familiarization prior to execution in aircraft.

b. One CI will act as flight lead and one CI will act as NFS observer. Link and instruct multiple devices simultaneously.

c. STK61 block events are considered academic in nature and are graded on a pass or fail standard.

3. Special Syllabus Requirements. None.

4. Discuss Items. QOD, lost sight procedures, in-flight emergency considerations.

5. Block MIF

STRIKE		**STK6102**
CTS Ref		
2	Initial Rendezvous	1
10	Spacer Pass	1
10	Pattern Communications	1
10	Weapons Pattern Management	1
10	Off-Target Rendezvous	1

Blk #	Media	Title	Events	Hrs	H/X
STK21	OFT	Strike EPs	1	1.3	1.3

1. Prerequisites. STK6102 (Strike).

2. Syllabus Notes. The NFS shall perform the following procedures per FTI, NATOPS, and SOP on this event: in-flight EPs (any), introduce hung ordnance approach, pitot static malfunction, simo run, emergency jettison, HUD failure, lost communications procedures, lost interval procedures, and ejection situations.

3. Special Syllabus Requirements. None.

4. Discuss Items. QOD and lost interval.

5. Block MIF

STRIKE		**STK2101**
CTS Ref	**Critical Items**	
0	Overall Sortie Performance	3+
10	Pattern Communications	3+
CTS Ref	**General Items**	
1	Mission Planning/Brief/Debrief	4+
1	Headwork/Situational Awareness	3+
1	Basic Air Work	4+
2	Admin	3+
7	Tac Admin	3+
10	Weapons Pattern Management	3+
2	Hung Ordnance Approach	3+
2	Abort Run	4+
2	HUD Failure	3+
2	Ground EPs	3+
2	Takeoff EPs	4+
2	In-Flight EPs	3+
2	Landing EPs	3+
2	Lost Communications	3+
2	Pitot Static Malfunction	3+
2	Simo Run	3+
2	Emergency Jettison	3+
2	Blown Tire During Takeoff/Landing	3+
5	Pattern Stall/Recovery	4+

Blk #	Media	Title	Events	Hrs	H/X
STK41	T-45C	Strike	4	3.6	0.9

1. Prerequisites

a. STK2101 (Strike EPs).

b. STK3106 (Strike).

c. DTF4102 (Division Tactical Formation).

2. Syllabus Notes

a. STK4101 shall be flown without ordnance loaded. NFSs shall be exposed to 30-/10-degree patterns. NFSs shall complete two off-target rendezvous and a hung ordnance straight-in. Flight brief shall be 2+00 prior to takeoff.

b. STK4102-04. NFS must deliver 4xMK-76 to complete each event. If 30- and 10-degree patterns are available, NFSs shall drop a minimum of 2x Mk-76 in the 10-degree pattern.

c. NFSs must complete three off-target rendezvous within block. At least one event shall be a four-plane; all events must be a three-plane minimum.

d. STK4102 shall introduce weapons preflight and hung ordnance checks.

e. At least one tape debrief is required. The debrief can assess another aircraft's tapes.

f. STK4101-02 shall utilize manual bombing, STK4103-04 shall utilize CCIP bombing.

3. Special Syllabus Requirements. None.

4. Discuss Items

STK4101
QOD, high-angle-off-tail rendezvous, and non-critical mission tasks.

STK4102
QOD, formation safety, master arm safety, and inadvertent weapons release.

STK4103
QOD, wind corrections, frag avoidance, and delivery validation.

STK4104
QOD, wind corrections, armament system management errors, and ordnance release troubleshooting.

5. Block MIF

STRIKE		STK4104
CTS Ref	**Critical Items**	
0	Overall Sortie Performance	3+
10	Pattern Communications	4+
10	Release Parameters	3+
10	Off-Target Rendezvous	3+
CTS Ref	**General Items**	
1	Mission Planning/Brief/Debrief	4+
1	Headwork/Situational Awareness	4+
1	Basic Air Work	4+
2	Admin	4+
7	Tac Admin	4+
10	Weapons Pattern Management	3+
10	Roll-in Mechanics	3+
10	Tracking/Dive Angle	3+
10	Error Corrections	3+
11	Off-Target Mechanics	3+
10	Accuracy	3+
3	Precautionary Approach	4+
3	Full-Stop Landing	4+
2	Hung Ordnance Approach	4+

Blk #	Media	Title	Events	Hrs	H/X
STK42	T-45C	Strike Solo	2	1.8	0.9

1. Prerequisites. STK4104 (Strike).

2. Syllabus Notes. A minimum of four bombs delivered is required to complete each event. CCIP shall be used.

3. Special Syllabus Requirements. None.

4. Discuss Items

STK42S1
QOD and FTI safety procedures.

STK42S2
QOD and FTI safety procedures.

5. Block MIF

STRIKE		STK42S2
CTS Ref	**Critical Items**	
0	Overall Sortie Performance	3+
10	Pattern Communications	4+
10	Off-Target Rendezvous	3+
CTS Ref	**General Items**	
1	Mission Planning/Brief/Debrief	4+
1	Headwork/Situational Awareness	4+
1	Basic Air Work	4+
2	Admin	4+
7	Tac Admin	4+
10	Weapons Pattern Management	3+
10	Accuracy	3+

Blk #	Media	Title	Events	Hrs	H/X
TAM31	OFT	Target Area Mechanics	2	3.0	1.5

1. Prerequisites. TAM1102 (Target Area Mechanics Exam).

2. Syllabus Notes. Introduction to TAM Execution Template communications. Execute a minimum of three Detached section attacks, concentrating on timing and attack parameters.

3. Special Syllabus Requirements. None.

4. Discuss Items

TAM3101
QOD, TAM Execution Template, Air-to-Surface Checklist, timing.

TAM3102
QOD, TAM Execution Template, Air-to-Surface Checklist, timing.

5. Block MIF

TARGET AREA MECHANICS		**TAM3102**
CTS Ref	**Critical Items**	
0	Overall Sortie Performance	3+
11	Communications	3+
11	Holding Procedures	3+
11	Air-to-Surface Checklist	3+
CTS Ref	**General Items**	
1	Mission Planning/Brief/Debrief	4+
1	Headwork/Situational Awareness	3+
1	Basic Air Work	3+
2	Admin	4+
11	Time-on-Target	3+
11	Dive Delivery Mechanics	3+

Blk #	Media	Title	Events	Hrs	H/X
TAM61	VTD	Target Area Mechanics Exposure	1	1.0	1.0

1. Prerequisites. TAM1102 (Target Area Mechanics Exam).

2. Syllabus Notes

a. Mission objective is Target Area Mechanics procedure rehearsal and familiarization prior to execution in aircraft.

b. CI will act as flight lead.

c. TAM61 block events are considered academic in nature, and are graded on a pass or fail standard.

3. Special Syllabus Requirements. None.

4. Discuss Items. QOD, lost sight procedures, in-flight emergency considerations.

5. Block MIF

TARGET AREA MECHANICS		**TAM6101**
CTS Ref		
11	Communications	1
11	Holding Procedures	1
11	Attached Attacks	1
11	Air-to-Surface Checklist	1

Blk #	Media	Title	Events	Hrs	H/X
TAM41	T-45C	Target Area Mechanics	2	2.2	1.1

1. Prerequisites

   a. TAM6101 (Target Area Mechanics).

   b. STK42S2 (Strike Solo).

   c. ON4102 (Operational Navigation).

2. Syllabus Notes

   a. Brief 2+00 prior to takeoff for TAM4101.

   b. No ordnance.

3. Special Syllabus Requirements. None.

4. Discuss Items. QOD, low-altitude flight safety, low-altitude section deconfliction, NORDO procedures, target area deconfliction, and lookout doctrine.

5. Block MIF

TARGET AREA MECHANICS		**TAM4102**
CTS Ref	**Critical Items**	
0	Overall Sortie Performance	3+
7	Tac Admin	4+
11	Communications	4+
11	Air-to-Surface Checklist	4+
CTS Ref	**General Items**	
1	Mission Planning/Brief/Debrief	4+
1	Headwork/Situational Awareness	4+
1	Basic Air Work	3+
2	Admin	4+
7	Tactical Lead	4+

MIF continued on next page.

TARGET AREA MECHANICS		TAM4102
CTS Ref	**General Items**	
9	Operational Navigation	4+
10	Strike Maneuvering	4+
11	Section Attack Mechanics	3+
11	Dive Delivery Mechanics	3+
11	Off-Target Rendezvous	3+
4	Precision or Non-Precision Approach	4+
3	Full-Stop Landing	4+

Blk #	Media	Title	Events	Hrs	H/X
TAM42	T-45C	Target Area Mechanics Solo	1	1.1	1.1

1. Prerequisites. TAM4102 (Target Area Mechanics).

2. Syllabus Notes. No ordnance.

3. Special Syllabus Requirements. None.

4. Discuss Items. QOD, egress and de-confliction, lost sight, target fixation.

5. Block MIF

TARGET AREA MECHANICS		**TAM42S1**
CTS Ref	**Critical Items**	
0	Overall Sortie Performance	3+
7	Tac Admin	4+
11	Communications	4+
11	Air-to-Surface Checklist	4+
CTS Ref	**General Items**	
1	Mission Planning/Brief/Debrief	4+
1	Headwork/Situational Awareness	4+
1	Basic Air Work	3+
2	Admin	4+
7	Tactical Lead	4+
10	Strike Maneuvering	4+
11	Section Attack Mechanics	3+
11	Dive Delivery Mechanics	3+
11	Off-Target Rendezvous	3+

Blk #	Media	Title	Events	Hrs	H/X
OCF32	OFT	Out-of-Control Simulator	1	1.0	1.0

1. Prerequisites. BFM1104 (Basic Fighter Maneuvering Exam).

2. Syllabus Notes. None.

3. Special Syllabus Requirements. None.

4. Discuss Items. QOD, runaway trim, engine flameout, ejection situations, locked-in compressor stall, air-start, and NATOPS Chapter 11.

5. Block MIF

OUT-OF-CONTROL FLIGHT		**OCF3201**
CTS Ref	**Critical Items**	
0	Overall Sortie Performance	3+
1	Mission Planning/Brief/Debrief	4+
2	Admin	4+
1	Basic Air Work	4+
CTS Ref	**General Items**	
1	Headwork/Situational Awareness	3+
5	High-AOA/Deep Stall Investigation/Rudder-Induced Dep	3+
5	70-Degree Nose-High Departure	3+
5	90-Degree Nose-High Departure	3+
5	110-Degree Nose-High Departure	3+
5	Lateral Stick Adverse Yaw Departure	3+
2	Stuck Throttle Approach	3+
2	Blown Tire During Field Landing	4+
2	Field Arrestment with Blown Tire(s)	4+
2	Air-start	4+
5	Spin/Spin Recovery	3+
5	Pedal Turns	3+
3	Precautionary Approach	4+
3	Full-Stop Landing	4+

Blk #	Media	Title	Events	Hrs	H/X
OCF42	T-45C	OCF/Basic Fighter Maneuvering Intro (OCF/1 V 0)	1	1.0	1.0

1. Prerequisites. OCF3201 (OCF Sim) within the previous 14 days.

2. Syllabus Notes

a. OCF4201 must be flown within 14 days of OCF3201. Stage currency extended to allow for BFM61 and BFM41 event execution.

b. Demonstrate flat scissors maneuvering, roller mechanics, break turn, deck transition, and miscellaneous 1 V 0 maneuvers.

c. OCF4201 updates dynamic flight currency.

d. BFM and OCF qualified IP required.

3. Special Syllabus Requirements. None.

4. Discuss Items. QOD, HUD air-to-air mode setup, departure recovery procedures, spin recovery procedures, departure recovery indications, unusual attitude recovery, and training rules.

5. Block MIF

OUT-OF-CONTROL FLIGHT		**OCF4201**
CTS Ref	**Critical Items**	
0	Overall Sortie Performance	3+
1	Mission Planning/Brief/Debrief	4+
2	Admin	4+
1	Basic Air Work	4+

MIF continued on next page.

OUT-OF-CONTROL FLIGHT		OCF4201
CTS Ref	**General Items**	
1	Headwork/Situational Awareness	3+
7	Tac Admin	3+
5	High-AOA/Deep Stall Investigation/Rudder-Induced Dep	3+
5	70-Degree Nose-High Departure	3+
5	110-Degree Nose-High Departure	3+
5	Lateral Stick Adverse Yaw Departure	3+
5	1v0 Maneuvers	3+
4	Low Oil Approach/PA	4+
3	Full-Stop Landing	4+

Blk#	Media	Title	Events	Hrs	H/X
BFM61	VTD	Basic Fighter Maneuvering Exposure	2	2.0	1.0

1. Prerequisites. BFM1104 (Basic Fighter Maneuvering Exam).

2. Syllabus Notes

a. Mission objective is BFM PADS setup, rehearsal and familiarization prior to execution in aircraft.

b. CI will act as lead.

c. BFM6101 will focus on offensive BFM PADS setup.

d. BFM6102 will focus on defensive BFM PADS setup.

e. BFM61 block events are considered academic in nature, and are graded on a pass or fail standard.

3. Special Syllabus Requirements. None.

4. Discuss Items. QOD, training rules, deck awareness, LAR considerations.

5. Block MIF

BASIC FIGHTER MANEUVERING		**BFM6102**
CTS Ref		
7	Tac Admin	1
12	Training Rules	1
13	Sight/Lookout Doctrine	1

Blk #	Media	Title	Events	Hrs	H/X
BFM41	T-45C	TACFORM Refresher/PADS Intro	1	1.0	1.0

1. Prerequisites. BFM6102 (BFM Exposure).

2. Syllabus Notes

a. This flight shall introduce BFM PADS setups for the horizontal scissors, rolling scissors, 6k and 9k sets, to include one offensive and defensive perch set, preferably demonstrated by the instructor pilot.

b. In order to take each demo past PADS setup and initial entry and reversal, all aircrew must be BFM current per CNATRA training rules.

3. Special Syllabus Requirements. None.

4. Discuss Items. QOD, deck awareness, departure recovery procedures, and training rules.

5. Block MIF

BASIC FIGHTER MANEUVERING		**BFM4101**
CTS Ref	**Critical Items**	
0	Overall Sortie Performance	3+
2	Admin	4+
7	Tac Admin	4+
CTS Ref	**General Items**	
1	Mission Planning/Brief/Debrief	4+
1	Headwork/Situational Awareness	3+
1	Basic Air Work	3+
3	Full-Stop Landing	4+

Blk #	Media	Title	Events	Hrs	H/X
BFM42	T-45C	Offensive Basic Fighter Maneuvering	3	3.0	1.0

1. Prerequisites

   a. OCF4201 (OCF/BFM Intro) within the previous 14 days.

   b. BFM4101 (TACFORM Refresher/PADS Intro).

2. Syllabus Notes

   a. Brief for BFM4201 shall be 2+00 hours prior to takeoff.

   b. NFSs shall lead the recovery portion of the flight at least once in the block.

   c. NFSs shall execute at least six total perch sets in block.

3. Special Syllabus Requirements. None.

4. Discuss Items

   BFM4201
   QOD, BFM concepts and definitions, departure recovery indications and procedures, training rules, deck awareness, and KIO procedures.

   BFM4202-3
   QOD, BFM concepts and definitions, deck transition timing, deck transition mechanics, on-deck maneuvering, and training rules.

5. Block MIF

BASIC FIGHTER MANEUVERING		BFM4203
CTS Ref	**Critical Items**	
0	Overall Sortie Performance	3+
2	Admin	4+
7	Tac Admin	4+
12	Training Rules	4+
13	Sight/Lookout Doctrine	3+
CTS Ref	**General Items**	
1	Mission Planning/Brief/Debrief	4+
1	Headwork/Situational Awareness	3+
1	Basic Air Work	3+
6	Formation Lead	3+
3	Precautionary Approach	4+
3	Full-Stop Landing	4+
13	Snap-Shot Drill	3+
13	Horizontal Scissors	3+
13	Rolling Scissors	3+
13	6k Perch Set	3+
13	9k Perch Set	3+

Blk #	Media	Title	Events	Hrs	H/X
BFM43	T-45C	Offensive Basic Fighter Maneuvering Solo	1	1.0	1.0

1. Prerequisites. BFM4203 (Offensive BFM).

2. Syllabus Notes. None.

3. Special Syllabus Requirements. None.

4. Discuss Items. QOD, BFM concepts and definitions, deck awareness, KIO procedures, unusual attitude recovery, and training rules.

5. Block MIF

BASIC FIGHTER MANEUVERING		**BFM43S1**
CTS Ref	**Critical Items**	
0	Overall Sortie Performance	3+
2	Admin	4+
7	Tac Admin	4+
12	Training Rules	4+
13	Sight/Lookout Doctrine	3+
CTS Ref	**General Items**	
1	Mission Planning/Brief/Debrief	4+
1	Headwork/Situational Awareness	3+
1	Basic Air Work	4+
13	Snap-Shot Drill	3+
13	Horizontal Scissors	3+
13	Rolling Scissors	3+
13	6k Perch Set	3+
13	9k Perch Set	3+

Blk #	Media	Title	Events	Hrs	H/X
BFM44	T-45C	Defensive Basic Fighter Maneuvering	2	1.8	0.9

1. Prerequisites. BFM43S1 (Offensive BFM Solo).

2. Syllabus Notes. NFSs shall perform a minimum of four perch sets in block.

3. Special Syllabus Requirements. None.

4. Discuss Items

BFM4401
QOD, BFM concepts and definitions, energy management, deck awareness, KIO procedures, unusual attitude recovery procedures, ballistic flight recognition, and training rules.

BFM4402
QOD, BFM concepts and definitions, and training rules.

5. Block MIF

BASIC FIGHTER MANEUVERING		**BFM4402**
CTS Ref	**Critical Items**	
0	Overall Sortie Performance	3+
2	Admin	4+
7	Tac Admin	4+
12	Training Rules	4+
13	Sight/Lookout Doctrine	3+
CTS Ref	**General Items**	
1	Mission Planning/Brief/Debrief	4+
1	Headwork/Situational Awareness	3+
1	Basic Air Work	4+
3	Precautionary Approach	4+
3	Full-Stop Landing	4+
13	Snap-Shot Drill	3+
13	Horizontal Scissors	3+
13	Rolling Scissors	3+
13	6k Perch Set	3+

Blk #	Media	Title	Events	Hrs	H/X
BFM45	T-45C	Defensive Basic Fighter Maneuvering Solo	1	0.9	0.9

1. Prerequisites. BFM4402 (Defensive BFM).

2. Syllabus Notes. The NFS shall perform a minimum of two perch sets.

3. Special Syllabus Requirements. None.

4. Discuss Items. QOD, training rules, deck transition and mechanics, ballistic profile recognition, and in-flight EPs.

5. Block MIF

BASIC FIGHTER MANEUVERING		**BFM45S1**
CTS Ref	**Critical Items**	
0	Overall Sortie Performance	3+
2	Admin	4+
7	Tac Admin	4+
12	Training Rules	4+
13	Sight/Lookout Doctrine	3+
CTS Ref	**General Items**	
1	Mission Planning/Brief/Debrief	4+
1	Headwork/Situational Awareness	3+
1	Basic Air Work	4+
13	Snap-Shot Drill	3+
13	Horizontal Scissors	3+
13	Rolling Scissors	3+
13	6k Perch Set	3+

Blk #	Media	Title	Events	Hrs	H/X
BFM46	T-45C	High Aspect Basic Fighter Maneuvering	2	1.8	0.9

1. Prerequisites

   a. BFM1106 (BFM 1 V 1 High-Aspect Exam).

   b. BFM45S1 (Defensive BFM Solo).

2. Syllabus Notes. None.

3. Special Syllabus Requirements. None.

4. Discuss Items. QOD, concepts and definitions, training rules, high-aspect BFM, one-circle engagement, two-circle engagement, energy management, controlling the merge, lost sight procedures, game-plan development, unique merges, and maintaining or regaining sight.

5. Block MIF

BASIC FIGHTER MANEUVERING		**BFM4602**
CTS Ref	**Critical Items**	
0	Overall Sortie Performance	3+
2	Admin	4+
7	Tac Admin	4+
12	Training Rules	4+
13	Sight/Lookout Doctrine	3+
CTS Ref	**General Items**	
1	Mission Planning/Brief/Debrief	4+
1	Headwork/Situational Awareness	4+
1	Basic Air Work	4+
3	Full-Stop Landing	4+
13	Snap-Shot Drill	1
13	Butterfly Set	3+
13	Abeam Set	3+

Blk #	Media	Title	Events	Hrs	H/X
BFM47	T-45C	High Aspect Basic Fighter Maneuvering Solo	2	1.8	0.9

1. Prerequisites. BFM4602 (High Aspect BFM).

2. Syllabus Notes. NFS shall execute one instrument approach in block.

3. Special Syllabus Requirements. None.

4. Discuss Items. QOD, BFM concepts and definitions, lost-sight game plan, maintaining or regaining sight, training rules, and in-flight EPs.

5. Block MIF

BASIC FIGHTER MANEUVERING		**BFM47S2**
CTS Ref	**Critical Items**	
0	Overall Sortie Performance	3+
2	Admin	4+
7	Tac Admin	4+
12	Training Rules	4+
13	Sight/Lookout Doctrine	3+
CTS Ref	**General Items**	
1	Mission Planning/Brief/Debrief	4+
1	Headwork/Situational Awareness	4+
1	Basic Air Work	4+
13	Snap-Shot Drill	1
13	Butterfly Set	3+
4	Precision or Non-Precision Approach	1
13	Abeam Set	3+

Blk #	Media	Title	Events	Hrs	H/X
FTX31	OFT	Fighter Intercepts	1	1.5	1.5

1. Prerequisites. FTX1102 (Fighter Intercepts Exam).

2. Syllabus Notes. Accomplish a minimum of three (3) intercepts.

3. Special Syllabus Requirements. None.

4. Discuss Items. QOD, timeline awareness, training rules, KIO procedures.

5. Block MIF

FIGHTER INTERCEPTS		**FTX3101**
CTS Ref	**Critical Items**	
0	Overall Sortie Performance	3+
2	Admin	3+
7	Tac Admin	3+
14	Timeline Adherence	3+
CTS Ref	**General Items**	
1	Mission Planning/Brief/Debrief	3+
1	Headwork/Situational Awareness	3+
1	Basic Air Work	3+
14	System Management	3+
14	Fighter Communications	3+

Blk#	Media	Title	Events	Hrs	H/X
FTX61	VTD	Fighter Intercepts Exposure	1	1.0	1.0

1. Prerequisites. FTX3101 (Fighter Intercepts).

2. Syllabus Notes

a. Mission objective is FTX rehearsal and familiarization prior to execution in aircraft.

b. CI will act as flight lead. NFS shall lead one (1) intercept, preferably a faded run.

c. FTX61 block events are considered academic in nature, and are graded on a pass or fail standard.

3. Special Syllabus Requirements. None.

4. Discuss Items. QOD, deck awareness, LAR considerations, faded run setup, NFS block, faded run execution, KIO procedures, formation lead considerations, and training rules.

5. Block MIF

FIGHTER INTERCEPTS		**FTX6101**
CTS Ref		
7	Tac Admin	1
14	Timeline Adherence	1
14	Fighter Communications	1
6	Formation Lead	1

Blk #	Media	Title	Events	Hrs	H/X
FTX41	T-45C	Fighter Intercepts	2	2.0	1.0

1. Prerequisites

a. FTX6101 (Fighter Intercepts Exposure).

b. BFM47S2 (High Aspect BFM Solo).

2. Syllabus Notes

a. Brief 2+00 hours prior to takeoff for FTX4101.

b. Execute OCF4286 (single-ship) or BFM4686 (section) warmup events prior to FTX41 if not dynamically current.

c. Accomplish a minimum of five (5) intercepts in block, four (4) of which shall be faded runs.

d. BFM shall not be executed unless all aircrew are dynamically current per CNATRA training rules.

e. NFS shall lead an intercept on each flight, preferably faded runs.

f. One instrument approach shall be conducted in block. An NFS leading a section approach is acceptable.

3. Special Syllabus Requirements. None.

4. Discuss Items. QOD, timeline awareness, faded run setup, NFS block, faded run execution, KIO procedures, formation lead considerations, and training rules.

5. Block MIF

FIGHTER INTERCEPTS		FTX4102
CTS Ref	**Critical Items**	
0	Overall Sortie Performance	3+
2	Admin	4+
7	Tac Admin	4+
14	3-D Formation Management	3+
14	Timeline Adherence	3+
CTS Ref	**General Items**	
1	Mission Planning/Brief/Debrief	4+
1	Headwork/Situational Awareness	4+
1	Basic Air Work	4+
12	Training Rules	4+
14	System Management	3+
14	Fighter Communications	3+
14	Employment	3+
14	Cold Ops	3+
14	Re-commit	3+
7	Tactical Lead	3+
7	Visual Mutual Support	3+
6	Formation Lead	3+
4	Precision or Non-Precision Approach	4+
3	Full-Stop Landing	4+

Blk #	Media	Title	Events	Hrs	H/X
FTX42	T-45C	Fighter Intercepts Solo	2	2.0	1.0

1. Prerequisites. FTX4102 (Fighter Intercepts).

2. Syllabus Notes

a. Accomplish a minimum of four (4) intercepts in block.

b. NFS shall lead at least one (1) intercept in each flight.

c. NFS shall brief and lead FTX42S2.

d. BFM shall not be executed unless all aircrew are dynamically current per CNATRA training rules.

3. Special Syllabus Requirements. None.

4. Discuss Items. QOD, timeline awareness, faded run setup, NFS block, faded run execution, KIO procedures, formation lead considerations, lost sight game plan, in-flight EPs, and training rules.

5. Block MIF

FIGHTER INTERCEPTS		**FTX42S2**
CTS Ref	**Critical Items**	
0	Overall Sortie Performance	3+
2	Admin	4+
7	Tac Admin	4+
14	3-D Formation Management	3+
14	Timeline Adherence	3+

MIF continued on next page.

FIGHTER INTERCEPTS		FTX42S2
CTS Ref	**General Items**	
1	Mission Planning/Brief/Debrief	4+
1	Headwork/Situational Awareness	4+
1	Basic Air Work	4+
12	Training Rules	4+
14	System Management	3+
14	Fighter Communications	3+
14	Employment	3+
14	Cold Ops	3+
14	Re-commit	3+
7	Tactical Lead	3+
7	Visual Mutual Support	3+
6	Formation Lead	3+

Chapter IX

Course Training Standards (CTS)

1. Purpose. These standards outline the tasks and proficiency required to graduate from this syllabus.

2. NFS Duties and Responsibilities

a. Plan the mission.

b. Ensure the aircraft is preflighted, inspected, and equipped for the assigned mission.

c. Operate the aircraft to accomplish the mission using sound judgment and airmanship.

3. General Standards

a. Unless otherwise specified, use ***Basic Air Work (BAW)*** standards for all items with altitude, airspeed, or heading parameters.

b. "Standard" equates to ***Good*** (G or 4).

c. Aircraft control must be smooth and positive. Performance may be within CTS and still not warrant a grade of ***Good*** if control inputs are delayed, erratic, imprecise, or inappropriate. Slight deviations in establishing or maintaining the proper or desired aircraft attitude or position may occur during the maneuver being performed.

d. Momentary deviations outside CTS that do not compromise flight safety are acceptable if subsequent corrections are timely.

e. Procedural knowledge and application must comply with applicable directives and allow efficient mission accomplishment. If individual tasks require pre-mission planning, the standards from ***Mission Planning*** apply.

4. Execution. The Maneuver Item File (MIF) regulates NFS progression to meet required standards prior to phase completion. Instructor Pilots shall evaluate NFS performance against these standards.

a. As NFSs progress through this syllabus, many maneuver items become combined in overarching CTS references. For example, "takeoff" is listed as a separate maneuver item early in training, but will fall under its overarching CTS reference "admin" as training focus shifts toward additional maneuver items.

b. The intent of this consolidation of maneuver items is to reinforce the importance of tying individual skills and maneuvers together into a singular overall effort.

5. Job Tasks. Specific performance and standards required are described as follows:

BEHAVIOR STATEMENT	STANDARDS
GRADED ITEM	
• A brief description of the behavior, required action, or conditions.	• The specific standards for the action. May be read as "The NFS aviator..."

6. Course Training Standards

BEHAVIOR STATEMENT	STANDARDS
0. Overall Performance Assessment	
• IP assessment of overall performance relative to block progress .	• Executes overall event mission, to include brief and debrief, to a level of performance commensurate with syllabus or block progression.
1. General	
Mission Planning/Brief/Debrief	
• Perform appropriate mission planning to include route selection,weather, NOTAMS, fuel-optimization, computing takeoff, climb, en route, descent, approach,and landing data: planning mission profile and alternate course of action where appropriate. • Attend and conduct pre and post-mission briefing and debriefing for simulator or aircraft event.	• Plans mission in a timely manner to meet training objectives, complete all applicable Navy and command forms correctly, and complies with all directives. • Applies CNAF M-3710.7 filing and approach criterion to planning andexecution of flight. • Aware of alternatives available, if flight cannot be completed as planned. • Briefs per NATOPS and command directives. • Asks questions, if necessary, to fully understand the mission overview and mission objectives, including ORM. • Clearly presents all information requested during briefing and debriefing. • Understands all CRM objectives and expectations for the mission. • Understands contingencies and plans to contend with them.

BEHAVIOR STATEMENT	STANDARDS
Mission Planning/Brief/Debrief (cont.)	
● Demonstrate knowledge of aircraft systems, procedures, and associated directives and instructions.	● Effectively compares mission results with briefed objectives. ● Displays professional attitude and ability to accept instruction. ● Demonstrates a thorough understanding of aircraft systems capabilities, aircraft directives, and local procedures. ● Knowledgeable of local working area WRT boundaries, altitudes, and significant landmarks without reference to in-flight guide or charts. ● Applies procedures from all applicable source guidance.
Headwork/Situational Awareness	
● Assess self and aircraft in relation to the dynamic environment of flight, threats, and mission forecast; then execute tasks basedon this assessment. ● Utilize CRM.	● Understands instructions, demonstrations, and explanations. ● Remains alert and spatially oriented. ● Correctly interprets in-flight events and applies strategies to proactively address them. ● Recognizes and avoids channelized attention. ● Effectively utilizes all CRM skills throughout all portions of training.
Basic Air Work	
● Perform general aircraft control and composite or instrument cross-check as appropriate.	● Maintains smooth positive aircraft control at all times. ● Ensures momentary deviations, ±5 seconds, do not exceed: ► Airspeed: ±5 percent. ► Altitude: ±100 feet. ► Heading: ±5 degrees. ► Course: ±1 dot or ½ scale. ► AOA: ±1 unit.

BEHAVIOR STATEMENT	STANDARDS
Basic Air Work (cont.)	
● Perform general aircraft control and composite or instrument cross-check in a partial panel situation.	● Avoids hazards (ground obstructions, terrain, aircraft, and severe weather). ● Smoothly transitions to and from partial panel instrument scan as required. ● Maintains course, altitude, and glideslope with minor deviations and appropriate error corrections for entirety of approach. ● Deviations do not jeopardize safety of flight.
Partial Panel Air Work	● Maintains positive control of the aircraft at all times with a smooth transition from full panel to partial panel scan. ● Ensures momentary deviations ±5 seconds, do not exceed: ► Airspeed: ±15 percent. ► Altitude: ±150 feet. ► Heading: ±10 degrees. ► AOA: ±1 unit. ● Deviations do not jeopardize safety of flight.
2. Admin	
Emergencies/EPs	
● Recognize system malfunction or emergency situation. ● Perform NATOPS immediate action emergency procedures.	● Expeditiously analyzes situation and systems and recognizes malfunction or emergency situation. ● Maintains control of aircraft while responding appropriately to malfunction or emergency. ● Maneuvers aircraft smartly to prevent degradation of situation with respect to external factors such as weather, traffic, etc.

BEHAVIOR STATEMENT	STANDARDS
Emergencies/EPs (cont.)	
● Perform NATOPS noncritical action emergency procedures to include: ► Analysis of hypothetical aircraft malfunctions. ► Simulated precautionary approaches and actual No-Flap Landing(s)s performed in the aircraft. ► Life support training, survival, and physiological training per NATOPS.	● Verbally states emergency NATOPS immediate action items in sequence, from memory, without error. ● Performs proper steps of emergency NATOPS immediate action items in sequence, from memory, without error. ● Performs proper steps to a satisfactory conclusion, effectively using NATOPS PCL to troubleshoot or complete NATOPS procedures. ● Incorporates effective CRM to secure additional assistance where applicable. ● Maintains situational awareness WRT local area and airfields while troubleshooting systems and responding appropriately to situation. ● Successfully recovers aircraft to suitable airfield or recognizes extremis situation and initiates ejection within safe parameters.
● Lost communications.	● Performs proper steps to a satisfactory conclusion, effectively using FIH to troubleshoot or complete lost communication procedures.
● Lost sight (Blind).	● Performs proper steps to a satisfactory conclusion, effectively using FTI procedures to create safe separation between aircraft.
Communication	
● Verbal	● Makes concise, timely transmissions and responses, using proper radio discipline with standard terminology. ● Makes required radio calls per FLIP requirements. ● Understands and prioritizes transmissions in a multiple communication(UHF, VHF and ICS) environment. ● Asks for and provides clarification when necessary. ● Maintains effective 2-way communication with other crew members.

BEHAVIOR STATEMENT	STANDARDS
Communications (cont.)	
● Visual	● Ensures visual signals are clearly visible to lead/wingman and per NATOPS, FTI, or flight briefing.
Ground Operations	
● Inspect and wear appropriate flight equipment. ● Perform exterior inspection, pre-start and pre-taxi checks to adhere to takeoff times within published tolerances. ● Coordinate checks with other aircrew for formation flight. ● Perform taxi to/from runway. ● Complete "Instrument," "Before Takeoff," and "After Landing" checklists. ● Perform the engine shutdown checklist. ● Perform post-flight inspection and administrative duties.	● Complies with NATOPS and command directives. ● Determines aircraft status and accepts or rejects aircraft based on NATOPS and command directives. ● Completes required checks correctly. ● Complies with NATOPS procedures and standardization tolerances. ● Ensures clearance of line personnel, ground equipment, and other aircraft using appropriate signals prior to activation of aircraft systems. ● Performs all checks, to include formation flight procedures per applicable directives. ● Taxies at speeds commensurate with traffic and surface conditions, following prescribed route and giving way to other aircraft as appropriate. ● Avoids hazards and ground obstructions. ● Completes per NATOPS procedures. ● Completes per NATOPS procedures. ● Completes all post-flight checks and administrative duties per NATOPSand applicable directives.
Flight Management	
● Perform in-flight planning and administrative functions, to include: ► General. ► Local course rules.	● Adjusts mission profile to comply with time/fuel limitations, as well as weather and area limits. ● Complies with established routes, altitudes, and procedures for operating in local airspace environment.

BEHAVIOR STATEMENT	STANDARDS
Flight Management (cont.)	
► Area management. ► Task management. ► Fuel management. ► Weather planning. ► In-flight checks. ► Route or destination change. ► Electronic Knee Board (EKB) Failure.	● Uses assigned airspace in an efficient manner with minimum delay between maneuvers. ● Remains within area boundaries with or without ground references. ● Prioritizes and accomplishes tasks in order of importance as it pertains to flight and mission accomplishment. ● Properly utilizes mission cross-check time based on terrain, taskload, and personal performance. ● Actively monitors fuel state throughout the mission. ● Complies with all established fuel requirements. ● Recognizes Joker or Bingo fuel within ±100 pounds of briefed quantity and makes timely call to IP or lead. ● Regulates flight profile, throttle, and configuration to optimize fuel consumption as appropriate for the mission profile and training objectives. ● Recognizes and applies CNAF M-3710.7 and FLIP weather minima required for selected type of approach to field. ● Completes all checklist items correctly and at proper point in mission, to include checking over other aircraft in the flight, per directives. ● Properly coordinates flight plan change through appropriate FSS or ATC facility using a DRAFT report or the IFR Supplement Change of Flight Plan formatting. ● If necessary, obtains new weather report along route of flight and at destination field. ● Calculates new fuel requirements along with time of flight. ● Successfully transitions to backup publications per applicable wing or squadron SOP.

BEHAVIOR STATEMENT	STANDARDS
Takeoff	
● Perform individual takeoff to include: ► Run-up check. ► Line-speed. ► Retracting gear and flaps. ► Accelerating to climb airspeed. ● Transition to instruments as required.	● Maintains position during engine run-up for static takeoff. ● Maintains runway centerline ±5 feet during takeoff. ● Rotates within -0 to +10 knots of computed rotation speed and maintains desired pitch attitude ±2 degrees. ● Establishes and maintains proper takeoff attitude at appropriate airspeed for existing conditions. ● Initiates gear and flap retraction when safely airborne and ensures fully retracted prior to exceeding 200 KIAS. ● Properly transitions to flight instruments as required for actual or simulated weather conditions.
Initial Rendezvous	
● Perform two- and four-ship takeoff as Wing per Formation FTI, to include: ► Section takeoff. ► Interval takeoff.	● Positions aircraft in appropriate lane of runway ±3 feet, on appropriate bearing line or "banana echelon." ● Achieves target interval ±1 second for brake release. ● Maintains appropriate lane of runway ±5 feet during takeoff roll. ● Lifts off no earlier than lead and maintains ±15 degrees of parade bearing. ● Configures on lead's signal, making smooth, positive control inputs; signals clean at appropriate time. ● Smoothly and expeditiously accelerates to appropriate rendezvous speed. ● Initiates cross to inside of expected turn within 5 seconds of aircraft clean, but not before interval.

BEHAVIOR STATEMENT	STANDARDS
Initial Rendezvous (cont.)	
	● Upon reaching target airspeed, expeditiously puts lead or interval on the horizon. ● Accomplishes timely rendezvous maintaining lead on horizon per CV,TACAN, or running rendezvous standards.
Departure	
● Safely maneuver aircraft out of airfield environment. ► IFR. ► VFR.	● Performs departure as published or directed. ● Complies with all restrictions. ● Achieves and maintains target climb schedule airspeeds ±10 KIAS or 0.02Mach at target altitudes ±1,000 feet. ● Initiates level-off at desired altitude per FTI. ● Promptly establishes cruise airspeed.
Navigation	
● Perform en route navigation to include: ► Perform visual navigation or dead reckoning procedures, to include chart interpretation.	● Complies with basic air work standards. ● Compensates for known wind drift as required. ● Identifies chart symbols with prominent landmarks along route. Navigates via dead reckoning or waypoint navigation, as applicable.
► Climbs and Descents.	● Maintains target airspeed ±10 knots. ● Levels off at desired altitude ±100 feet per FTI. ● Complies with all restrictions. ● Establishes a valid intercept. ● Maintains course ±5 degrees, 1 dot, or ½ scale.
► Intercept and maintain course - perform VOR or TACAN course intercepts inbound, outbound, or immediately after station passage,and maintain course.	● Establishes valid arc intercept, utilizing appropriate lead turn as needed. ● Maintains arc ±0.2 mile. ● Establishes valid arc-to-radial intercept. ● Complies with basic air work standards. ● Compensates for known wind drift as required. ● Maintains target airspeed ±10 knots.

BEHAVIOR STATEMENT	STANDARDS
Navigation (cont.)	
► Arcing - Perform VOR/DME and TACAN radial-to-arc intercepts and maintain arcs. ► Non-system point-to-point.	● Makes initial turn in the proper direction. ● Performs steps to TACAN or VOR/DME point-to-point per FTI. ● Corrects initial turn and maintains heading +10 degrees to arrive at the desired point ±0.5 NM. ● Complies with basic air work standards.
► System point-to-point.	● Compensates for known wind drift as required. ● Maintains target airspeed ±10 knots. ● Makes initial turn in the proper direction. ● Enters proper fix and all required navigational information into GINA and proceeds direct using TACAN or waypoint offset procedures. ● Arrives at the desired point ±0.2 NM.
► STAR - Perform standard arrival (STAR) procedure per FLIP publication.	● Establishes and maintains valid course intercepts 1 dot or ½ scale, ±5 degrees. ● Establishes valid arc or radial intercepts and maintains arcs ±0.5 mile. ● Meets all altitude and airspeed restrictions.
Arrival	
● Perform a descent and traffic entry, to include: ► En route descent. ► Max range descent. ► Field break.	● Executes as published or directed. ● Complies with all restrictions and directives. ● Analyzes internal and external factors to select most effective method of descent (en route or max range). ● Utilizes LAW effectively to observe platform and subsequent altitude restrictions. ● Observes "minute to live" rule (unless scenario or circumstances specifically dictate otherwise).

BEHAVIOR STATEMENT	STANDARDS
Arrival (cont.)	
	● Establishes proper interval for pattern entry. ● Maintains break altitude ±100 feet until established on downwind. ● Configures in adequate time to perform landing and AOA and airspeed checks prior to approach turn 90-degree position.
Holding	
● Perform high- and low-altitude VOR/TACAN holding as described by controller or per FLIP document.	● Performs published standard entry procedures and maintains designated pattern per Instrument NATOPS and FTI. ● Complies with holding pattern limits: ► Uses proper voice procedures. ► Maintains holding airspeed ±5 KIAS.
3. Landing Pattern	
● Perform entry into visual landing pattern (pattern entry to the start) to include: ► Visual straight-in. ► Downwind entry. ► Overhead pattern (left-hand FCLP-type).	● Configures in adequate time to perform landing and AOA and airspeed checks prior to glideslope acquisition or final descent. ● Makes timely corrections for glideslope, AOA, and lineup deviations. ● Applies crosswind corrections adequately to maintain centerline both on final and during touchdown. ● Configures in adequate time to perform landing and AOA and airspeed checks prior to 180 position. ● Makes timely corrections for glideslope, AOA, and lineup deviations. ● Applies crosswind corrections adequately to maintain centerline both on final and during touchdown. ● Maintains pattern altitude ±50 feet on downwind. ● Makes appropriate crosswind corrections on downwind to arrive at proper abeam distance.

BEHAVIOR STATEMENT	STANDARDS
3. Landing Pattern (cont.)	
	● Initiates approach turn with appropriate extension off of abeam to achieve proper groove length (15-18 seconds). ● Manages energy state and AOB while making timely corrections to deviations throughout approach turn to arrive at the start within ±5 degrees of centerline, on-speed, with appropriate VSI, and with the ball centered to mid-high (3-4 balls) on the lens.
Landing	
● (Start to touchdown) Perform touch-and-go or full-stop landing to include the following: ► Touch-and-go. ▪ Full-flap. ▪ Half-flap. ▪ No-flap. ▪ Crosswind. ► FCLP-type landing (IFLOLS).	● References optical landing system, if available, to achieve safe approach glideslope. ● Touches down at proper pitch attitude, maintains proper ground track, uses crosswind controls as required. ● Touches down in prescribed landing zone per NATOPS and local procedures. ● Touches down with no greater than -600 fpm rate of descent for flap configurations other than full. ● Performs graded touch-and-go or full-stop landing utilizing IFLOLS lens. ● Adequately manages energy state during wings-level transition to maintain reasonable VSI, AOA, and lineup control. ● Makes timely and appropriate corrections to maintain or correct back to optimum glideslope, AOA, and lineup. ● Applies crosswind corrections adequately to maintain centerline both on final and during touchdown.

BEHAVIOR STATEMENT	STANDARDS
Landing (cont.)	
► Roll-and-go. ▪ Full-flap. ▪ Half-flap (sim short-field arrest). ▪ No-flap. ► Night landing at field without a lens. ► Full-stop.	● Consistently touches down with a stable, centered-to-high ball, on-speed, and on centerline. ● Maintains runway alignment using aileron, rudder, and nose-wheel steering to track down runway. ● Recognizes groundspeed checkpoints and executes go-around at target airspeed ±5 KIAS, ±200 feet of target runway remaining. ● If required, performs instrument-to-visual scan while maintaining glideslope and centerline. ● Applies crosswind corrections adequately to maintain centerline both on final and during touchdown. ● Adequately manages energy state during approach to landing with proper AOA and VSI control. ● Touches down in prescribed landing zone per NATOPS and local procedures. ● Applies appropriate crosswind corrections and maintains runway alignment using aileron, rudder, and nose-wheel steering. ● Applies braking smoothly and effectively to meet deceleration schedule. ● Adjusts braking to achieve appropriate line speeds.
Wave-off	
● Perform wave-off procedures.	● Immediately executes wave-off procedures when required or directed, maintaining landing attitude and AOA until safe climb established. ● Maintains safe lateral separation from interval aircraft in VFR pattern.

BEHAVIOR STATEMENT	STANDARDS
Precautionary Approach	
• Perform precautionary approach per NATOPS, FTI and local SOP and course rules, to include: ► Overhead. ► Abeam. ► Straight-in. ► Modified. ► Bird Strike or Dirty PA. ► BASH mitigation procedure. • Performs precautionary instrument approach per NATOPS, FTI and local SOP and course rules, to include: ► Low oil approach. ► Min or emer fuel approach.	• Properly coordinates maneuver with ATC. • Effectively manages airspace for entry, including appropriate voice reports. • Effectively manages energy state via configuration and maintains profile without manipulation of throttle. If simulating FOD or bird damage with associate high-power engine surge (locked-in compressor stall), momentarily selects IDLE in order to clear. • Utilizes target airspeed and altitude checkpoints (±15 knots, +300, -200feet) to effectively maintain profile. • Manages flare adequately to touch down in first third of runway or prior to A-gear, if required. • Safely achieves flight with flying airspeed, mil power, and speed-brakes retracted during touch-and-go. • Properly coordinates maneuver with ATC. • Effectively manages energy state via configuration to maintain adequate approach profile.
LSO-Controlled Pattern	
• Perform graded carrier landing under LSO control to field carrier box or CV landing area, to include following: ► CV Pattern.	• Executes FCLP or CV pattern entry from takeoff or catapult launch per CV NATOPS and FTI. • Maintains pattern altitude on downwind ±50 feet. • Makes appropriate crosswind correction on downwind to arrive at proper abeam distance. • Initiates approach turn with appropriate extension off abeam to achieve proper groove length. • Manages energy state and makes timely corrections to deviations around approach turn without assistance.

BEHAVIOR STATEMENT	STANDARDS
LSO-Controlled Pattern (cont.)	
► Start position.	● Intercepts acceptable glideslope, centerline and groove length without LSO assistance.
► AOA control.	● Maintains on speed with only minor deviations (±1 unit).
► Glideslope control.	● Maintains glide-path with average deviations and corrections without LSO assistance.
► Power Control.	● Controls throttle movements for proper correction of glide-path deviations without LSO assistance.
► Lineup control.	● Intercepts and tracks centerline with average deviations and correctionswithout LSO assistance.
► Error detection and correction.	● Detects and corrects in a timely manner deviation in glide-path, lineup,and AOA without assistance.
► Response to LSO calls.	● Responds in a safe and timely manner to glide-path, power, attitude, lineup, and wave-off calls from LSO.
► Bolter and touch-and-go technique.	● Simultaneously moves throttle to MRT, retracts speed brakes and rotatesto optimum AOA immediately upon touchdown without LSO assistance.
► Wave-off technique.	● Immediately executes wave-off procedures when required or directed.
CVN Operations	
● Safely operate T-45C in and around aircraft carrier.	● Properly controls and maneuvers aircraft per T-45C NATOPS, CV NATOPS, and FTI procedures.
4. Approach	
High-Altitude Penetration	
● Perform a VOR, VOR/DME, or TACAN penetration (arc and radial intercept) from IAF to FAF, as published in FLIP document or local procedures.	● Complies with published penetration course, arc, and altitudes. ● Complies with basic air work standards. ● Establishes valid intercepts. ● Maintains course ±5 degrees, 1 dot, or ½ scale. ● Establishes valid arc and radial intercepts. ● Maintains arcs ±0.5 NM.

BEHAVIOR STATEMENT	STANDARDS
Precision Approach	
● Perform precision, full panel, partial panel, or no-gyro approaches as published in FLIP or local procedures, to include: ► ILS approach. ► PAR approach. ▪ No-Gyro PAR. ► Transition from one-half flap approach setting to full flaps for landing.	● Complies with published approach and NATOPS procedures. ● Maintains target AOA or final approach airspeed ±1 unit AOA or ±5KIAS during final descent. ● Arrives at DA in position to maintain a normal visual glide-path to the runway and land safely. ● Maintains CDI and GSI within 1 dot, ½ scale deflection. ● Maintains ±3 degrees of assigned heading (except gyro out) and does not achieve multiple "well above" or "well below" glide-path calls. ● Prior to DA, configures to full flaps and reviews landing checks complete to confirm the configuration change. ● Recalculates and slows to the new full-flap target AOA ±1 Unit or ±5KIAS while maintaining appropriate glideslope to touchdown.
Non-Precision Approach	
● Perform non-precision, full panel, partial panel, or no-gyro approaches as published in FLIP or local procedures, to include: ► Localizer approach or BC localizer. ► TACAN or VOR/DME approach.	● Complies with published approach and NATOPS procedures. ● Arrives at and maintains MDA -0, +100 feet at or prior to VDP. ● Arrives in position to maintain a normal visual glide-path to the runway and land safely. ● Begins timing within 5 seconds, if appropriate. ● Maintains target AOA or final approach airspeed ±1 unit AOA or ±5KIAS after FAF. ● Maintains CDI within 1 dot, ½ scale deflection. ● Maintains target AOA or final approach airspeed ±1 unit AOA or ±5KIAS after FAF. ● Maintains final approach course ±1 dot, ½ scale, or 5 degrees.

BEHAVIOR STATEMENT	STANDARDS
Non-Precision Approach (cont.)	
► ASR approach.	● Maintains target AOA or final approach airspeed ±1 unit AOA or ±5KIAS during and after descent to MDA. ● Maintains ±3 degrees of assigned heading (Except No-Gyro). ● Does not exceed 1 call of "well left or right of course" and complies with controller's instructions in a timely manner. ● Observes "minute to live" rule during descent to MDA.
► Transition from one-half flap approach setting to full flaps for landing.	● Prior to VDP or arriving in position to maintain a normal visual glide path to the runway, configures to full flaps and reviews landing checks complete to confirm the configuration change. ● Recalculates and slows to the new full-flap target AOA ±1 Unit AOA or ±5 KIAS and then maintains appropriate glideslope to touchdown.
Circling Approach	
● Perform a circling approach and maneuver as published in FLIP document or local procedures.	● Accomplishes per Instrument FTI and Instrument NATOPS. ● Prior to circling maneuver maintains course and altitude per non-precision approach standards. ● During maneuver, maintains circling MDA -0 feet, and maintains visual reference to the airport until acquiring visual glide-path. ● Positions aircraft for a safe landing. ● Once visual reference with the runway environment is acquired, appropriately transitions from an instrument scan to a visual scan while beginning the circling maneuver as published, as instructed by ATC, or in an appropriate manner to safely and efficiently execute the maneuver.

BEHAVIOR STATEMENT	STANDARDS
Circling Approach (cont.)	
	● Remains within the clear zone for the approach category. ● If required, executes appropriate missed approach instructions for the approach flown. ● Executes circling maneuver on the appropriate side of the airfield.
Missed Approach	
● Perform a missed approach and partial panel missed approach. ● Perform appropriate climb-out.	● Complies with FLIP document and ATC instructions for missed approach or climb-out instructions. ● Completes per Instrument FTI and Instrument NATOPS.
5. MOA Maneuvers	
● Perform maneuvers as described in the FTI or as directed, to include: ► Vertical recovery. ► Minimum radius turn.	● Executes per FTI descriptions, to include: ► Attains stabilized target entry airspeed ±5 knots. ► Smoothly applies back stick to achieve 17 units without pitch-buck. ► Elevates nose to and maintains attitude at 60 degrees (±3 degrees) until recovery. ► Initiates recovery at target airspeed ±5 knots. ► Begins maneuver with sufficient altitude excess to complete maneuver. ● Executes per FAM FTI descriptions, to include: ► Attains stabilized target entry airspeed ±5 knots. ► Maintains MRT throughout maneuver. ► Smoothly applies back stick to achieve 17 ±1 unit. ► Maintains ±5 knots throughout maneuver. ► Prevents excessive nose “ballooning” during reversals (100-feet max).

BEHAVIOR STATEMENT	STANDARDS
5. MOA Maneuvers (cont.)	
	► Completes reversal and final rollout ±5 degrees of target heading. ► Begins maneuver with sufficient altitude excess to complete maneuver.
● Perform instrument training maneuvers as described in Instrument FTI or as directed, full or partial panel, to include: ► Climbs and descents. ► Level speed changes. ► Timed turns. ● A brief description of the behavior, required action, and/or conditions. ► Turn pattern. ► Vertical S maneuvers. ▪ S-1 pattern. ▪ S-3 pattern. ► Slow flight maneuver.	● Effectively utilizes power to maintain airspeed ±10 knots. ● Maintains target VSI ±200 fpm. ● Levels off at desired altitude ±100 feet per FTI. ● Maintains altitude ±100 feet. ● Achieves and maintains target airspeed ±5 knots. ● Maintains standard or one-half standard turn rate to achieve desired heading change in appropriate time period, ±5 seconds. ● Uses indicated airspeed to appropriately determine AOB. ● Monitors turn needle and adjusts AOB as required to maintain standard or one-half standard turn rate. ● Effectively utilizes power to maintain airspeed ±5 knots. ● Maintains altitude ±100 feet. ● Performs turn reversals at target heading ±5 degrees. ● Maintains VSI ±200 fpm. ● Maintains ±5 KIAS of desired airspeed. ● Maintains AOB ±5 degrees. ● Reverses direction or level off ±100 feet of desired altitude. ● Maintains timing ±5 seconds. ● Makes timely and appropriate corrections for deviations. ● Reconfigures aircraft at appropriate airspeed, maintaining ±100 feet of target altitude.

BEHAVIOR STATEMENT	STANDARDS
5. MOA Maneuvers (cont.)	
	● Maintains target airspeed ±5 knots or on-speed AOA ±2 units. ● Establishes target ROD ±200 fpm.
Aerobatics	
● Perform aerobatic maneuvers per Instrument FTI, to include: ► Aileron roll. ► Wingover. ► Barrel roll. ► Loop. ► One-half Cuban eight. ► Immelmann. ► Split-S. ● Perform maneuvers listed above in a visual environment per FTI. In addition, perform squirrel cage.	● Verbalizes and attains target entry parameters (±5 knots, ±100 feet) prior to beginning the maneuver. ● Flies in a smooth, positive, and coordinated manner. ● Achieves and maintains target G load ±1 G and AOA ±2 units during overhead maneuvers. ● Executes rolling maneuvers at target attitude ±5 degrees. ● Exits maneuver at original entry parameters ±200 feet, ±10 knots, ±10 degrees. ● Plans maneuver entries to remain within area boundaries. ● Ensures primary emphasis during aerobatic maneuvers is on use of outside references. ● Efficiently links series of maneuvers.
Stall Series	
● Perform approaches to stall, full stalls, and recoveries per FTI, to include the following: ► Power-off stall. ► Break turn stall. ► Landing attitude maneuver. ► Landing attitude stall. ► Approach turn stall. ► Accelerated stall.	● Effectively trims aircraft for level flight and on-speed prior to commencing maneuver. ● Maintains altitude ±100 feet and VSI 0±200 fpm prior to stall. ● Recognizes approach-to-stall indications and recovers per NATOPS and FTI procedures, with no loss of altitude (recovery complete when two positive rates of climb established). ● Recognizes full-stall indications and recovers per NATOPS and FTI procedures with minimum loss of altitude ≤500 feet (recovery complete when two positive rates of climb established). ● Prevents entry into secondary stall; recognizes secondary stall, if entered, and recovers properly. ● Does not exceed gear or flap limitation airspeeds.

BEHAVIOR STATEMENT	STANDARDS
Unusual Attitude Recoveries	
● Perform recoveries per appropriate FTI for: ► Nose-high recovery. ► Nose-low recovery.	● Uses correct instrument flight references throughout recoveries. ● Recovers to level flight expeditiously without stalling or exceeding aircraft limitations. ● Recovers to level flight without excessive altitude loss, stall, or exceeding aircraft limitations. ● Recovery is complete when the descent is stopped.
OCF Recognition/Recovery	
● Performs OCF maneuvers per FTI, to include: ► High AOA, deep stall investigation. ► 70-, 90-, 110-degree departures. ► Lateral stick adverse yaw departure.	● Demonstrates in-depth knowledge of NATOPS OCF procedures and prohibited maneuvers. ● Correctly enters prescribed syllabus maneuvers per OCF FTI. ● Correctly applies recovery control inputs and procedures per OCF FTI.
6. Formation	
● Perform two- and four-ship formation as the Wingman per Formation FTI, to include: ► Parade or fingertip. ► Turns.	● Complies with Formation FTI. ● Effectively passes signals to successive wingmen while smoothlymaintaining position. ● Maintains parade position per FTI: wingtip separation -0, +5 feet, step-down ±5 feet, and bearing line ±10 degrees, using smooth, positive control inputs. ● Smoothly and positively corrects back to position within 5 secondswithout prompting from IP. ● Stacks level with lead ±5 feet and maintains fore and aft references duringroll-in, rollout, and VFR turn-away position. ● Stacks level with lead or interval ±5 feet.

BEHAVIOR STATEMENT	STANDARDS
6. Formation (cont.)	
► Cross-under or shuffle. ► Lead change.	● Maintains fore and aft references during roll-in, turn, and rollout. ● Crosses below lead or interval's jet wash (+0 to –20 feet) with constant trackcrossing rate, achieving target nose and tail clearance no farther aft than one aircraft length. ● Refuses or assumes lead within two seconds of initiation, using appropriate visual, voice, or light signals. ● Maintains target altitude ±100 feet and airspeed ±5 knots while in lead. ● Passes lead utilizing appropriate visual, voice, or light signals. ● Positively maneuvers aircraft to establish wingtip separation -0, +10 feet and step-down ±5 feet, and no further aft than cruise bearing line per FTI.
Cruise	● Smoothly and positively corrects back to position within 5 seconds without prompting from IP. ● Makes appropriate corrections with throttle, AOB, and pitch as required.
Admin Cruise	● As briefed for administrative or transit portions of the flight, initially establishes the aircraft on the lead's 30-degree bearing line with appropriate nose-to-tail separation and step down as per FTI. ● Cross-under as required per Formation FTI. ● Maintains appropriate lane during reversals. ● Smoothly and positively corrects back to position within 5 seconds without prompting from IP.
Tail Chase	
● Perform tail chase exercise per FTI.	● Utilizes proper lead, pure, and lag pursuit to maintain 2,000-4,000 feet in trail of lead. ● Maintains a 500-foot bubble from lead at all times.

BEHAVIOR STATEMENT	STANDARDS
Rendezvous	
► Breakup and Rendezvous. ► CV Rendezvous. ► Running Rendezvous. ► TACAN Rendezvous. ► Division Rendezvous.	● Maintains target airspeed ±5 knots during breakup turn and while in trail. ● Rolls out 1,000 ±200 feet in trail of lead or interval for 2-second break. 4- and 6- second breaks will increase trail distance per FTI. ● Maintains visual situational awareness to all aircraft ahead, with safe separation from interval. ● Expeditiously maneuvers to bearing line. ● Maintains a stable plane-of-motion, co-altitude with lead or interval. ● Recognizes and makes corrections without prompting to deviations in bearing line, fuselage alignment, and airspeed control while maintaining positive closure. ● Controls closure at the in-close position to effect smooth cross-under to echelon. ● Maintains situational awareness to all aircraft ahead with safe separation and closure to lead or interval. ● Maintains proper step-down ±150 feet below lead's altitude until on bearing line. ● Properly utilizes communication to control lead's lighting at night. ● Expeditiously establishes aircraft on lead's altitude. ● Positively corrects to bearing line and maintains a consistent controlled rate of closure throughout. ● During breakup and rendezvous, does not exceed maximum AOB for position per the FTI. ● Executes cross-under per the FTI at a speed that the aircraft could safely join into an open slot between two aircraft. ● Recognizes unsafe or excessive parameters and expeditiously initiates maneuver.

BEHAVIOR STATEMENT	STANDARDS
Rendezvous (cont.)	
► Underrun.	● Immediately responds to underrun command given by lead or IP. ● Day: Expeditiously arrives at perch position as defined in FTI ±50 feet. ● Night: Arrives outside lead's turn at 500 feet (±100 feet) below lead's altitude.
Recovery	
► Section break. ► Division break.	● Establishes aircraft in FTI parade position prior to the numbers or as briefed. ● Sets the briefed interval. ● Keeps lead on horizon. ● Arrives in trail of lead while configuring aircraft for landing. ● Established in FTI parade prior to the numbers or as briefed. ● -2 sets the briefed interval. ● -3 and -4 match break interval. ● Keeps lead on the horizon throughout break. ● Arrives in trail of preceding aircraft while configuring aircraft for landing.
Section Approach	
● Perform two-ship approach procedures while at altitude or under controlling agency as: ► Section lead.	● Complies with approach procedures and standards being flown. ● Lands in center of appropriate side of the runway. ● Maintains runway alignment after landing. ● Detaches wingman at appropriate time in a safe position for landing.

BEHAVIOR STATEMENT	STANDARDS
Section Approach (cont.)	
► Section wing.	● Performs landing checklist prior to signaling lead, and signals lead at appropriate time. ● Properly manages configuration and energy state to effect safe landingwhen detached. ● Rejoins safely and expeditiously within two miles during touch-and-go and rejoin. ● Maintains parade position parameters per Formation FTI during missed approach, matching lead's configuration changes via hand or radio signals. ● During Flight Separation on Final, gains adequate separation once detached, configures aircraft appropriately, and visually transitions to an appropriate glide-path prior to landing.
Formation Lead	
● Perform two-ship formation as Lead per Formation FTI, to include:	● Complies with Formation FTI and course rules, considering airspace andweather to plan maneuvers. ● Completes profile in a smooth manner without exceeding wingman'scapabilities and without degrading flight safety. ● Maintains a smooth, stable platform, avoiding abrupt power changes andmaintaining >80 percent rpm while monitoring -2. ● Utilizes proper communications and signals as lead. ● Maintains visual awareness of wingman.
► Departure.	● Monitors wingman during initial join-up. ● Communicates with ATC to effect join-up as necessary.
► Arrival.	● Effectively manages formation in order to arrive at proper parameters for briefed or contingency RTB profile. ● Communicates with ATC to effect brief or contingency RTB.

BEHAVIOR STATEMENT	STANDARDS
7. Tactical Admin	
● Perform tactical flight maneuvering and administrative items to include: ► G-warm. ► Armament system management. ► Combat (FENCE) checks. ► Knock-It-Offs (KIO). ► G and Fuel checks. ► PADS. ► Perform rejoins from combat spread. ► Battle damage checks. ► Deck Awareness.	● Performs G-awareness turns per CNAF M-3710.7U and TAC FTI. ● Ensures proper anti-G straining technique and proper anti-G suit operation. ● Maintains G-loading within NATOPS limits. ● Manages armament system to ensure proper program for current mission. ● Ensures proper system management after exercise has been terminated. ● Completes combat checks per FTI, and expeditiously reports FENCE check completion. ● Makes and responds to KIO calls per training rules. ● Safely maneuvers aircraft to de-conflict with other aircraft while returning to pre-briefed position. ● Knows and expeditiously maneuvers aircraft to proper position, altitude,and distance for next expected syllabus maneuver. ● Correctly reports post-maneuvering maximum "g" attained and current fuel state prior to arriving at PADS and without prompting from IP or Lead. ● Attains briefed engagement start parameters within tolerances: ► Airspeed: ±10 KIAS. ► Range: ±0.1 NM. ► Altitude: ±100 feet. ► Bearing: ±10 degrees. ● Expeditiously maneuvers to an appropriate rejoin. ● Maintains positive separation from flight members throughout rejoin. ● Completes battle damage checks per FTI, expeditiously reports completion. ● Maneuvering does not result in flight below the hard deck.

BEHAVIOR STATEMENT	STANDARDS
Tactical Lead	
► Perform two-ship lead tactical maneuvering. ► TACFORM lead.	● Ensures formation remains within assigned airspace. ● Executes turns per FTI, maintaining or regaining visual contact and mutual support with wingman. ● Performs target attacks per FTI. ● Maintains appropriate flight deconfliction. ● Complies with TACFORM FTI or as briefed. ● Considers airspace and weather to plan/execute all maneuvers. ● Executes turns/maneuvers per FTI. ● Utilizes proper communications and signals as tactical lead. ● Maintains visual awareness of wingman: ► Uses visual signals as briefed/published. ► Maintains or quickly regains visual contact and mutual support. ● Utilizes appropriate altitude/airspeed excursions as lead. ● Maintains target airspeed ±10 knots in turns, and altitude ±100 feet, utilizing appropriate deviations to correct for known air work errors.
Visual Lookout	
● Maintain an effective visual lookout doctrine in both a single-and multi-plane environment.	● Maintains or expeditiously gains or regains visual contact of lead, adversary, and wing through lookout or communications. ● Provides visual mutual support when visual. ● Demonstrates awareness of potential ground or airborne hazards. ● Positively responds to and maneuvers to avoid observed flight hazards.

BEHAVIOR STATEMENT	STANDARDS
8. Tactical Formation	
Combat Spread	
► Perform two- and four-ship tactical maneuvering in Combat Spread, toinclude the following:	● Maintains position per FTI or as briefed. ● Accomplishes responsibilities, including clearing, as briefed. ● Recognizes and complies with visual signals as briefed or published. ● Executes turns per FTI to roll out in combat spread; if not in proper position, make timely positive corrections. ● Maintains or quickly regains visual contact and mutual support.
Tac Wing Position	
► Perform two- and four-ship tactical maneuvering from the 'Tac Wing' position, to include the following:	● Maintains position per FTI or as briefed. ● Accomplishes responsibilities, including clearing, as briefed. ● Executes turns per FTI; if out of position, make timely corrections.
Tac Air Work	
► Perform two- and four-ship tactical maneuvering to include the following: ▪ Shackles/Shackle Exercise Tac turns. ▪ In-place turns/In-place Turn Exercise. ▪ Cross turns. ▪ Division wall. ▪ Fluid Four.	● Maintains target airspeed ±10 knots in turns, utilizing appropriate deviations to correct for known position errors.

BEHAVIOR STATEMENT	STANDARDS
Error Corrections	
► Combat spread.	● Utilizes appropriate altitude and airspeed excursions to maintain or regain bearing line ±10 degrees, abeam distance 0.8 to 1.0 NM, with 1,000 feet of step-up or down from lead. ● Regains position within 20 seconds out of turns. ● Executes briefed Combat Spread maneuvers from various PADS, to include 0.5-1.5 NM abeam and between 200 and 350 knots at co-altitude.
Timing/Sight Picture	
► Perform two- and four- ship tactical maneuvering from Combat Spread position, to include the following:	● Executes turns per FTI; if not in proper position, make timely positive corrections.
Gunsight Tracking Exercise	
► Perform gunsight tracking exercise per TACFORM FTI.	● Executes comm per TAC FTI. ● Recognizes attack window and executes attack window entry mechanics to arrive in the control zone with range, angles, and closure under control. ● Maneuvers aircraft per TAC FTI to achieve valid employment opportunity.
Advanced Tac-Form Maneuvering	
► Tactical maneuvering to include: ▪ Off-heading Tac turns. ▪ Co-altitude maneuvering.	● Maintains target ±10 knots in turns,utilizing appropriate deviations to correct for known position errors.

BEHAVIOR STATEMENT	STANDARDS
9. Operational Navigation (ON)	
● Perform low-level procedures, to include: ► Route entry. ► Altitude control. ► Time control. ► Course control. ► In-flight computation. ► Chart interpretation. ► Turns. ► Ridge crossing. ► Route abort or exit.	● Accomplishes required ATC coordination; visually identifies route entry; complies with entry time requirements; effectively maneuvers into route structure. ● Maintains target altitude -0, +200 feet AGL per FLIP AP/1B, unless obstacles or safety dictate otherwise. Avoids abrupt altitude changes. Ensures altitude and obstacle clearance per regulatory guidance. ● Maintains awareness of time, using appropriate adjustments to arrive at final checkpoint ±10 seconds of preplanned or amended ETA computed at route entry. ● Maintains planned course ±2 NM. Reaches each checkpoint within a ±2NM radius. Ensures flight remains within route borders. ● Computes appropriate adjustments to ensure course, time, and altitude standards are achieved. ● Identifies chart symbols with prominent landmarks along route. Navigates via dead reckoning or waypoint navigation, as applicable. ● Turns to maintain or achieve course control standards; maintains level to slightly climbing; turns per briefing or contract, ensures de-confliction with wingman or lead. ● Executes ridge crossing per briefing and regulations. Maintains briefed altitude (-0 to +300 feet) and above minimum airspeed. Does not exceed maximum bank.

BEHAVIOR STATEMENT	STANDARDS
9. Operational Navigation (ON) (cont.)	
► Weather response.	● Maintains safe, positive control. ● Per FLIP and regulatory guidance; coordinates with ATC; monitors wingman. ● Recognizes degrading weather on the route and aborts the low altitude environment to stay VFR. ● If unable to remain VFR and IFR conditions imminent, safely climbs avoiding terrain per route study, and coordinates with appropriate controlling agency.
Dead Reckoning Navigation	
► Perform visual navigation procedures and chart interpretation.	● Identifies chart symbols with prominent landmarks along route. ● Navigates via dead reckoning or waypoint navigation, as applicable.
10. Strike	
Strike Maneuvering	
● Maneuver own ship within section or division in air-to-ground environment into a position from where ordnance or simulated ordnance can be safely delivered.	● Flies all maneuvers per FTI or applicable stage lecture.
Pattern Communications	
	● Makes the appropriate pattern calls in timely manner.
Weapons Pattern Management	
● Spacer pass. ● System management. ● Interval. ● Pattern.	● Maintains FTI double-interval cruise position. ● Makes proper trim inputs, maintains balanced flight prior to pattern entry. ● Properly sets up stores page for loaded weapons and patterns to be flown. ● Selects appropriate target waypoint with run-in line setup. ● Maintains pattern altitude and visual contact with interval or remains atsanctuary pattern altitude until visual contact is regained.

BEHAVIOR STATEMENT	STANDARDS
Weapons Pattern Management (cont.)	
	● Maintains proper position in pattern and elevates to target pattern altitude once sight is regained. If required, maintains interval and rejoins at the prebriefed position. ► Airspeed: ±10 KCAS. ► Abeam distance: ±0.2 DME. ► Pattern altitude: ±100 feet.
Roll-in Mechanics	
● Roll-in.	● Utilizes appropriate AOB, application of G, and AOA to arrive at preplanned aim off angle for a specified delivery pattern, ±0.1 DME, adjusted for wind if necessary.
Release Parameters	
● System. ● Acquisition. ● Aim-point. ● Parameters. ● Safety.	● Basic A/G switchology to include: A/G mode selected, proper weapon selected, delivery mode, and master arm. ● Locates the proper target. ● At release, the target is under one-half of the CCIP cross. ● Dive angle ±5 degrees (<20 degrees planned) or ±10 degrees (≥20 degrees planned). ● Planned airspeed ±50 knots. ● NLT release: from -0 to +200 feet of planned release. ● G at release: 0.8-1.0. ● Applies 4 G within 2-3 seconds and maintains until velocity vector is above the horizon.
Off-Target Mechanics	
● Dive recovery.	● Executes proper off-target procedures per FTI to arrive within parameters of the abeam position following a delivery profile. ● Promptly recognizes recovery altitude and smoothly maneuvers to avoid low or rolling pullout or overstress.

BEHAVIOR STATEMENT	STANDARDS
Off-Target Rendezvous	
● Off-target Rendezvous.	● Expeditiously climbs and turns to sanctuary altitude. ● Maintains sanctuaryaltitude ±100 feet. ● Gains and maintains sight of all preceding aircraft. ● Once on sanctuary altitude, expeditiously maneuvers to bearing line. ● Once on bearing line, maintains ±10 degrees of bearing line and +100 feetof lead's aircraft. ● Timely transmission of 'visual or blind' status.
Error Corrections	
► Strike Tracking and Error Corrections.	● Maintains within ±15 degrees of planned run-in heading. ● Maintains within ±5 degrees of target dive angle. ● Executes error sensitivity techniques for deviations from the plannedrelease parameters.
Accuracy	
● Depending on CEP accuracy, achieve the MIF.	● Calculations for CEP require a minimum of four bombs dropped. ● If CEP is ≤50 feet, grade as a 5. ● If CEP is 51-100 feet, grade as a 4. ● If CEP is 101-150 feet, grade as a 3. ● If CEP is ≥151 feet, grade as a 2.
Hung Ordnance Approach	
● Hung ordnance checks. ● Hung ordnance approach.	● Executes visual inspection for unexpended ordnance as well as anintegrity check of other flight members. ● Executes proper hung ordnance procedures per FTI or flight brief.

BEHAVIOR STATEMENT	STANDARDS
11. Target Area Mechanics	
Air-to-Surface Checklist	
● Accuracy. ● Timeline.	● Correctly completes all items of the Air-to-Surface checklist per FTI. ● Completes all items of the Air-to-Surface checklist with the exception of 'Master ARM' to 'ARM', prior to 8 NM. ● Places 'Master ARM' to 'ARM' prior to the RIP.
Target Area Holding	● Performs assigned holding procedures and maintains parameters: ► Altitude: +/- 150 feet. ► Groundspeed: +/- 10 knots.
Time-on-Target	
● Time-on-Target. ● Course corrections.	● Actual TOT within 30 seconds of issued TOT. ● Adjusts holding as required to correct for timing.
Target Area Mechanics Comm	
● TAM Read-backs.	● Target elevation and coordinates are read back off of the WYPT DATA sublevel of the MFCD. ● Read-backs are clear and conducted in the correct order, with noextraneous data. ● Read-backs contain no incorrect information.
Section Attack Mechanics	
● Timeline Adherence. ● Action Mechanics. ● Egress and De-confliction.	● Initiates action for section same-side attack at 6.0 +/- 0.1 NM to the target. ● Initiate action maneuver in the correct direction. ► AOA: 14 +/- 1 unit. ► Groundspeed: +/- 10 knots. ► Action offset heading: 90 +/- 5 degrees. ► Offset wings-level timing: 15 +/- 1 second. ● Egresses target in appropriate direction, correcting back to assigned altitude. Maintains or regains situational awareness of Lead aircraft, and executes rejoin into assigned formation in timely manner.

BEHAVIOR STATEMENT	STANDARDS
12. Training Rules	
● All environments. ► BFM and A/G. ► Minimum Altitude Awareness. ● BFM training rules.	● Demonstrates a thorough knowledge of training rules. ● Correctly applies training rules to varying scenarios. ● Maintains briefed altitude de-confliction game plan. ● Initiates or responds to knock-it-off procedures correctly and appropriately. ● Following a knock-it-off transmission, ceases maneuvering and properly maneuvers the aircraft to maintain safety of flight and SA. ● Maintains proper separation from aircraft. ● Recognizes flight below the hard deck, floor, or minimum altitude and responds correctly per training rules. ● Adheres to BFM training rules. ● Recognizes soft deck parameters and responds per training rules. ● Responds to de-confliction radio calls in a timely and appropriate manner. ● De-conflicts correctly when radio transmissions are not made.
13. BFM	
● Demonstrates CNATRA training rules knowledge and adherence. ● SSD.	● Recognizes and acknowledges pending and current deviations from training. Begins on PADS, adheres to training rules, quickly and smoothly solves for POM, range, and lead with valid shots, executes timely reversals, and maintains proper geometry. ● Utilizes clear and concise communications during shots and able to recognize and solve for maneuvers. ● ***Defensive***: Begins on PADS, adheres to training rules, and provides a stable, predictable platform during the non-maneuvering runs to allow shooter to maximize training.

BEHAVIOR STATEMENT	STANDARDS
13. BFM (cont.)	
● Horizontal scissors.	● Executes timely reversals, maintains proper geometry, and performs timely maneuvers to defeat impending shots from wingman. ● ***Offensive***: Begins on PADS, executes proper throttle mechanics on countdown, and aggressively maneuvers nose up and LV aft. ● Quickly and efficiently stabilizes at target airspeed of 130-150 knots. ● Executes timely reversals and LV placement to remain offensive and maneuver to employ weapons. ● Recognizes shot opportunities and if bandit redefines, aggressively follows and executes proper deck transition. ● ***Defensive***: Begins on PADS, executes proper throttle mechanics on countdown, and aggressively maneuvers nose up and LV aft. ● Quickly and efficiently stabilizes at target airspeed of 130-150 knots. ● Executes timely reversals and LV placement to either neutralize the fight or gain an offensive advantage. ● Recognizes impending WEZs and defends appropriately and redefines when the opportunity presents itself using the proper deck transition.
● Rolling scissors.	● ***Offensive***: Begins on PADS, executes proper throttle mechanics on start, and aggressively maneuvers to gain proper corkscrew altitude. ● Executes timely corkscrew with proper mechanics, and minimal altitude loss. ● Recognizes winning or losing and employs weapons in LAR. ● Applies constant pressure to adversary, recognizes the redefinition, aggressively follows, and transitions appropriately.

BEHAVIOR STATEMENT	STANDARDS
13. BFM (cont.)	
● 6k and 9k perch. ● LAR recognition. ● WEZ recognition. ● Deck awareness. ● Defensive break turn. ● AWE timing recognition. ● Energy management.	● ***Defensive***: Begins on PADS, executes proper throttle mechanics on start,and aggressively maneuvers to defend and force the overshoot. ● Executes proper and timely corkscrew mechanics and recognizes winningor losing. ● Recognizes impending WEZs and defends appropriately. ● Redefines and utilizes appropriate deck transition. ● ***Offensive***: Proper start and entry per FTI, timely communications, and Fox-2 taken to begin set. ● Recognizes attack window, and executes proper attack window entry. ● Captures proper airspeed and works to employ weapons without sacrificing positional advantage. ● Recognizes employment opportunities and continues to apply pressure to adversary throughout set, remaining offensive. ● ***Defensive***: Proper start and entry per FTI, timely reversal to set up impending shot, and executes proper defensive break turn mechanics. ● Captures at appropriate time and keeps sight of adversary throughout engagement. ● Recognizes impending WEZs and defends appropriately. ● Recognizes redefinition opportunities and denies threat employment opportunities. ● Knows and is able to execute valid shots from the CNATRA Sidewinderand Gun envelopes. ● Recognizes a pending WEZ and applies appropriate DBFM mechanics to deny shot. ● Able to assess the bandit's nose or range. Makes a timely "Chaff, Flares" call andexecutes a defensive break turn into the bandit. ● References the 10-percent rule to execute a proper deck transition and arrives slightly above the hard deck with appropriate energy.

BEHAVIOR STATEMENT	STANDARDS
13. BFM (cont.)	
● Fight redefinition recognition and selection. ● Butterfly set and abeam set.

BEHAVIOR STATEMENT	STANDARDS
13. BFM (cont.)	
● Attain or maintain proper aircraft energy state for a given tactical situation. ● High-aspect BFM.	● Demonstrates knowledge and understanding of T-45C EM diagram. ● Effectively utilizes control and throttle movements to gain and sustain best performance available for a given tactical situation. ● Correctly maneuvers to maintain or change energy state based on one-circle or two-circle engagements. ● Applies the fundamentals of high-aspect BFM per the FTI. ● Uses aggressive out-of-plane maneuvering and lift vector placement to become offensive or maintain neutral position with the bandit. ● Applies the proper one-circle or two-circle mechanics. ● Recognizes transition to offensive or defensive and engages appropriately.
14. Fighter Intercepts	
3D Formation Management	
► Position Keeping. ► Energy Management. ► ATA Management. ► Mechanics. ► Error Correction.	● Able to maintain expected position in Combat Spread per the TACFORM FTI, and adjust formation as required for tactical employment during crank and recommit. ● Maintains correct airspeed throughout intercept +/- 15 knots. ● During crank, cold ops, and recommit, maintains correct ATA +/- 5 degrees (or heading assigned by flight lead). ● Executes shot, drag defense, and recommit mechanics per FTX FTI. ● Errors are corrected in a timely manner. ● During drag defense and recommit, attempts are made to fix formation errors via manipulating the timing of turns.

BEHAVIOR STATEMENT	STANDARDS
Fighter Intercept Comm	
● Timeliness. ● Accuracy.	● Fighter communications are made at the correct ranges per the Fighter timeline described in the FTX FTI. Communications do not negatively affect timeliness of Fighter maneuvers. ● Communications are made correctly per the FTX FTI. Sort is correctly added to shot communications with regard to wingman position in the formation.
System Management	
► Sanitization. ► Meld. ► Cold Ops.	● Prior to executing each intercept, all systems are set up per the sanitization set as described in the FTX FTI. ● Once targeting is received, all systems are set up per the meld system setup as described in the FTX FTI. ● During cold ops, all systems are set up per the 'WOR' checklist as described in the FTX FTI.
Cold Ops	
► Formation Management. ► Checklist. ► Altitude Management.	● Drag defense mechanics are executed to arrive with the threat at the 6 o'clock position (or heading assigned by flight lead) +/- 5 degrees. ● System setup per the "WOR" checklist in the FTX FTI. ● Altitude is managed during drag defense and cold ops to remain inside the assigned airspace, decelerate no slower than 330 knots, and arrive as close to 15,000 feet MSL as fighter tempo allows, maintaining combat spread.
Recommit	
► Position Keeping. ► Mechanics. ► ATA Management.	● Attempts to fix position errors during recommit turn. ● Executes MIL-powered, no slower than 330 knots, level turn during recommit per FTI +/- 5 degrees. Executes two shots, then directs the section to defend via appropriate communications no later than MAR.

BEHAVIOR STATEMENT	STANDARDS
Employment	
► Shot Mechanics. ► Energy Management.	● Shot is executed with 2-3 G's +/- 0.5 G's. ● Shot is executed 30-degrees nose-high +/- 5 degrees. ● Shot is executed within the Shot Window. ● After shot and during transition to crank, energy and AOA is managed inorder to not exceed 17 units AOA.
Timeline Adherence	
► Timeline Adherence.	● All maneuvers are executed no earlier than the correct range +0.2 NM, and no later than the correct range.

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Chapter X

Master Materials List

1. Individually Issued Materials

TITLE	IDENTIFICATION	QTY PER NFS
a. T-45C Combined MPTS Flight Training Curriculum	CNATRAINST 1542.167D	1
b. Flight Training Instructions (FTI)	CNATRA PAT PUB P-1204 through P-1291 as applicable	9
c. DOD FLIP Publications		
(1) En route IFR Supplement U.S.		3
(2) En route High Altitude Chart (H1, H2)		6
(3) Terminal High Altitude Instrument Approach Procedures (NW, SW, E)		6
d. TRAWING In-Flight Guide	Locally produced and issued	
e. Aviation Training Jacket	CNATRA-GEN 1542/10A	1
f. Pilot Training Summary	CNATRA 1542/95	1
g. Jacket Review	CNATRA-GEN 1542/66	1
h. Academic Lesson Guides		
(1) Aviation NFS Indoc (ASI)	CNATRA P-1277	1
(2) Aerodynamics (AERO)	CNATRA P-1279	1
(3) Engineering (ENG) Book 1 and 2	CNATRA P-1278	1
(4) Instrument Navigation (INAV)	CNATRA P-1282	1
(5) Meteorology (METRO)	CNATRA P-1280	1
(6) Operational Navigation (ON)	CNATRA P-1224	1

(7) Instrument Rating Flight Procedures	CNATRA P-1245	1

2. Support Materials

TITLE	IDENTIFICATION	QUANTITY
a. T-45C NATOPS Flight Manual	NAVAIR A1-T45AC-NFM-000	255
b. T-45C Pocket Checklist	NAVAIR A1-T45AC-NFM-500	255
c. T-45C NATOPS Flight Manual (performance charts)	NAVAIR A1-T45AC-NFM-300	255
d. NATOPS Instrument Flight Manual	Stock No. 0437LP9001019	50
e. NATOPS General Operating Instruction	CNAF M-3710.7	25
f. Aeronautical Information Manual (AIM)	FAA Publication	100
g. Flight Clothing	(Identification and quantity listed in CNATRAINST 10126.1; cost listed in NAVSUP PUB. 4100.)	
h. Aviation Training Forms are generated by the Training and Learning Management System (T/LMS)		
i. T-45C Standard Operating Procedures (SOP)	COMTRAWINGONEINS T 3710.7T/ COMTRAWINGTWOINS T 3710.7R (Locally produced and issued)	1
j. Lecture Guides	CNATRA PAT PUBS	18

3. Aircraft and Major Training Devices

TITLE	IDENTIFICATION
a. Aircraft	T-45C
b. Instrument Flight Trainer (IFT) Nonvisual Simulator	Device 2F137 or 2F137C
c. Operational Flight Trainer (OFT) Visual Simulator	Device 2F138C, D, or E
d. Virtual Training Device	T-45C VTD

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